Appendix B — Argumentation

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1 Introduction

This appendix records the argument of the main text as an explicit, *defeasible* argument map. The claim being audited is the thesis itself, as the argument graph states it: that CIF can be engineered from its *performance targets* instead of from a tuned per-gap schedule; that its yield follows a closed-form law with no fitted constant and its selectivity a derived law with a single calibrated prefactor; that the two are *provably decoupled*, so the size threshold and the filtrate fraction target are independent design *inputs* rather than coupled outcomes; and that all of this holds inside a validity envelope the design tool checks: it refuses a design that leaves the plateau or the Reynolds gates, and it reports, without refusing, a design over its length budget. The plain-English rendering compresses those four to three (engineerable from targets, on decoupled yield and cutoff laws, inside a checked envelope), and this appendix argues them in that order. Both performance axes are therefore in scope, not the cutoff alone. The map has two storeys accordingly: the lower one is the cutoff law (C1–C3′, with C5–C7 corroborating into it and the yield-side separation lemma C4 drawn in but deliberately *disconnected* from its spine), and the upper one is what that disconnection buys: the design map the separation, with the plateau schedule, makes locally invertible (C8), and the five-leg accrual over the operating law, the separation lemma, the map's inversion, its realization and the declared scope, that is the thesis. C4 is the hinge between them: drawn low, joined to nothing in the cutoff chain, and a premise of everything above it. C9, the transport lemma, belongs to neither storey: it upgrades C1's steric premise and, within the model's premises, closes C3d's residual list, without lying on any path to the thesis.

This appendix's purpose is auditing rather than exposition: each load-bearing claim is stated as a separate node, together with the support it rests on, the named reasoning scheme that licenses it, and, most importantly, the residual risk, i.e. the critical questions that remain unanswered. The closed-form derivations the chain invokes are proved line by line in Appendix A; the symbolic results are reproduced by the reference code `cif_cutoff.py` (its `verify_all()` self-tests) and the empirical and CFD results by that program's frozen Part II regression stamps, both recorded in the derivation record [8]. This appendix is the connective tissue between them: it does not re-prove the mathematics, it certifies that every link in the argument has been examined and states exactly what is, and is not, yet settled. In brief: the cutoff law is *proven in form* and *empirically solid in its apparent prefactor*; the separation $F \perp\!\!\!\perp f_{\mathrm{gap}}$ is proven outright and F's absolute magnitude independently validated against 3-D simulation; and the target-to-geometry map inverts, inside a declared and machine-checked envelope. What is *not* settled is stated node by node rather than summarised away. The argument graph carries 10 undischarged critical questions distributed over 7 of its 68 argument nodes. Two of those deserve naming here. The lumped residual $\sqrt{c}\,C$, tied to the unpublished gap count N, is the one the *interpretation* of the cutoff prefactor turns on (C3d): it is one item on that list rather than the whole of it. And exactly one of the ten is a *live rival* rather than a residual: the $f_{\mathrm{eff}}(d)$ account of the cutoff knee, which C3c argues against and does not eliminate, and which the graph marks with its own status for that reason. Ten undischarged questions and one live rival are two different counts of two different things; both are carried below. The full justification of that assessment is the business of the table below.

1.1 How to read the table

Each node is a *defeasible* argument in the sense of Walton, Reed & Macagno [16] (hereafter WRM 2008): a claim follows from its premises *unless* one of the governing scheme's critical questions (CQs) is answered against it. The appendix instantiates 13 distinct schemes (twelve of WRM's, plus *accrual* as a local tag the thesis section declares as such), of which defeasible *modus ponens* is the most common and Argument from Expert Opinion

governs C1. Each node names its own scheme. A node is therefore never simply "true" or "false"; it carries a calibrated status and an explicit list of what could still defeat it. The statuses below are the ones the argument graph defines, and the table uses no others; each says for itself whether it leaves a critical question open.

- **proven.** A deductive identity (symbolic or circuit-theoretic); no open critical question.
- **warranted.** Strong defeasible support, with every critical question material to the claim answered and residual risk minor or none. A qualifier ("on CIF data," "on the form," "uniform-f_{eff}-robust," "conditionally") names the scope within which the warrant holds; it narrows a warrant without being a status of its own.
- **resolved.** A definitional or adjudicative question, settled.
- **validated.** An empirical *magnitude* confirmed against independent data. This is a positive finding, not a weakening: it says the number was checked, not that the claim is in doubt. C4 carries it beside *proven*, because the separation and the magnitude are two different claims.
- **warranted as a band.** Warranted only as a band or consistency check, not as a discriminating test (C7).
- **warranted (existence).** The *existence* claim is warranted; the mechanism is contested separately (C3c's existence half).
- **warranted; rival open.** Warranted, with one named rival left explicitly open (C3c's mechanism half).
- **partially closed.** Some factors of the claim are settled (often by direct simulation) while at least one named quantity or critical question is still open (C3b, C3d, C8).
- **defeated.** A rival kept visible so that its defeat is explicit, rather than deleted. Four nodes carry it.
- **open rival.** A rival that is *not* defeated: the appendix's one live open question, and the only status the graph marks toggleable. Exactly one node carries it: the $f_{\text{eff}}(d)$ account of the knee, which C3c argues against and does not eliminate. It is what the Dung projection reports as the single two-way fork under full stress.

A status is *qualified* whenever something remains open, and this appendix never prints a bare "warranted" or "proven" over an open critical question; where it once did, the table now carries the qualifier and the per-node section names the open item by the identifier the graph uses for it. The partition is the graph's: a node with no undischarged exception carries *proven*, *warranted*, *resolved* or *defeated*; one with an undischarged exception must carry a qualified status.

CQ numbering is local. The CQ numbers in this appendix are *this appendix's*, not WRM 2008's, and the two differ. For Argument from Expert Opinion this appendix numbers CQ1 expertise, CQ2 assertion/backup, CQ3 consistency, CQ4 trustworthiness/independence, CQ5 the steric criterion; WRM 2008 (p. 15) numbers CQ1 expertise, CQ2 field, CQ3 opinion, CQ4 trustworthiness (bias as 4.1), CQ5 consistency, CQ6 backup evidence. For C2's analogy CQs the mapping is local CQ1 (relevant differences) = Walton CQ3, local CQ2 (base-case truth) = Walton CQ1, local CQ3 (contrary instances) $\approx$ Walton CQ4 (counter-analogy); Walton CQ2 has no local-numbered counterpart because it is discharged by the similarity premises rather than by a numbered question. The local numbering is retained ("CQ5" is the caption adjudication and is cited a dozen times), and the mapping is stated here instead.

Where a node bundles a proven identity with an empirical calibration, so that part of it is deductive and part defeasible (e.g. C2's exact cubic together with its fitted prefactor, or C6's algebraic exponents together with their measured values), the status names the *weaker*, defeasible component; the proven part is recorded in the per-node detail. That rule is about the parts of a *single* node. It does not carry to an *accrual*, whose premises are other nodes: an accrual adds no inference of its own beyond the conjunction, so it inherits a weaker premise's conditions as conditions *of its statement* rather than as its own status. The thesis is the case in point, and it says so where it is argued.

The four columns are the node label; the claim in one line; the status, with the governing Walton scheme and the convergent support; and the open critical questions that constitute the node's residual risk.

1.2 The backbone

The argument has two storeys. The lower one is a single chain with corroborating side-branches and one deliberately disconnected lemma; the upper one spends that chain on an engineering claim:

$$\text{C1 (mechanism)} \longrightarrow \text{C2 } (\delta = w_c\sqrt{f_{\text{gap}}/3}) \longrightarrow \text{C3}' \, (d_{\text{crit}} = \kappa_{\text{app}}\,\delta),$$

$$\text{with C5, C6, C7 corroborating} \quad \text{and} \quad \text{C4: } F \perp\!\!\!\perp f_{\text{gap}} \text{ a separation lemma;}$$

$$\{\text{C2, C3a, C4, S1}\} \longrightarrow \text{C8 (design map)} \longrightarrow \{\text{C3}', \text{ C4, C8, scope}\} \longrightarrow \text{THESIS,}$$

$$\text{with C9 (transport lemma) underpinning C1 and C3d.}$$

Mechanism (C1) fixes that the cutoff is set by the diverted-lamina width; the hydrodynamic conversion (C2) turns that width into a closed form; the operating law (C3′, decomposed into C3a–C3d) promotes it to a particle-diameter cutoff and calibrates its prefactor; the Xia scaling (C5), the exponents (C6), and the Dinh consistency band (C7) corroborate; and the filtrate-fraction identity (C4) is proved *independent* of the gap fraction, so device yield carries no cutoff information. On the upper storey, that independence is what makes the target-to-geometry map locally invertible (C8), and the thesis is an accrual over five legs: the operating law, the separation lemma, the locally invertible map, its realization, and the declared scope; the display above writes C8 once for the last two, which the graph carries as two premises. C9 is the odd one out: it derives $\kappa_{\text{steric}} = 2$ from the flow model rather than from citation, which *upgrades* C1's premise without adding a branch, and it is therefore attached to the chain without lying on any path to the thesis.

Figure 1 renders both storeys in one view, which is the map to read first. Each of the upper-storey nodes then has a detailed map of its own: Figure 11 for the transport lemma, Figure 12 for the design map, and Figure 13 for the thesis and its five legs, which is also where the corroborating branches are shown for what they are: nodes with no support path to the thesis at all.

1.3 Notation and conventions

The subject law is

$$d_{\text{crit}} = \kappa_{\text{app}}(d/w_c,\ \text{depth})\ w_c\sqrt{f_{\text{gap}}/3}, \qquad \delta = w_c\sqrt{f_{\text{gap}}/3},$$

where δ is the deep-channel skim width: the leading term of the exact dividing-streamline inversion of the skim cubic $f_{\text{gap}} = 3x^2 - 2x^3$ (the Inglis ε-cubic, [12, p. 656]; $x = \delta/w_c$). Here κ_{app} is the *apparent* prefactor, which maps that idealized skim width onto the measured cutoff diameter; it is defined against the *nominal* design f_{gap} and the parabolic δ. κ_{app} **is a plateau-region constant, not a universal one.** It is constant only for $d/w_c \gtrsim 0.035$, where $\kappa_{\text{app}} = 4.91$; below that the prefactor becomes a function of the cutoff itself and the law above becomes *implicit* in d_{crit}, which is why C7's bracket device, below the knee at its initial gap fraction, has no stable fixed point above the sub-steric foot to solve for. The prefactor ramps through that knee to a *sub-steric (recovery-suppressed)* foot $\kappa_{\text{app}} \approx 1.34$ ($d/w_c < 0.012$), which is *below* even the *depth-corrected* steric floor $\kappa_{\text{steric}} K^{\text{CFD}}_{\text{depth}} \approx 1.782$ at the shallowest probed aspect ratio (the bare steric floor is $\kappa_{\text{steric}} = 2$; the two are distinguished wherever both appear), because small-particle recovery is incomplete ($\delta_{\text{eff}} < \delta_{\text{nominal}}$; C3c). Every numerical use of κ_{app} in this appendix is a plateau-region use unless it says otherwise. The plateau prefactor factors as

$$\kappa_{\text{app}} = \kappa_{\text{steric}} \cdot K_{\text{in-plane}} \cdot K_{\text{depth}} \cdot \sqrt{c} \cdot C = 2 \cdot 1 \cdot R(\alpha) \cdot \sqrt{c} \cdot C,$$

with $\kappa_{\text{steric}} = 2$ (cutoff radius = lamina width, hence $D_c = 2b$, where b is Inglis's first-streamline width, re-lettered in the second convention below; corroborated by [12, p. 656] and [5, p. 917] main text, and *derived* from the flow model in C9 rather than taken on citation), $K_{\text{in-plane}} = 1$ (single-gap CFD, [7]), and $R(\alpha) = K_{\text{depth}}$ the finite-depth factor [6, p. 875]. These three, and the residual $\sqrt{c}\,C$, are each carried in the derivation record [8], which is the numerical authority for their values.

Five conventions are fixed here to forestall confusion.

- **C7 and C8 are the *graph's* numbers, not the prospectus's.** The labels C1–C9 in this appendix are the argument graph's node identifiers: the graph took them from this appendix, and holds the canonical `Graph-*` numbering. Two of them collide with prospectus contribution numbers: in *this appendix* C7 is the Dinh held-out consistency band and C8 is the design map, whereas the *prospectus* C7 is

the targets-to-photomask tool and its C8 the machine-enforced validity envelope. Different objects, the same numerals. The graph resolves this by namespacing rather than renumbering, and asks for explicit `Graph-`/`Prospectus-` prefixes wherever either is cited. This appendix adopts the resolution and narrows the prefix: the two section headings that carry the collision say "(graph)" in the heading itself, and everywhere else here an unqualified C7 or C8 is the graph's, because a reader of this appendix meets a prospectus number in exactly one place, the thesis section's scope note, which is also where the reason the prospectus pair carries no argument node at all is recorded.

- **First-streamline width vs. the size-scaling exponent.** Two distinct quantities are written with the letter β/b across these literatures and must not be conflated. Inglis (2006) denotes the first-streamline width by the Greek letter β (his Fig. 1 caption: "β is the width of the first streamline," [12, p. 655, Fig. 1 caption]), with critical diameter $D_c = 2\beta$ [12, p. 656, eqn (1)]. To avoid collision with the CIF size-scaling exponent $\beta = \frac{1}{2}$ $(= \partial \ln d_{\mathrm{crit}} / \partial \ln f_{\mathrm{gap}})$ used throughout this manuscript, we render Inglis's width as lowercase Latin b wherever it appears; thus his $D_c = 2\beta$ is written $D_c = 2b$. The re-lettering is purely notational and leaves $\kappa_{\mathrm{steric}} = 2$ unaffected.

- **The lumped residual, two equivalent conventions.** The data-fixed excess of the apparent prefactor over the steric value can be written with the depth factor $R(\alpha)$ either pulled out or absorbed; both denote the same number. Keeping $R(\alpha)$ explicit (the convention of this appendix and Appendix A), $K_{\mathrm{depth}}\sqrt{c}\,C = \kappa_{\mathrm{app}}/\kappa_{\mathrm{steric}} = 2.4569$ (depth-folded, and α-independent), so that dividing out the *measured* $K_{\mathrm{depth}}^{\mathrm{CFD}}$ leaves the depth-separated $\sqrt{c}\,C = \kappa_{\mathrm{app}}/(\kappa_{\mathrm{steric}} K_{\mathrm{depth}}^{\mathrm{CFD}}) = 2.7574/2.5999/2.5044$ at $\alpha = 0.80/1.40/2.69$: a triplet, not a range, and each value belongs to its own α. The reference code `cif_cutoff.py` [8] instead *folds* K_{depth} into the residual and reports $\sqrt{c}\,C = \kappa_{\mathrm{app}}/\kappa_{\mathrm{steric}} = 2.4569$; the two are the same quantity with K_{depth} either pulled out (here) or absorbed (there), and the glossary's rule applies: *never print the two conventions as one range.*

- **Aspect ratio.** Throughout this appendix $\alpha \equiv h/w_c$, depth over central-channel width, and it is *not* sorted to $\alpha \geq 1$. The reference mathematics takes the same ratio as a parameter literally named `beta` (`c2(beta)`, `K_depth(beta)`, evaluated at 0.80/1.40/2.69): when reading code, `beta` is this appendix's α, the same quantity under a different name, *not* a reciprocal. When reading this appendix, β is never an aspect ratio; it is the cutoff-law scaling exponent. One further correspondence with the glossary, stated so that a reader moving between them is not misled: this appendix's d/w_c is the glossary's $\widetilde{d}$, the same quantity under a different spelling. The steric prefactor, the glossary's κ_{steric}, is printed the same way here.

- **Depth factor: three triplets, kept apart.** Appendix A carries *three* depth-factor triplets and this appendix follows its symbols. Two are taken over the aspect ratios probed by direct simulation ($\alpha = 0.80/1.40/2.69$): $K_{\mathrm{depth}}^{\mathrm{th}}$, the *analytic* rectangular-duct factor [6, p. 875] (0.8879/0.9500/0.9794), and $K_{\mathrm{depth}}^{\mathrm{CFD}}$, the *measured* single-gap CFD factor (0.891/0.945/0.981). They agree to at most 0.531%, and where only the band matters this appendix writes $R(\alpha) \in [0.89, 0.98]$ for either. The *third* triplet is different in kind, and it is where this appendix's one acknowledged assumption lives: the analytic factor evaluated not at those aspect ratios but at the three Gifford Fig. 3 *calibration widths.* That figure's caption reports the widths and nothing else: "Effect of central channel width ($w_c = 100$ μm, 125 μm, 150 μm)" [1, p. 4500, Fig. 3 caption]. So the calibration array has *no stated depth.* The 80 μm used to form α is the depth of the *separate* Fig. 4 demonstration device: "The depth of the device was 80 μm $\pm$ 3 μm; the width of the central channel was 100 μm" [1, p. 4500, Fig. 4 caption]: the caption specifically, because the body sentence on that same page states the depth without the tolerance. Carried across the three widths it gives $\alpha = 0.8000/0.6400/0.5333$ and $K_{\mathrm{depth}}^{\mathrm{th}} = 0.888/0.849/0.811$, an 11–19% correction, *not* negligible, and largest at the widest channel. The true array depth is not known and is not inferred here (the glossary, "Calibration depth"), and 80 μm is not the only depth the paper reports: the four high-ratio devices of its Fig. 5 are "Four 125 μm-deep CIF devices with values of f_{gap} ranging 2.2–10.0 $\times 10^{-4}$ were tested" [1, p. 4501, Fig. 5 caption], at the *same* 100 μm central channel as the Fig. 4 device, and the Fig. 6 platelet-enrichment device is "The depth of the device was 150 μm $\pm$5 μm; the width of the central channel was 125 μm" [1, p. 4502]. Because K_{depth} rises monotonically towards 1 with α, either of those deeper assumptions would make the correction both *smaller* and *more uniform* across the three widths. Taking the shallowest reported depth is therefore the *conservative* reading, and, if the array is at least

that deep, the 11–19% above is an upper bound rather than an estimate. A number from one triplet must never be quoted against another. Throughout, K_{depth} is folded into the *apparent* prefactor as a per-device interpretive factor; it is *not* applied to the data, which collapse more tightly on the clean parabolic δ, so it does not shift the fitted κ_{app} and enters only the residual decomposition.

2 Claim-by-claim argumentation table

Table 1 is the proof summary: every node of the backbone, its status under its Walton scheme, and its residual risk. The per-node premises, verbatim source quotations, and full critical-question lists are collected in the detailed treatment that follows this summary.

Table 1: Claim-by-claim argumentation map for the whole argument: the cutoff law (C1–C7), the transport lemma (C9), the design map (C8) and the thesis. Status levels are defined in §1.1 and taken from the argument graph's own legend: *proven*, *warranted*, *resolved* and *defeated* carry no open critical question; *validated*, *warranted as a band*, *warranted (existence)*, *warranted; rival open*, *partially closed* and *open rival* are the graph's *qualified* statuses, each of which may stand over an undischarged exception, and the right-hand column says which. That partition is the graph's own (`meta.convention`), not this appendix's. A status may pair two of these where a node bundles two claims, as C4 does: *proven* for the separation and *validated* for the magnitude, the latter carrying the node's one open exception.

Node	**Claim**	**Status (Walton scheme; support)**	**Open critical question (residual risk)**
C1	The cutoff is set by the diverted-lamina width, not the gap width ($a \lesssim w$, with Dinh's w being our δ; $d_{\mathrm{crit}} \approx 2\delta$).	**Warranted.** Expert Opinion (WRM 2008), convergent over five peer-reviewed sources spanning *three* independent groups (the graph's own `accrual_over` list): the originating CIF lab (Gifford [1]; Lezzar [4]; Dinh [5]: all from the Shevkoplyas lab) plus two independent confirmations (Yamada & Seki [11]; Inglis & Herman 2013 [13]).	$a < w$ vs $a < w/2$ *resolved* $\rightarrow \kappa_{\mathrm{steric}} = 2$ (Dinh *main text*, [5, p. 917], + Inglis's $D_c = 2\beta$, our $2b$, [12, p. 656]); the Fig. 1 "about half" caption [5, p. 916] is *deselected*, not refuted. Residual: none on the mechanism.
C2	$\delta = w_c\sqrt{f_{\mathrm{gap}}/3}$, the leading term of the exact cubic $f = 3x^2 - 2x^3$.	**Warranted on the form.** A proven, code-checked identity (deductive), reinforced by Analogy (DLD, [12, p. 656]) and Expert Opinion (Inglis & Herman's identical near-wall-lamina integral, [13, p. 1726]); exact, Inglis-identical, CFD-confirmed ($\delta_{\mathrm{skim}} \propto \sqrt{f}$, $R^2 \geq 0.9999$, and agreement with the exact rectangular-duct field to $\leq 0.6\%$, both [7]).	None on form. The cutoff is fit on the parabolic δ; the finite-depth factor $R(\alpha)$ is folded into the apparent κ, not applied to the data.
C3a	$d_{\mathrm{crit}} \propto$ lamina width ($\kappa \equiv d_{\mathrm{crit}}/\delta$).	**Warranted.** Expert Opinion over four sources (Xia assumption (b), [2]; Inglis $D_c = 2b$, [12, p. 656]; Dinh's C1 criterion, [5]; Inglis & Herman's design assumption, [13]) *and* Argument from Example: the measured exponents select the form, $\beta = 0.51$ excluding the Hele-Shaw $\beta = 1$. The second leg is what keeps the operating law, C8 and the design map from resting on expert testimony alone: they survive the withdrawal of every source.	The *value* of the constant $\rightarrow$ C3b–C3d.

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Table 1 (continued)

Node	Claim	Status (Walton scheme; support)	Open critical question (residual risk)
C3b	Plateau $\kappa_{\text{app}} = 4.91$ $[4.75, 5.08]$, $\beta = 0.51$, $\gamma = 1.02$.	**Partially closed: warranted on CIF data; the cross-paper anchor is a lead, not a corroboration.** Argument from Example (the error-weighted fit generates the premise): CV 4.0%; CIs inflated by $\sqrt{\chi^2_{\text{red}}}$ ($\chi^2_{\text{red}} = 3.8$), hence conservative; survives a 2000-draw digitization Monte Carlo ($P(0.4<\beta<0.7) = 1.00$) and is invariant across three independent digitizations (the primary and two re-digitizations), which return κ_{app} in $[4.91, 4.92]$ and reject the global law every time. None of that depends on the Xia comparison.	`xia_anchor_unverified`. The recalculation is real and its inputs are quoted, but it is *not* a held-out test, for the two reasons CQ2 separates, after a caveat that is the graph's own: the $\sim 140\,\mu$m figure is the channel *depth* while the central-channel *width* tapers $300 \to 150\,\mu$m, so the literal device aspect ratio is uncertain, which is why `cif_cutoff_future_work.py` flags the anchor UNVERIFIED. It is also circular: Xia [2] takes its $7\,\mu$m cutoff from *the same Gifford Fig. 3* [1, p. 4500] this plateau was fitted from, and Gifford is a Xia co-author. Decisively, the 5.2% agreement *is* the depth correction: $1/K_{\text{depth}}(1.40) - 1 = 5.3\%$, and $4.9138/0.9500 = 5.17$, so the comparison is equally consistent with a depth-robust prefactor and with a depth correction of that size and discriminates between them not at all. Reported as a **lead**; the fit and the exponents stand on the Gifford three-width data alone.
C3c	The cutoff knee is real.	**Warranted on CIF data (existence); mechanism rival open.** Argument from Sign for the *existence*, Abduction for the *mechanism* (Doyeux, [14, p. 359]); the digitization undercutter is closed separately, by Evidence-to-Hypothesis in falsification form. The global single power law is rejected at reduced $\chi^2 = 30.6$, and κ_{app} ramps from 4.91 to 1.34 as $d/w_c \to 0$. Both figures are *binned means*: the $n = 8$ plateau bin ($d/w_c \geq 0.035$) and the $n = 3$ foot bin ($d/w_c < 0.012$); the fifteen individual points span 1.20–5.20, which is the span and never the ramp. Foot and rejection robust across all three digitizations.	`more_reliable` (existence) and `more_coherent_explanation` (mechanism). *Mechanism*: the knee is attributed to a finite-particle capture-regime change, deferred to future finite-particle work; the $f_{\text{eff}}(d)$ alternative (the content of `more_reliable`) is bounded but not eliminated; DLD is silent, its *apparent* prefactor κ_{app} running 2.040–2.659 with no knee and never below the steric floor $\kappa_{\text{steric}} = 2$. Those two bounds are *two computations of the same DLD prefactor, not two laboratories*, and they run in opposite senses: Davis's eqn 2-22 falls with ε from 2.659 to 2.459 [15, eqn (2-22)], while the exact Inglis cubic rises from 2.040 to 2.449 [12, p. 656].
C3d	Physical decomposition of κ_{app}.	**Partially closed.** Numerical verification (CFD, [7]; deductive computation) + Established Rule + Expert Opinion (Doyeux, [14, p. 359]) + Abduction. Settled: $\kappa_{\text{steric}} = 2$, $K_{\text{in-plane}} = 1$ (CFD, Runs A/B/C: no in-plane funnelling), $R(\alpha) \in [0.89, 0.98]$. Data-fixed residual $\kappa_{\text{app}}/\kappa_{\text{steric}} = R(\alpha)\sqrt{c}\,C = 2.4569$ (depth-folded); dividing out the *measured* $K^{\text{CFD}}_{\text{depth}}$ leaves the depth-separated $\sqrt{c}\,C = 2.7574/2.5999/2.5044$ at $\alpha = 0.80/1.40/2.69$ open, *a triplet and not a range* (§1.3), each value belonging to its own α (per-gap over-extraction × cumulative finite-particle displacement, the Doyeux mechanism), tied to the unpublished gap count N. Every factor is carried in the derivation record [8].	`more_coherent_explanation` (the mechanism) and `degenerate_from_published_data` (the split: single-lineage data fix the *product* $\sqrt{c}\,C$ and not its division into c and C). Needs finite-particle work + a micrograph gap count. *Not* the single-gap solve (done). *Not* the F-identity (decoupled, C4): reading κ out of F inverts the C4 lemma.

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Table 1 (continued)

Node	Claim	Status (Walton scheme; support)	Open critical question (residual risk)
C5	The skim law derives the corrected Xia scaling $f_{\text{gap}} \propto w_c^{-2}$.	**Warranted.** Evidence-to-Hypothesis (verification form): the law predicts the *corrected* direction and excludes the published original [2], confirmed by the 2022 author correction [3, p. 1].	Shares premises with C2/C3a: a consilience (one relationship), not a fully independent test.
C6	$\beta = \frac{1}{2}$, $\gamma = 1$.	**Warranted; uniform-f_{eff}-robust.** Defeasible Modus Ponens on the C2 identity and Argument from an Established Rule for the shift-invariance (both deductive, Appendix A, Result 6), corroborated by Argument from Example twice over: the CIF plateau fit ($\beta = 0.51$, $\gamma = 1.02$) and Davis's independent DLD device family (exponent 0.48, P-C6.5). The shift-invariance is *algebraically exact, not approximate*: identical at $c = 1, 2, 3.5$.	—
C7	The prediction for Dinh's $f^*_{\text{gap}} = 1.04 \times 10^{-4}$ device sits inside the measured bracket: band $[1.25, 2.17]\,\mu\text{m} \subset (1.0, 4.7)\,\mu\text{m}$ [5, ESI, Fig. S1].	**Warranted as a band.** Numerical computation for the band (deductive), checked against data that was not fitted (Evidence to a Hypothesis). Corroboration only, not a discriminating test.	None *open*: the graph records no open exception here and every critical question is answered. What the answers establish is how little the band can be asked to carry: it is doubly sensitive to depth and f_{eff} (at its initial gap fraction that device sits below the knee, where $d_{\text{crit}} = \kappa\,\delta$ has no stable fixed point *above the sub-steric foot*), and it is weak (4.7× bracket). The interval $c_{\text{Dinh}}/c_{\text{Gifford}} \in [0.64, 4.69]$ over which the band stays inside $(1.0, 4.7)\,\mu\text{m}$ is a *tolerance*, not a plausibility range: it is 7.3× wide because the bracket is, and CQ4 reads the same insensitivity as weakness. Anchor on plateau devices, not Dinh.
C4	$F \perp\!\!\!\perp f_{\text{gap}}$ (separation lemma): $F = 2R_c/(R_s + 2R_c)$ depends only on outlet geometry.	**Proven** (the separation); **validated** (F's magnitude). The decoupling $F \perp\!\!\!\perp f_{\text{gap}}$ is a deductive current-divider identity, carried by Argument from an Established Rule (the hydraulic–electric analogy), and carries no open critical question; F's absolute magnitude is a separate, empirical claim, confirmed to within 1σ by COMSOL [7] on Dinh V1/V3 in the companion preprint [19], and it is that claim the residuals at right attach to. Neither bears on the decoupling.	The 9.3% (V1) / 12.6% (V3) 1-D-vs-3-D residual is on F itself (closed by 3-D outlet-bifurcation physics) and does *not* couple F to f_{gap}; `sibling_3d_junction_uncertainty_on_sqrt_c`, a sibling 3-D-junction effect, contributes a $\sim$5%-scale uncertainty on $\sqrt{c}$, distinct from F's own residual and the node's one open critical question.
C9	Transport lemma: $\varphi' = (\varphi - f)/(1 - 2f)$ makes the centreline a *repelling* fixed point, so selection is a hard wall constraint rather than a statistical criterion; with centre exclusion (P-C9.1′) that gives $\kappa_{\text{steric}} = 2$ exactly.	**Proven.** Defeasible Modus Ponens on a closed-form migration map, with Argument from an Established Rule for its three further consequences; the critical position reproduces to six decimals ($\varphi_{\text{crit}} = 0.216643$). Deliberately *not* load-bearing: no support path runs from $\kappa_{\text{steric}} = 2$ to the thesis.	None. Within the model's premises, the lemma also *closes* C3d's residual list to three model gaps (the flat-lamina idealisation, the wall porosity ϕ, and finite-particle displacement), which answers C3d's exhaustiveness question while leaving its identifiability question open.

continued on next page

Table 1 (continued)

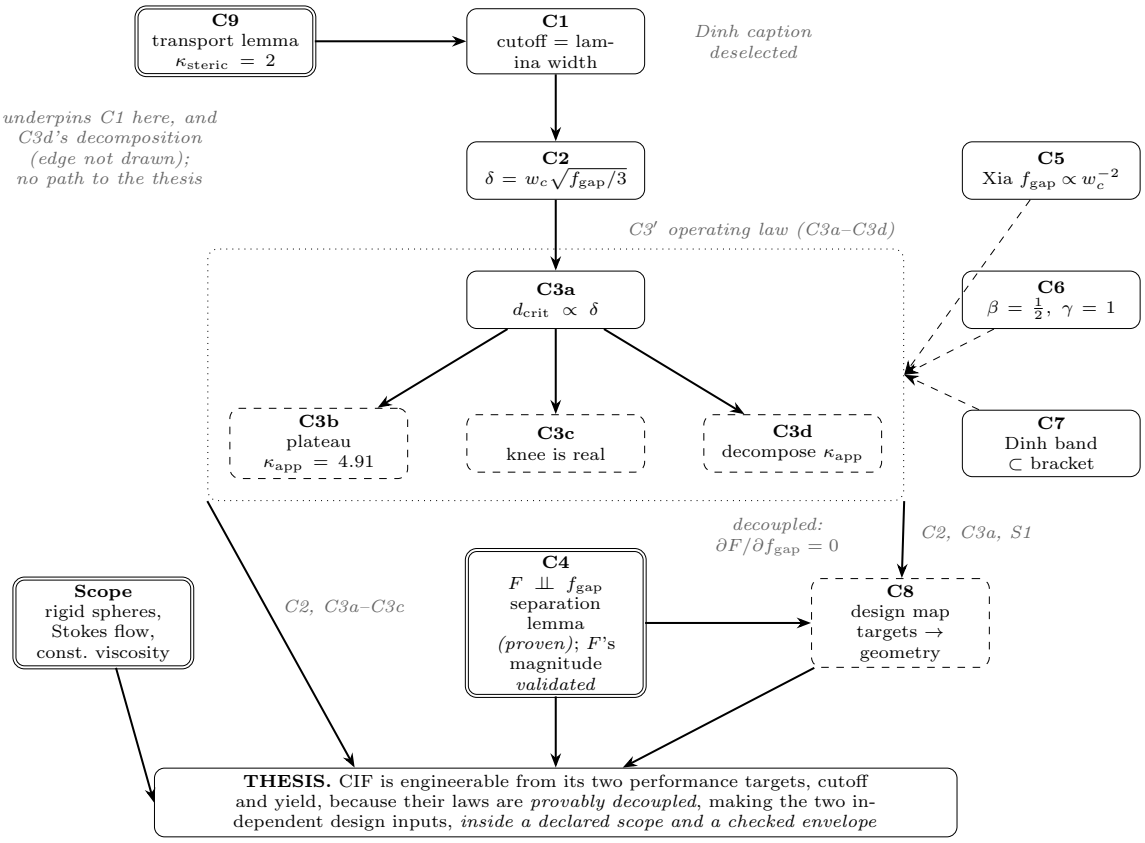
Node	Claim	Status (Walton scheme; support)	Open critical question (residual risk)
C8	The design map: the target Jacobian is lower-triangular ($\partial d_{\text{crit}}/\partial N = 0$) with non-zero diagonal, so targets invert locally to geometry and forward substitution realizes both on a device inside the envelope.	**Partially closed.** The *inversion* (C8a) is proven, resting on C2, C3a, C4 and the S1 schedule, and is reached a second time, by construction rather than by argument, through an executable forward march ($N \cdot D^2$ holds to 0.29% across five cutoff targets while N spans 1446–6562). The *realization* (C8b) is Argument from an Established Rule and rests on the inversion plus three granted assumptions; the 1.73-decade decoupling sweep supports C4, not C8b.	Three undercutters, all *withheld* rather than excluded: `below_knee`, `runout_mismatch`, `pinned_row_count`. The first has published counterexamples: at their wide ends the plateau law puts Dinh's three designs at $u = d/w_c = 0.029$–0.032 (the bracket device at 0.029), and Xia's RS-7 lower still at 0.023 on the taper Xia prints (C5), against the floor u_{lo} set here to the plateau-bin edge 0.035, so it is the guard that holds the defeater dormant. The *need* for the guard is derived: below the plateau edge the operating law is *implicit* in d_{crit} (C7), so a target no longer fixes the cutoff. Its *value* is not: u_{lo} is an empirical bin edge, an external input (CQ7).
Thesis	CIF is engineerable from its two performance targets rather than from a tuned per-gap schedule: yield follows a closed-form law with no fitted constant and selectivity a derived law with a single calibrated prefactor, and the two are *provably decoupled*, so cutoff and yield are independent design inputs rather than coupled outcomes.	**Warranted, conditionally.** Accrual over five legs: the operating law (C2, C3a–C3c), the separation lemma (C4), the locally invertible map (C8a), realizability (C8b) and the declared scope. C1 and C9 establish $\kappa_{\text{steric}} = 2$, which underpins the mechanism and C3d's decomposition; neither is a premise of the operating law, and neither reaches this node. Evaluated under preponderance of the evidence at the weakest leg's tier.	Inherited, not new: C8's three envelope boundaries, plus the open evidentiary questions of C3b–C3c and C4. Scope is carried as a premise (rigid spheres in Stokes flow), so platelet discoidness, deformability, cell crowding and whole blood are *declared* out of scope, not overlooked.

3 Net assessment

The cutoff law is **proven in form**: the exact skim cubic, its identity with the Inglis critical-streamline cubic, the $\beta = \frac{1}{2}$ / $\gamma = 1$ exponents, and the recovery of the corrected Xia scaling are all established symbolically and reproduced byte-for-byte by the reference code [8], with the Xia recovery entering as a *consilience* rather than an independent test, since it shares its premises with C2 and C3a (C5). It is **empirically solid in its apparent plateau prefactor** ($\kappa_{\text{app}} = 4.91$, 95% CI $[4.75, 5.08]$, CV 4.0%), robust to a 2000-draw digitization Monte Carlo, and returning κ_{app} in $[4.91, 4.92]$ across three independent digitizations of the calibration figure (the primary and two re-digitizations). The cross-paper comparison with Xia [2] ($\kappa_{\text{app}} = 5.17$, a 5.2% agreement) once stood here as corroboration and is now reported as a **lead**: its $\sim 140\,\mu$m figure is the channel *depth* while the central-channel *width* tapers $300 \to 150\,\mu$m, so the device aspect ratio is uncertain; Xia reads its $7\,\mu$m cutoff from the same Gifford Fig. 3 [1, p. 4500] this plateau was fitted from; and, decisively, that 5.2% is numerically the $140/80\,\mu$m depth correction itself, so the agreement cannot discriminate between a depth-robust prefactor and a depth correction of exactly that size (C3b, CQ2). It carries no weight in this assessment, which is why the sentence above rests on the CIF calibration data alone. Its physical decomposition is **partially closed**: $\kappa_{\text{steric}} = 2$, $K_{\text{in-plane}} = 1$ (CFD-settled, [7]), and $R(\alpha) \in [0.89, 0.98]$ over the three CFD-probed aspect ratios (the analytic duct law [6, p. 875] extending to 0.811 at the widest calibration channel, $w_c = 150\,\mu$m, which is the lowest aspect ratio of the calibration triplet) are settled, so the data-fixed residual is $\kappa_{\text{app}}/\kappa_{\text{steric}} = R(\alpha)\sqrt{c}\,C = 2.4569$. Dividing out the measured $K^{\text{CFD}}_{\text{depth}}$ leaves a single open quantity, the depth-separated triplet $\sqrt{c}\,C = 2.7574/2.5999/2.5044$ at $\alpha = 0.80/1.40/2.69$, a triplet and not a range (§1.3). That quantity is per-gap over-extraction times cumulative finite-particle displacement, tied to the unpublished gap count N. Every constant here is carried in the derivation record [8]. The exponents (by shift-invariance, C6), the knee's *existence* (a size-dependent f_{eff} is itself size-dependence and so cannot defeat the variation it would merely

reattribute), and the Dinh consistency band (which stays inside the bracket across $c_{\mathrm{Dinh}}/c_{\mathrm{Gifford}} \in [0.64, 4.69]$) all survive the f_{eff} confound; only the absolute prefactor's *decomposition* and the knee *mechanism* depend on it, and closing it is the task of finite-particle simulation plus a micrograph gap count, *not* of the single-gap solve (done) and *not* of the F-identity, which is provably decoupled (C4).

Two scope conditions frame the whole assessment. First, κ_{app} is *apparent*: it is defined against the *nominal* design f_{gap} and the parabolic δ, so the open residual $\sqrt{c}\,C$ rescales only the decomposition, never any reported result. Second, the law is derived for *rigid* spheres, and the graph carries that restriction as a premise: deformability, cell crowding and whole blood are declared out of scope. The restriction is not a formality. Davis reports that deformable red cells "do not always behave as if they have a clear hard sphere threshold" [15, §5.2.5], and observes departures in *both* directions and with flow speed. Cells fast enough to "fit" into the first streamline stop displacing [15, §6.1.1], while at lower speeds cells displace in devices designed to retain them [15, §6.1.2]. He declines to attribute either to deformability alone, writing of the spherical white cells that "the cells and beads also differ in density, electrostatic interactions and surface chemistry" [15, §6.2.1]. No signed correction is therefore available to carry deformable cells inside this law, which is precisely why they are excluded rather than corrected for.



Open critical questions (residual risk). *C3b: the cross-paper anchor is circular* and *its 5.2% agreement is numerically the depth correction, so it discriminates nothing and is reported as a* lead*; the fit itself is unaffected. C3c: the $f_{\text{eff}}(d)$ alternative is bounded, not eliminated, and the knee* mechanism *is deferred to future finite-particle work. C3d: the lumped residual, depth-separated $\sqrt{c}\,C = 2.7574/2.5999/2.5044$ over $\alpha = 0.80/1.40/2.69$ (a triplet, not a range), tied to the unpublished gap count N. C4: a* 9.3% *(V1) /* 12.6% *(V3) 1-D-vs-3-D residual on F's* accuracy *(it does not couple F to f_{gap}). C8: three undercutters,* withheld *rather than excluded:* `below_knee`, `runout_mismatch` *and* `pinned_row_count`*, which is what the dashed border on C8 records, and which Figure 12 argues in full.*

Figure 1: Argument map for the whole argument: the visual companion to the backbone above. The *lower storey* is the cutoff law: C1→C2→C3′ down the spine, with C5, C6 and C7 corroborating into it and C4 drawn deliberately disconnected from it. The *upper storey* spends that disconnection: because F carries no cutoff information, the two design targets are independently specifiable, which is what makes the design map C8 locally invertible, and the thesis is the accrual of the operating law, the separation lemma, the design map (its inversion C8a and its realization C8b, drawn here as one node) and the declared scope: the thesis's five legs, in four arrows. The two edges leaving the C3′ cluster are labelled, because C3d interprets the prefactor's magnitude and is a premise of neither consumer; an unlabelled cluster arrow would assert that it is. C9 stands to one side of the spine on purpose: it derives $\kappa_{\text{steric}} = 2$ and so *underpins* C1, and C3d's decomposition, of which only the C1 edge is drawn; no support path runs from it to the thesis. Each upper-storey node has a detailed map of its own (Figures 11, 12 and 13); this one shows how they connect. Border style encodes status: a single border marks a *warranted* node, a dashed border (C3b, C3c, C3d, C8) a node with something still open, and a doubled border (C9, C4, Scope) a *proven* identity or declaration. C4's doubled border covers the separation itself; F's absolute *magnitude* is a separate, empirical claim, which is where C4's residual at the foot of this figure attaches. Those three words are this map's convention and this map's alone; the per-node figures that follow use the dashed border for whatever *that* figure declares in its own caption (an adjudicated question in Figure 2, an excluded rival in Figure 6, a withheld undercutter in Figure 12), and each says which. Solid arrows are support; dashed arrows are corroboration, with one exception the spine makes unavoidable. C3a→C3b, C3a→C3c and C3a→C3d are premise edges the graph carries; C1→C2 and C2→C3a are *reading order*, not graph edges: `mechanism` is not a premise of `c2_skim`, whose premise is the parabolic gap profile; `skim_law` is not a premise of `c3a`, whose premise is expert proportionality; and C2 and C3a enter the operating law as siblings rather than in series. The chain is how the argument is read and how the physics composes; the graph's support structure branches at C3′. Unlike Appendix A's Figure 1, which is a derivation *dependency* graph, this map shows the *support and defeat* structure: each node's grounds and the open critical questions that could still defeat it.

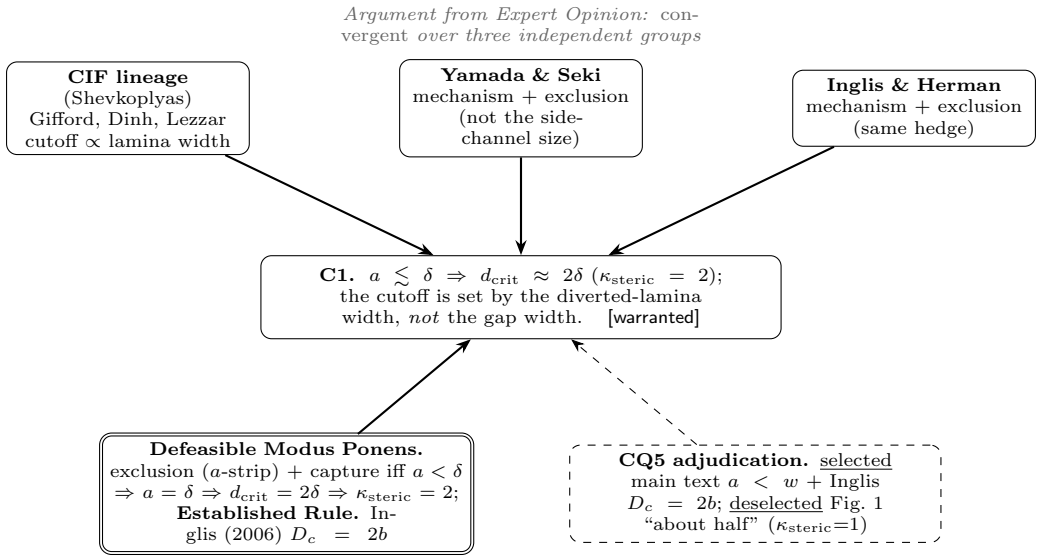


Figure 2: Argument graph for C1. Three independent groups converge on the mechanism (Expert Opinion); a short deductive strand fixes $\kappa_{\text{steric}} = 2$ by Defeasible Modus Ponens from geometric exclusion plus the capture criterion, with Argument from an Established Rule on Inglis's $D_c = 2b$; CQ5 adjudicates the lone caption-vs-main-text discrepancy. Border style follows the overview map: single border = warranted claim, doubled border = deductive identity, dashed box = an open/adjudicated critical question.

C1 — Mechanism: the cutoff is set by the diverted-lamina width

Summary. Five peer-reviewed sources, spanning *three* independent research groups, assert that the CIF size cutoff is set by the width of the near-wall lamina diverted into the side channels, not by the gap width. A standard geometric-exclusion premise (a particle's centre is barred from a wall strip equal to its radius) converts "the lamina reaches the centre" into the capture criterion $a \lesssim \delta$, so the cutoff diameter is $d_{\text{crit}} \approx 2\delta$ and the steric prefactor is $\kappa_{\text{steric}} = 2$. The corpus holds one internal discrepancy, between Dinh's figure caption ("about half") and its main text ($a < w$); it is adjudicated to $\kappa_{\text{steric}} = 2$ by the main text's reasoning and by the Inglis-2006 $D_c = 2b$ foundation, with the caption *deselected* rather than overruled. The node is **warranted**: the mechanism carries no residual; only the *value* of the prefactor (settled here at $\kappa_{\text{steric}} = 2$, its excess deferred to C3d) is argued downstream.

Claim. A particle is carried into the side (filtrate) channel iff its effective radius is at most the diverted-lamina width, $a \lesssim \delta$ (Dinh's w; Inglis's first-streamline width b); the cutoff diameter is $d_{\text{crit}} \approx 2\delta$. The cutoff is governed by the lamina width, not the gap width.

Scheme. Argument from Expert Opinion (WRM 2008, § Argument from Expert Opinion), *convergent over five peer-reviewed sources spanning three independent groups*, with a deductive strand beneath it: Defeasible Modus Ponens on geometric exclusion plus the capture criterion, and Argument from an Established Rule on Inglis's $D_c = 2b$.

Premises (paper-stated, verbatim).

P-C1.1 (Gifford 2014, [1, p. 4503]).

"the size cutoff is determined primarily by the width of the streamlines diverted into the side (filtrate) channels." ⇒ cutoff ∝ lamina (streamline) width, not gap width.

P-C1.2′ (Dinh 2024, [5, p. 917]; lower branch).

"For particles with a radius smaller than the width of the extracted flow lamina ($a < w$), we would expect their distribution between the filtrate and retentate to approximate that of the fluid volume." ⇒ $a < \delta \Rightarrow$ tracer-like.

P-C1.2″ (Dinh 2024, [5, p. 917]; upper branch).

"we would expect particles with radii exceeding the width of the extracted flow lamina ($a > w$) to remain

in the middle channel." $\Rightarrow a > \delta \Rightarrow$ retained. With P-C1.2′ the threshold sits at $a = \delta$, hence $d_{\text{crit}} = 2\delta$.

P-C1.3 (Yamada & Seki 2005, [11, p. 1234]; geometric exclusion).
"the center position of a particle cannot be present within a certain distance from the sidewalls, which is equal to the particle radius." $\Rightarrow$ the centre is excluded from a wall strip of thickness a; capture requires the lamina to reach the centre, i.e. $a \lesssim \delta$.

P-C1.3′ (Yamada & Seki 2005, [11, p. 1234]; mechanism).
"When the relative flow rates distributed into side channels at a branch point are sufficiently low ... only a small portion of the liquid near the sidewalls is withdrawn from the main stream. In this flow state, particles whose diameter is larger than a specific value would never go through the side channels, even if they are flowing near the sidewalls in the main channel, and even if the particle size is smaller than the cross sectional size of the side channel." $\Rightarrow$ the third independent group asserts the mechanism itself and negates the gap explicitly: retention is set by the narrow withdrawn lamina, not by the side channel's own cross-section. The condition Yamada states, a sufficiently small relative flow rate into each side channel, is the regime CIF is built for: its per-gap flow fraction f_{gap} is small by design, which is why a CIF device needs many gaps (P-C1.4). In the source this is the sentence that P-C1.3 immediately explains, on the same page.

P-C1.4 (Lezzar 2022, [4, p. 2 (Sci Rep PDF pagination)]).
"It is the magnitude of this flow fraction—not the width of the gaps—that determines the size cutoff" (and the design reports a 6 μm critical diameter, requiring "thousands of gaps"). $\Rightarrow$ an explicit statement that the flow fraction (hence the lamina), not the gap, sets the cutoff.

P-C1.5 / IH-Q1 (Inglis & Herman 2013, [13, p. 1724]; independent).
"Assuming that particles ... are geometrically restricted away from walls by a distance equal to their radius, the cutoff size is determined by the width of the boundary layer in the main channel that is drawn up into each branch", carrying the same hedge CIF carries, "a frequent assumption in the microfluidic particle separation literature, that is not always true." $\Rightarrow$ an independent branch-flow group states the C1 mechanism and the geometric-exclusion premise in one sentence.

Logical inference. P-C1.3 (exclusion strip of thickness a) together with P-C1.2′/2″ (capture iff $a < \delta$) gives the marginal captured particle $a = \delta$, hence $d_{\text{crit}} = 2a = 2\delta$; i.e. the steric prefactor relating cutoff diameter to lamina width is $\kappa_{\text{steric}} = 2$. This is the object proved as Result 5 of Appendix A and held in `steric.KAPPA_STERIC` in the reference code; it feeds CQ5 below (where it is defended against the caption) and C3d (where its excess $\kappa_{\text{app}}/\kappa_{\text{steric}} = 2.4569$ is decomposed).

Critical questions.

CQ1: Expertise (Position-to-Know).
All sources are domain experts on this device class: Gifford, Dinh, Lezzar are primary CIF/plasma-skimming papers; Inglis & Herman (2013) and Inglis (2006) are an independent branch-flow/DLD group; Yamada & Seki (2005) originated hydrodynamic filtration. *Answered: yes.*

CQ2: Assertion (Evidence).
Each source actually asserts the mechanism; verbatim P-C1.1–P-C1.5. *Answered: yes.*

CQ3: Consistency among experts.
No primary source contradicts the lamina-width (not gap-width) mechanism: all three groups state it directly (Gifford/Lezzar/Dinh, with Lezzar contrastive against the gap width; Inglis & Herman; and Yamada & Seki, contrastive against the side channel's own cross-section, P-C1.3′). Yamada & Seki additionally supply the geometric-exclusion sub-premise the deductive strand rests on (P-C1.3). *Answered: yes.*

CQ4: Trustworthiness / independence (bias CQ).
The convergence is not an artifact of a single laboratory. Three of the five mechanism sources (Gifford, Dinh, Lezzar) share the Shevkoplyas group, as does Xia in the wider corpus, so within that lineage they *corroborate but do not independently establish the mechanism*; it is *independently* asserted by two further groups, Yamada & Seki (2005) and Inglis & Herman (2013), and the geometric-exclusion sub-premise is a standard result in the size-separation literature (stated crisply by Yamada & Seki; the same centre-position picture underlies pinched-flow fractionation, cf. [18]). The two independent groups, not the lineage, carry the claim. *Answered: yes. (Inglis 2006 and Inglis & Herman 2013 share a lead author and are one group, not two; the three independent groups are the Shevkoplyas lineage, Yamada & Seki, and Inglis.)*

CQ5: Internal consistency of the steric criterion (RESOLVED $\rightarrow \kappa_{\text{steric}} = 2$).
The criterion is $a = \delta$ ($\kappa_{\text{steric}} = 2$), not $a = \delta/2$ ($\kappa_{\text{steric}} = 1$). Dinh states it two ways: the Fig. 1 caption (verbatim) "Particles with an effective radius smaller than about half the width of the extracted flow lamina w are carried into the side channels by the filtrate; larger particles remain in the middle channel and become progressively more concentrated" [5, p. 916, Fig. 1 caption], which taken literally gives $\kappa_{\text{steric}} = 1$; versus the main text (P-C1.2′/2″, $a < w$ / $a > w$), which gives $\kappa_{\text{steric}} = 2$. The discrepancy is settled in favour of the main text because (i) the main text is the substantive discussion and supplies the reasoning, whereas the caption is a one-line paraphrase that even hedges ("*about* half"); and (ii) the main text matches the theoretical foundation CIF is built on: Inglis (2006) defines the critical radius as equal to the first-streamline width and hence $D_c = 2b$ (C2), and the CIF "extracted lamina" *is* Inglis's "first streamline." The caption is therefore *deselected* (it may describe a stricter "fully entrained" threshold; the marginal particle is never drawn in Dinh's schematic), not declared wrong. This *confirms*, rather than moves, the $\kappa_{\text{steric}} = 2$ baseline: had the caption been adopted, the C3d residual would sit at $\kappa_{\text{app}}/\kappa_{\text{steric}} = 4.9138$ rather than 2.4569. *Answered:* $\kappa_{\text{steric}} = 2$.

Residual risk. None on the mechanism. The $a < w$ vs $a < w/2$ wording is resolved to $\kappa_{\text{steric}} = 2$ (CQ5), now with independent third-party support (Inglis & Herman, P-C1.5). The *magnitude* of the prefactor in excess of κ_{steric} is not a C1 question; it is decomposed at C3d.

Status. Warranted.

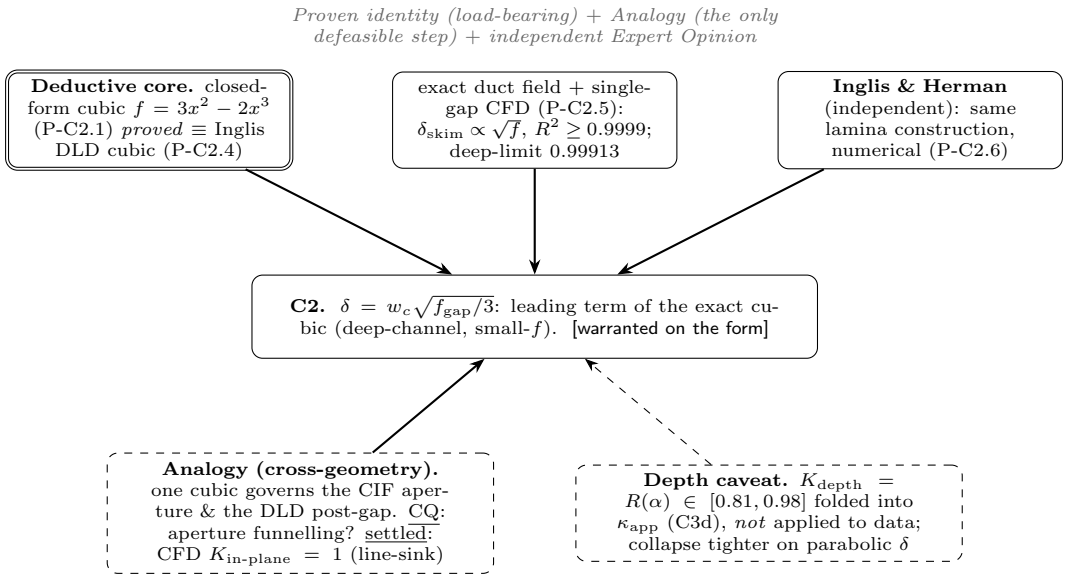


Figure 3: Argument graph for C2. The deductive core, the closed-form cubic and its *proved* identity to the Inglis DLD cubic (doubled border), is the load-bearing claim; the exact-duct/CFD recovery and the independent Inglis & Herman construction corroborate it. The single defeasible step is the *cross-geometry* transfer of one cubic between a wall-aperture and a post-gap; its critical question (aperture funnelling) is settled by the single-gap CFD ($K_{\text{in-plane}} = 1$). The depth factor is a bounded caveat folded into κ_{app}, not a defeater of the form. Border style follows the overview map: doubled = deductive identity, single = warranted/corroborating, dashed = critical question or caveat.

C2 — Hydrodynamic conversion: $\delta = w_c\sqrt{f_{\text{gap}}/3}$

Summary. Integrating the parabolic gap profile over the near-wall lamina yields the exact flow-fraction cubic $f = 3x^2 - 2x^3$ ($x = \delta/w_c$), whose small-f inverse is $\delta = w_c\sqrt{f/3}$ to within 0.5% across the Gifford calibration/design band ($f_{\text{gap}} \leq 5.76 \times 10^{-4}$; worst case 0.47%) and within 0.9% across the wider numerical sweep band ($f_{\text{gap}} = 3.63 \times 10^{-5}$ to 1.93×10^{-3}; worst case 0.86%). The band is quoted with the figure throughout, because the design-band and sweep-band worst cases differ by nearly a factor of two (0.47% against 0.86%), and each figure is the small-f inversion error with the parabola itself taken exact. This cubic is *proved* identical to the Inglis (2006) DLD critical-streamline cubic, unifying the two device classes. The same near-wall-lamina construction is *independently applied* (numerically, with no closed form) by Inglis & Herman (2013) over the same rectangular-duct field, and the cubic's deep-channel form is recovered by an exact rectangular-duct calculation and by single-gap CFD. The only correction at real, finite aspect ratios is the bounded depth factor $K_{\text{depth}} = R(\alpha)$, CFD-measured to be $[0.89, 0.98]$ over the probed range $\alpha = 0.8$–2.69 and extended by the analytic duct law to a 0.81 floor at $\alpha = 80/150 = 0.5333$; it is folded into the apparent prefactor rather than applied to the data (the cutoff data collapse more tightly on the clean parabolic δ). The node is **warranted on the form**: the identity is the load-bearing claim, the cross-geometry transfer is the only defeasible step, and the single-gap CFD closes it ($K_{\text{in-plane}} = 1$).

Claim. In the deep-channel, small-f limit the diverted lamina width is $\delta = w_c\sqrt{f_{\text{gap}}/3}$, the leading term of the exact cubic $f = 3x^2 - 2x^3$ ($x = \delta/w_c$).

Scheme. A deductive core, *a proven, code-checked identity* (defeasible modus ponens in the umbrella sense; no critical questions of its own), comprising (i) the closed-form cubic and its inversion (P-C2.1) and (ii) the *proved* algebraic identity of the CIF and DLD cubics (P-C2.4). This is reinforced by an *Argument from Analogy* for the physical transfer of the *single* cubic across two different geometries (a wall-aperture vs. the space between two posts). That is the step the Analogy-CQs interrogate and the single-gap CFD settles ($K_{\text{in-plane}} = 1$). It is reinforced further by an independent *Argument from Expert Opinion* (Inglis & Herman's identical near-wall-lamina construction). The identity is therefore the load-bearing claim; the analogy bears only on whether one equation governs both geometries, not on the algebra itself.

Notation. Inglis (2006) prints the first-streamline width as Greek β throughout: in the defining sentence ("which we define as β"), in eqn (1) ($D_c = 2\beta$), in the cubic eqn (5), and in the Fig. 1 caption and in-figure label; Latin "b" does *not* appear as that symbol anywhere in the paper (confirmed against the typeset page, not the OCR, which silently flattens $\beta \to b$). Per the convention fixed in the overview's Notation section, this manuscript renders the symbol as Latin b; quoted Inglis phrases keep his glyph, and our restatements use b. The identity $D_c = 2b$ (Inglis's $D_c = 2\beta$) gives $\kappa_{\text{steric}} = 2$, independent of the glyph.

Premises.

P-C2.1 (exact cubic: symbolic, code-checked).
Integrating the width-wise parabolic velocity profile $u \propto s(w_c - s)$ across the gap gives the cumulative near-wall flow fraction $f(x) = 3x^2 - 2x^3$ exactly ($x = \delta/w_c$), with total flux $Q_{\text{tot}} \propto w_c^3/6$. Inverting in the small-f limit,

$$\delta \;=\; w_c\sqrt{f/3} \;+\; \tfrac{1}{9}\,w_c f \;+\; \cdots\,,$$

with coefficients $a_1 = 1/\sqrt{3} = 0.577350$ and $a_2 = 1/9 = 0.111111$ (the program asserts both exactly; Appendix A also lists $a_3 = 0.053458$). The leading term is *exact as* $f \to 0$; its relative error grows as $\frac{1}{3\sqrt{3}}\sqrt{f} \approx 0.192\sqrt{f}$, reaching $\approx 0.2\%$ at $f = 1.04 \times 10^{-4}$, a worst case of 0.47% across the Gifford calibration/design band ($f_{\text{gap}} \le 5.76 \times 10^{-4}$; Gifford's own per-gap design bound is $f_{\text{gap}} < 5 \times 10^{-4}$ [1, p. 4498]), and a worst case of 0.86% across the wider numerical sweep band, which reaches $f_{\text{gap}} = 1.93 \times 10^{-3}$. `[code: skim.cubic, skim.delta_leading, skim.a1/a2, skim.verify_symbolic/_numeric; Appendix A Results 1--2]`

P-C2.2 (Xia 2016, verbatim).
The centre-channel cross-sectional velocity profile is "approximately parabolic" [2, art. 35943, p. 10 (Sci Rep PDF pagination)], which licenses the parabolic premise of P-C2.1. On the same page Xia add that the diverted streamline width is "roughly proportional to the effective c.d.," the proportionality C3a relies on.

P-C2.3 (Inglis 2006: the DLD construction, faithfully rendered).
The critical particle radius "is equal to the width of the first streamline" [12, p. 656]; Inglis denotes that width β (rendered b here; see *Notation*) and writes $D_c = 2\beta$ [eqn (1)], i.e. $D_c = 2b$. The streamline-width construction is, in our notation,

$$\int_0^b u(x)\,\mathrm{d}x \;=\; \varepsilon \int_0^g u(x)\,\mathrm{d}x, \qquad u(x) = \frac{g^2}{4} - \Big(x - \frac{g}{2}\Big)^2,$$

which reduces to the cubic $\big(b/g\big)^3 - \frac{3}{2}\big(b/g\big)^2 + \varepsilon/2 = 0$ [12, eqns (3)–(5)]. Inglis obtains this "by assuming a conventional parabolic flow profile through the gap, with zero velocity at the post sidewalls" [12, p. 656].

P-C2.4 (identity: symbolic, code-checked).
Computing the Inglis integral $\int_0^b u = \varepsilon\int_0^g u$ (with $u = g^2/4 - (x - g/2)^2$) yields $\varepsilon = 3y^2 - 2y^3$ ($y = b/g$), *algebraically identical* to the CIF cubic $f = 3x^2 - 2x^3$. `[code: steric.eps_of_y, asserted equal to skim.cubic in steric.verify_symbolic; Appendix A Results 3--4]` Hence the CIF skim cubic and the DLD critical-streamline cubic are the same equation: a CIF gap and a DLD gap obey one streamline-width law, and the leading inversion gives $b = g\sqrt{\varepsilon/3}$, identical in form to $\delta = w_c\sqrt{f/3}$. This cross-domain identity, which the prior DLD and CIF literatures do not state, is the structural keystone that lets results proven for one device transfer to the other.

P-C2.5 (exact duct field: numerical, code-checked).
The exact rectangular-duct (depth-integrated) velocity field [17, eqn (15)] recovers δ to 0.09% in the deep limit ($\delta_{\text{general}}/\delta_{\text{parab}} \to 0.99913$ at $\alpha = 50$, confirmed by re-running the routine), and single-gap CFD confirms $\delta_{\text{skim}} \propto \sqrt{f_{\text{phys}}}$ to $R^2 \ge 0.9999$ with δ_{skim} on the exact duct law to $\le 0.6\%$ (the validator's guard tolerance is 0.8%; the *achieved* agreement is $\le 0.6\%$, max 0.59% on Run B for δ_{skim} itself). The CFD-measured depth factors are $R(0.80/1.40/2.69) = 0.891/0.945/0.981$ against the analytic duct law's 0.8879/0.9500/0.9794; the analytic triple is $\sqrt{3/c_2(\alpha)}$ evaluated by the kernel's `K_depth`. `[code: f_general, duct.K_depth, verify_cfd; Methods §4.3]`

P-C2.6 / IH-Q2 (Inglis & Herman 2013, verbatim: independent use of the construction).
The cutoff flux is "calculated by integrating the analytical solution to the Navier–Stokes equation … from $W/2 - R_c$ to $W/2$ … on the depth-averaged velocity, $\bar{u}(x)$, in a rectangular channel ($d \times w$)" [13, p. 1726].

An independent group thus integrates the same near-wall lamina over the same rectangular-duct field to relate flow fraction to cutoff (numerically, with no closed form). They attribute that duct field to the *same* source this node cites, [17]: Brody's eqn (15) is their ref. 17, so the duct-field provenance is shared, not merely parallel.

Distinction from Inglis & Herman (sets up the novelty statement). Inglis & Herman integrate the analytical depth-averaged duct field for the cutoff flux and solve the device numerically (COMSOL), with no closed-form cutoff law, and report that analytical resistor-network approaches "have so far failed us" [13, p. 1725]. CIF's new contribution is the closed-form parabolic-limit cubic (P-C2.1), its exact identity to the Inglis-2006 DLD cubic (P-C2.4), the cross-domain unification the prior literatures do not state, and (separately, C4) the closed-form F-identity that *succeeds* where their network route failed.

Empirical support for the $\sqrt{f}$ law. Inglis's own illustrative values confirm the $\sqrt{f}$ scaling this node asserts: "For $\varepsilon = 0.1$ the predicted critical particle diameter is $0.4g$ … For $\varepsilon = 0.01$ … $0.12g$" [12, p. 656]. These match the *exact* cubic root ($d_{\mathrm{crit}}/g = 0.392$ at $\varepsilon = 0.1$, 0.118 at 0.01, which Inglis rounds to 0.4 and 0.12); the leading law $2\sqrt{\varepsilon/3} = 0.365$ and 0.115 captures the $\sqrt{\varepsilon}$ scaling but runs a few percent low at these DLD-scale shift fractions (6.8% at $\varepsilon = 0.1$, 2.0% at 0.01); the $+f/9$ term of P-C2.1 removes most of that (leaving 1.1% and 0.10%), and the gap shrinks to sub-percent across the much smaller CIF f-range. Inglis attributes the "surprisingly large critical size at small ε" to the slow near-wall fluid forcing a wider first streamline [12, p. 656]: the parabolic-vs-plug enhancement that distinguishes the $\sqrt{f}$ regime from a flat profile. Davis's empirical $d_{\mathrm{crit}} = 1.4\,G\varepsilon^{0.48}$ [15, eqn (2-22)] (his G is the DLD gap, Inglis's g; he quotes $0.46\,G$ at $\varepsilon = 0.1$) is the same square-root-like law fit to data ($\varepsilon^{0.48} \approx \sqrt{\varepsilon}$). Because the prefactor these imply is constant across $\varepsilon = 0.01$–0.5 (no regime change), this evidence supports the $\sqrt{f}$ law and, secondarily, the $\kappa_{\mathrm{steric}} = 2$ floor, and *not* a knee (C3c).

Critical questions. The cubic identity itself (P-C2.1, P-C2.4) is deductive and carries no critical questions; the questions below test the single *defeasible* step, the cross-geometry transfer of the cubic, and the bounded depth caveat, not the algebra.

Verification-CQ: does the cubic hold against data?

The cubic is a proven identity on a stated premise: P-C2.1 (symbolic proof) + P-C2.4 (Inglis-identical) + P-C2.5 (exact-duct + CFD recovery). The construction was validated two ways by Davis. His Femlab finite-element simulations find the parabolic profile "a reasonable estimation for the velocity profile between two walls" [15, §2.3.2], and his measured critical diameters over "about 20 different devices" ($G = 1.3$–$38\,\mu$m, $\varepsilon = 0.005$–0.5, beads 0.9–22 μm) [15, §2.4] fit the square-root-like law. It was validated experimentally by Inglis (particles 2.3–22 μm [12, p. 655]), and it is now reproduced by the CIF single-gap CFD in full 3-D geometry. Scheme: Argument from Evidence to Hypothesis, verification form (the graph's `c2_davis_fem` and `c2_cfd`). → *answered: yes.*

Bounds: how large are the deviations?

Deviations are bounded. The leading-δ form is *exact as* $f \to 0$ and its relative error *grows* with f (it is $\approx 0.192\sqrt{f}$, hence largest at the top of the range), but stays bounded: $\approx 0.2\%$ at $f = 1.04 \times 10^{-4}$, worst case 0.47% across the design band and 0.86% across the numerical sweep band; the only other deviation is from non-parabolic (finite-depth) profiles, bounded by $K_{\mathrm{depth}} = R(\alpha) \in [0.81, 0.98]$ (CFD-anchored at the upper end, analytic at the lower). Scheme: none. This is the deductive core's own error budget, not a critical question; the core carries none, as stated above. → *answered: bounded.*

Analogy-CQ1: relevant differences (DLD ↔ CIF).

The single cubic transfers across the two geometries despite the difference: a CIF gap is a finite aperture in a channel wall, not the space between two posts, but the single-gap CFD shows this aperture is a line-sink with no funnelling ($K_{\text{in-plane}} = 1$), so the cubic transfers intact and there is no aperture-funnelling enhancement of the prefactor, which is what settles the funnelling question at C3d. Scheme: Argument from Analogy, difference CQ. → *answered: the form transfers; no funnelling enhancement.*

Analogy-CQ2: base-case truth.

The DLD base case is true, triply: Inglis experimental data + Davis FEA + CIF CFD all confirm the parabolic streamline-width cubic. Scheme: Analogy, base-truth CQ. → *answered: yes.*

Analogy-CQ3: contrary instances.

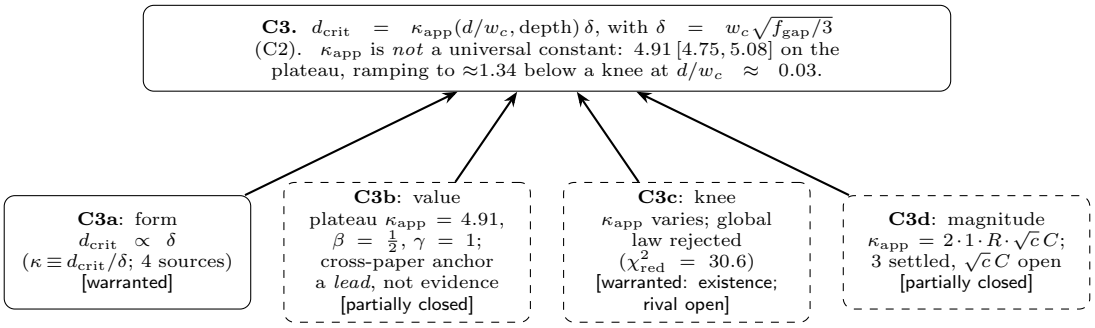
The analogy excludes the right cases: plug / electroosmotic flow is correctly excluded because CIF is

pressure-driven with no-slip walls. Davis quantifies the contrast through a *variable* enhancement parameter ζ, "a function of ε and post geometry" [15, eqn (2-19)], obtained as the cube-root of his eqn (2-21): at $\varepsilon = 0.1$ the cubic has a single real root $\zeta = 1.96$, giving $d_{\mathrm{crit}} = 3.92\,G\varepsilon \approx 0.39\,G$, "about twice as wide as" the plug-flow baseline $d_{\mathrm{crit}} = 2\,G\varepsilon$ for "even width stream lines" [15, p. 22]. Crucially ζ is ε-dependent (it $\to 1$ in the plug limit), which is *why* the parabolic law is $\sqrt{\varepsilon}$-like rather than linear, consistent with this node's $\delta \propto \sqrt{f}$ and with Davis's own $\varepsilon^{0.48}$ fit. Note that ζ is Davis's *streamline-width enhancement*, *not* κ_{steric}: the steric factor $\kappa_{\mathrm{steric}} = 2$ is pure $D = 2\delta$ geometry and is profile-*independent*; what the profile changes is the $f \leftrightarrow \delta$ map (parabolic $\delta \propto \sqrt{f}$ vs. plug $\delta \propto f$). Scheme: Analogy, contrary-case CQ. $\to$ *answered: yes.*

Expert-CQ (Inglis & Herman: the independent strand).
The corroborating expert source is credible, independent, and consistent: Inglis & Herman 2013 is peer-reviewed (*Lab on a Chip*), an independent (non-Shevkoplyas) branch-flow group, and applies the *identical* near-wall-lamina construction over the same duct field (P-C2.6); their result (no closed form) does not conflict with CIF's: it is the numerical counterpart of the same integral. Scheme: Expert Opinion (expertise, independence, consistency CQs). $\to$ *answered: yes.*

Residual risk. None on the form (exact, Inglis-identical, CFD-confirmed). One caveat: "deep parabolic" is a slight stretch at real aspect ratios. CIF runs at $\alpha = 0.5$–2.7, below Davis's stated validity for ignoring depth: "*a ratio of depth E to gap size G (aspect ratio) of at least 5 ... the depth dimension can be ignored, resulting in a two dimensional calculation*" [15, §2.3.2]. Both ratios are depth/(parabola-span), so the comparison is like-for-like. That is precisely the small $K_{\mathrm{depth}} = R(\alpha)$ correction the duct law gives and the CFD confirms ($[0.89, 0.98]$ over $\alpha = 0.8$–2.69; analytic floor 0.81 at $\alpha = 80/150 = 0.5333$). The cutoff still empirically prefers the parabolic δ (C3b, Methods §4.2): the collapse is tighter on the clean δ, so $R(\alpha)$ is folded into the apparent κ_{app} rather than applied to the data.

Status. Warranted on the form.



C3a fixes the form, *C3b the plateau* value *and exponents, C3c shows the value* varies *(the knee), C3d* interprets *the value. Single border = warranted; dashed = at least one item still open.*

Figure 4: Structure of node C3. The operating law rests on four sub-arguments: proportionality (C3a), the plateau value and exponents (C3b), the reality of the knee (C3c), and the decomposition of the prefactor's magnitude (C3d). Only C3a is closed. The other three are drawn dashed, and for three different reasons: C3b's cross-paper anchor is circular *and* non-diagnostic, so it is reported as a lead; C3c's *existence* claim stands while its *mechanism* has a live rival; and C3d's lumped residual $\sqrt{c}\,C$ is a named open quantity. None of the three touches the plateau fit itself, which is settled on the CIF calibration data alone.

C3 — The operating law: $d_{\text{crit}} = \kappa_{\text{app}}(d/w_c, \textbf{depth})\,\delta$

Summary. The operating law converts the diverted-lamina width δ (fixed by C2 as $\delta = w_c\sqrt{f_{\text{gap}}/3}$) into a particle-diameter cutoff through a single prefactor, $d_{\text{crit}} = \kappa_{\text{app}}\,\delta$. The central empirical finding is that this prefactor is *not* the universal constant the early literature assumed. On the large-particle plateau ($d/w_c \gtrsim 0.035$) it is tightly determined at $\kappa_{\text{app}} = 4.91$ on the CIF calibration data, and it carries the $\frac{1}{2}$ and 1 scaling exponents the law predicts (C3b; exponents derived in Appendix A, Result 6). The cross-paper comparison that once stood beside those numbers is now reported as a *lead*: it is same-figure, and its agreement is numerically the depth correction (C3b, CQ2). Below a knee at $d/w_c \approx 0.03$ it ramps down to a *sub-steric (recovery-suppressed)* foot of ≈1.34, a size dependence that a single power law cannot fit and that, correctly normalised, deterministic lateral displacement (DLD) theory does not share, marking it CIF-specific and finite-particle in origin (C3c). Decomposing the plateau value settles three of its five factors: the steric factor ($\kappa_{\text{steric}} = 2$, Appendix A Result 5), in-plane funnelling ($K_{\text{in-plane}} = 1$, by single-gap CFD), and a depth factor ($K_{\text{depth}} = R(\alpha)$); it isolates a single lumped residual $\sqrt{c}\,C$ that the published data cannot split and that is provably *not* recoverable from the device-level filtrate fraction F (C3d).

Claim. $d_{\text{crit}} = \kappa_{\text{app}}\,\delta$, where κ_{app} is not a universal constant: $\kappa_{\text{app}} = 4.91$ [4.75, 5.08] on the plateau ($d/w_c \gtrsim 0.035$), ramping to ≈1.34 below a knee at $d/w_c \approx 0.03$. (This replaces the refuted "$\kappa \approx 5$ constant, globally $\sqrt{f}$" claim.) The lamina width $\delta = w_c\sqrt{f_{\text{gap}}/3}$ is fixed by C2; $\kappa_{\text{steric}} = 2$ and the $\frac{1}{2}$/1 exponents are derived in Appendix A (Results 5, 6). Four sub-arguments follow, each a defeasible argument in the sense of WRM 2008.

C3a — Proportionality ($d_{\text{crit}} \propto$ lamina width)

Summary. Four peer-reviewed sources state the cutoff scales linearly with the diverted-lamina width, so the operating law can be written $d_{\text{crit}} = \kappa\,\delta$ with κ a (possibly size-dependent) proportionality constant; the dispute is confined to κ's value.

Claim. The cutoff diameter scales linearly with the lamina width; $\kappa \equiv d_{\text{crit}}/\delta$.

Scheme. Argument from Expert Opinion (WRM 2008), convergent over four primary sources, + Argument from Example (the measured exponents select the form). The expertise, assertion, and backup-evidence critical questions for the expert-opinion scheme are discharged at C1 (the same convergent peer-reviewed base) and are inherited here; the live critical question at C3a is consistency.

Premises.

P-C3a.1 (Xia 2016, verbatim).

"the width of the streamline that is diverted into a side channel at each (filtration) gap is roughly proportional to the effective c.d. of the device at that gap" [2, Methods, p. 10 (Sci Rep PDF pagination), assumption (b)]. Xia defines "c.d." as the *critical diameter*: "particles above a certain size (or 'critical diameter', c.d.)" [2, p. 2 (Sci Rep PDF pagination)]. So assumption (b) states $\delta \propto d_{\text{crit}}$, equivalently $d_{\text{crit}} \propto \delta$, the claim.

P-C3a.2 (Inglis 2006).

$d_{\text{crit}} = 2b$ [12, eqn (1)]: the critical diameter is twice the first-streamline width b, hence linear in it. (Inglis prints this width as Greek β; we render it b throughout; see the C2 Notation paragraph. This premise is P-C2.3; with the CIF extracted lamina identified with Inglis's first streamline, $b \leftrightarrow \delta$.)

P-C3a.3 (Dinh 2024).

The C1 criterion, particles with $a < w$ "approximate ... the fluid volume" while those with $a > w$ "remain in the middle channel", is a linear-in-w threshold [5, p. 917] (P-C1.2′/2″). (Dinh's w is our δ.)

P-C3a.4 / IH-Q2′ (Inglis & Herman 2013, verbatim).

The design assumption "a boundary stream width of R_c ... gives a cutoff size of R_c" [13, p. 1726]: cutoff radius linear in lamina width. This is the design-assumption sentence; it is distinct from the adjacent integration sentence used at C2 (registry tag IH-Q2) and carries its own registry number, IH-Q2′.

Critical question.

CQ: Consistency.

The four sources agree the cutoff scales with the lamina width (premises P-C3a.1–4); all four state a linear cutoff–lamina-width relation and no primary source contradicts it. Only the value of the proportionality constant is disputed (→ C3b–d). Scheme: Expert Opinion, consistency CQ. → *answered: yes.*

Status. Warranted.

C3b — Plateau value and exponents

Summary. A conservative weighted fit to three-width calibration data places the plateau prefactor at $\kappa_{\text{app}} = 4.91$ (CV 4.0%) with the predicted $\frac{1}{2}$ and 1 exponents inside their confidence intervals, and the result is invariant to digitization noise and to three independent digitizations (the primary and two re-digitizations). This node also carries a *downgrade*. The cross-paper comparison with Xia's devices, long reported here as a held-out corroboration at $\kappa_{\text{app}} = 5.17$, is a real recalculation from quoted data but it is not held out and it is not diagnostic: Xia takes its cutoff from the same Gifford figure this plateau was fitted from, and (the stronger reason) the 5.2% by which the recomputed mean exceeds the plateau *is* the 140/80 μm depth correction. It is reported as a *lead*. The fit stands on the CIF calibration data alone, and it does so without help.

Claim. On the large-particle plateau, $\kappa_{\text{app}} = 4.91$ [4.75, 5.08] (CV 4.0%), with cutoff exponents $\beta = 0.51$ [0.41, 0.61] (model $\frac{1}{2}$) and $\gamma = 1.02$ [0.84, 1.20] (model 1). The model exponents $\frac{1}{2}$ and 1 are derived in C6 / Appendix A (Result 6). The cross-paper figure $\kappa_{\text{app}} \approx 5.17$ (Xia) is reported below as a *lead* and is not part of this claim (CQ2).

Scheme. Argument from Example. (The weighted least-squares fit is premise-generation, a deterministic computation, not a defeasible scheme; the generalisation from the calibration sample is the Argument from Example.)

Premises (derived).

P-C3b.1 (weighted fit).

A weighted least-squares fit (error-bar-weighted, with confidence intervals scaled by $\sqrt{\chi^2_{\text{red}}} = \sqrt{3.8} \approx 1.95$, hence conservative) on the verified Gifford Fig. 3 data (15 points across three width curves; plateau subset $d/w_c \geq 0.035$, $n = 8$) gives $\beta = 0.51$ [0.41, 0.61], $\gamma = 1.02$ [0.84, 1.20], plateau $\kappa_{\text{app}} = 4.91$ [4.75, 5.08], CV 4.0%. A threshold-sensitivity sweep gives $\beta \in [0.51, 0.59]$ across cuts 0.030–0.045, so the result is not an artifact of the plateau definition. `[code: cif_cutoff.py B3/B4]`

P-C3b.2 (cross-paper consistency check: *not* a held-out test).

Recalculation from three Xia devices, using their stated cutoff and f_{gap} values, gives $\kappa_{\text{app}} = 5.03/5.16/5.32$, mean 5.17, at 140 μm depth against the Gifford plateau 4.9138 fitted at an assumed 80 μm: an agreement

of 5.2% across papers and depths. All three device triples are quoted data: OR-7 at $f_{\mathrm{gap}} = 5.8 \times 10^{-4}$ for a $7\,\mu$m cutoff, and RS-8 and RS-9 at 7.2×10^{-4} and 8.6×10^{-4} for 8 and $9\,\mu$m, all at $w_c(\mathrm{ref}) = 100\,\mu$m, are each stated verbatim in the source [2, pp. 6 and 10 (Sci Rep PDF pagination)]. **The recalculation is real and rests on quoted data.** What it is *not* is a held-out prediction, for two reasons that CQ2 separates, and the appendix states them rather than leaving them to be found. `[code: B5; glossary A.7]`

P-C3b.3 (digitization robustness).
The plateau β/γ survive a 2000-draw digitization Monte Carlo (β median 0.51, $P(0.4 < \beta < 0.7) = 1.00$) and are invariant across three independent digitizations of Fig. 3 (the primary and two re-digitizations) ($\kappa_{\mathrm{app}} \in [4.91, 4.92]$, $< 1\%$ spread; $\beta \in [0.50, 0.51]$; $\gamma \in [1.01, 1.02]$; the digitizations agree to a median of 0.4% and a worst-point 2.1% on f^*). `[code: B2b/B2c]`

Critical questions.

CQ1: Sample / computation adequacy.
The plateau κ and exponents are well-determined and not digitization artifacts: CV 4.0% across three widths; conservative (χ^2-inflated) confidence intervals; a threshold sweep that keeps $\beta \in [0.51, 0.59]$ across cuts 0.030–0.045; survival of the Monte Carlo and of three independent digitizations; and all numbers reproducing byte-for-byte on execution. None of this depends on the Xia anchor. Scheme: Argument from Example. → *answered: yes.*

CQ2: Is the anchor independent, and is the agreement diagnostic? (`xia_anchor_unverified`)
Neither, and this is the node's one open question. There are two separate reasons, **and the second is the stronger one**. It is worth saying at the outset what is *not* in question: the recalculation is real and its inputs are quoted, so nothing here impugns the arithmetic or the data.

First, it is not independent. Xia reads its cutoff from the same Gifford figure the plateau was fitted from, and Gifford is a Xia co-author, so the anchor and the fit share a *figure* rather than merely a literature. Xia's own sentence supplies the chain: "*OR-7 was patterned using* $w_c = 100\,\mu$m *and* $f_{\mathrm{gap}} = 5.8 \times 10^{-4}$*, a combination of parameters we found corresponding to a c.d. of 7*" μm "*previously*", citing Gifford 2014 and pointing at the specific figure: "*e.g. see Fig. 3 of Gifford et al.*" That is the figure this plateau is fitted from. Reproducibility is not independence.

Second, and decisively: the agreement is numerically the depth correction itself. Xia's devices are $140\,\mu$m deep against the plateau's assumed $80\,\mu$m, and $1/K_{\mathrm{depth}}(1.40) - 1 = 5.3\%$, the same correction as the 5.2% agreement above, the two printed figures differing only because they fall either side of a rounding boundary at full precision. Equivalently, $4.9138/0.9500 = 5.17$ reproduces the recomputed mean to three significant figures. The observed agreement is therefore *equally consistent* with a depth-robust prefactor and with a depth correction of exactly that size: it discriminates between them not at all. This is the reason that matters, and note that it does not depend on settling where 5.17 came from. Whether the number is a genuine recalculation that happens to land on the depth correction, or was synthesised from it, the comparison carries the same information either way, which is none.

What survives. The plateau value, the exponents and their intervals are fixed by the Gifford three-width fit and were never computed from this comparison, so none of them moves. What cannot be claimed is that they are corroborated *across laboratories* or *across depth*. The exponents' independent corroboration is C6's Davis cross-check (an unrelated laboratory and device family), and the depth factor's is the single-gap CFD of C3d. → *open: reported as a* ***lead****, not as a cross-paper agreement, and carrying no evidential weight in this appendix.*

Residual risk. One open item, named: `xia_anchor_unverified`. It bears on the cross-paper comparison only. The fitted plateau value, the exponents and their intervals do not rest on it, and nothing downstream (the operating law, C6's exponents, C7's band) is recomputed as a result, because none of them ever took an input from it. Closing it would need a cutoff determination for Xia's devices made independently of Gifford Fig. 3, which the primary paper does not report; it will not be closed from the literature.

Status. Partially closed: warranted on CIF data; the cross-paper anchor is a lead, not a corroboration.

C3c — The knee is real (defended on CIF data alone)

Summary. A single power law cannot fit the data ($\chi^2_{\text{red}} = 30.6$) and κ_{app} swings 3.7× from plateau to foot, so the prefactor is genuinely size-dependent: a knee. DLD theory, correctly re-normalised, has no knee (constant κ_{steric}), so the feature is CIF-specific. Because a size-independent Stokes-flow solve cannot produce a size-dependent knee, and the CFD finds no flow funnelling, the knee is attributed to a finite-particle capture-regime change (mechanism class: Doyeux's finite-sphere displacement). Existence is warranted now; the mechanistic shape is deferred to finite-particle work.

Claim. κ_{app} is not constant: a single global power law is rejected, and κ_{app} falls from 4.91 (plateau) to ≈1.34 (foot) through a knee at $d/w_c \approx 0.03$. The knee is attributed to a finite-particle capture-regime change; its existence is warranted on CIF data, its mechanism deferred to future finite-particle work.

Scheme. Argument from Sign for the *existence* + Abduction for the *mechanism*; the digitization undercutter is closed separately, by Evidence-to-Hypothesis in falsification form.

Premises (derived).

P-C3c.1 (global misfit).
The global single power law is rejected at $\chi^2_{\text{red}} = 30.6$ (frozen regression test: it must be rejected); a constant plateau-κ mispredicts the foot by 269% (frozen test 2). `[code: B3/B8]`

P-C3c.2 (the ramp).
κ_{app} ramps from 4.91 (plateau) to 1.34 (foot, $d/w_c < 0.012$), a 3.7× swing; the foot κ is 1.33–1.34 across all three independent digitizations. `[code: B4]`

P-C3c.3 (DLD is silent).
In the CIF normalisation $\kappa_{\text{app}} = d_{\text{crit}}/\left(g\sqrt{\varepsilon/3}\right)$, DLD theory gives a near-constant *apparent* prefactor, Davis 2.459–2.659 (recomputed from eqn 2-22), Inglis 2.040–2.449 (exact cubic root), across $\varepsilon = 0.01$–0.5: no knee, and never below the steric floor. *These are κ_{app}, not κ_{steric}*, which is exactly 2 (C9); the distinction matters because the CIF knee drops *below* 2 in the apparent prefactor, a regime DLD never probes, and it is the apparent prefactor that both literatures report. The Davis/Inglis small-ε values sit just above $\kappa_{\text{steric}} = 2$, supporting the steric floor of C2 and C9, not the knee. `[code: B7]`

P-C3c.4 (the finite-particle mechanism class).
Doyeux 2011 exhibits the relevant mechanism: a finite sphere is displaced relative to the fluid dividing line, and the attraction they measure is "directed towards the low flow rate branch" [14, abstract], i.e. toward the (low-flow) side channel. (The classical Zweifach–Fung reading points the other way; it is stated, and answered from the same abstract, at P-C3d.6.) This displacement is a finite-size effect, absent for a point tracer and growing with particle radius, so it raises the effective cutoff most for the largest particles (high κ_{app} on the plateau) and vanishes for the smallest, where κ_{app} falls toward and then below the steric floor to the sub-steric foot (P-C3c.5), reproducing the knee's direction; the precise shape $\kappa_{\text{app}}(d/w_c)$ is deferred to future finite-particle simulation.

P-C3c.5 (a known size-dependent extraction effect: bounded, and not the knee's cause).
Dinh reports that "recovery of the smallest 2 fL PLTs ($a \approx 1\,\mu$m) in the filtrate was consistently lower than the percent of extracted fluid volume" [5, p. 917], so size-dependent departures from tracer behaviour do occur. Two considerations bear on whether this effect could be the knee. *First, its sign would lower, not raise, κ_{app} at small d:* suppressing the recovery of the smallest particles narrows the effective lamina they sample ($\delta_{\text{eff}} < \delta_{\text{nominal}}$), and since $\kappa_{\text{app}} = \kappa_{\text{steric}}\,(\delta_{\text{eff}}/\delta_{\text{nominal}})$, the apparent prefactor falls. Crowding therefore *shares* the knee's direction and could be confounded with it, so its sign alone cannot rule it out. *Second, and decisively, it is a concentration-/flow-dependent many-body effect, not an intrinsic cutoff property:* Dinh's own sweep shows the 2 fL filtrate-recovery difference varying with operating conditions (23.2 → 9.8 percentage points as the flow rate rises from 30 to 90 μL/min) [5, p. 917], so it is minimal in the dilute, single-particle regime the rigid-bead calibration assumes. Crowding thus neither produces the calibration knee nor supplies the size-dependent f_{eff} the alternative hypothesis (CQ2) would need.

Critical questions.

CQ1: Strength of the sign (`more_reliable`).
The misfit is strong and quantified: $\chi^2_{\text{red}} = 30.6$ (30.6–31.4 across digitizations); plateau/foot differ by 3.7×;

no plausible digitization error bridges this (foot κ 1.33–1.34, global rejection 30.6–31.4 in every digitization). Scheme: Argument from Sign, strength CQ. The graph carries this as an *open* exception on `c3c_exists`, not because the sign is weak, but because "is there a more reliable indicator?" cannot be closed by the same data that raised it. → *answered on the evidence available: strong.*

CQ2: Alternative explanations of the sign (`more_coherent_explanation`).
The live alternative, that the apparent knee is an effective per-gap fraction f_{eff} that varies (with device or with particle size) rather than a steric regime change, is bounded but not eliminated. (i) The exponents are shift-invariant, so any *uniform* c leaves β, γ unchanged (C6 / Appendix A Result 6), disposing of a uniform f_{eff} but not a size-dependent one; (ii) Doyeux's finite-particle displacement (P-C3c.4) supplies a *positive* steric mechanism that reproduces the knee's direction, which a size-dependent f_{eff} would have to mimic without an independent driver; (iii) DLD is silent (P-C3c.3); (iv) the one identified size-dependent extraction effect, crowding (P-C3c.5), is concentration-gated and minimal in the dilute calibration, so the alternative lacks an operative driver there. Scheme: Abduction, more-coherent-explanation CQ (the exception this question is named for, and the scheme of `c3c_mech`). → *answered: the size-dependent f_{eff} alternative is bounded, not ruled out, but it contests only the **mechanism**; it cannot defeat the knee's **existence**, because a size-dependent f_{eff} is itself a form of size-dependence (see "Containment" below). The steric reading of the mechanism is favoured; the mechanism is deferred to future finite-particle work.*

CQ3: Mechanism honesty.
The knee is not "3-D funnelling." Two independent reasons: (a) a Stokes-flow solve is size-independent by construction, so it cannot produce a size-dependent knee at all; (b) the CFD measured $K_{\text{in-plane}} = 1$, so the flow does not funnel (C3d, Methods §4.3). Scheme: Established Rule (size-independence of Stokes flow) + numerical verification (CFD). → *answered: the mechanism is finite-particle and deferred; the "funnelling" reading is retracted.*

Residual risk. Two open exceptions, named in the graph: `more_reliable` on the knee's *existence* and `more_coherent_explanation` on its *mechanism*. The mechanism requires future finite-particle simulation; the size-dependent f_{eff} alternative is bounded but not eliminated.

Status. Warranted on CIF data (existence); mechanism rival open.

Containment of the $f_{\text{eff}}(d)$ alternative. The only *rival* still open in this appendix is one hypothesis: a *size-dependent* effective flow fraction $f_{\text{eff}}(d)$ in place of the finite-particle steric account. (An open *rival* is not the same object as an open critical question: `feff_arg` is the single node in the graph whose status is *open rival*, whereas the 10 undischarged critical questions of §1.1 are exception atoms spread over 7 nodes, two of them on this very node.) Its reach is exactly two nodes. It undercuts (a) the knee *mechanism* (this node's abductive attribution) and (b) the $\sqrt{c}\,C$ *decomposition* (C3d). It does *not* touch the operating law (C3′), the plateau exponents $\beta = \frac{1}{2}, \gamma = 1$ (C6, shift-invariant to any uniform c; the size-dependent case is delegated to this node), or the knee's *existence*. Existence is immune for a structural reason: a size-dependent $f_{\text{eff}}(d)$ is *itself* a form of size-dependence of the apparent prefactor, so it *co-signs* "κ_{app} varies" rather than defeating it; the contest is over the *mechanism* of the variation, never its reality. Granting the exception therefore flips only the mechanism and the decomposition to open; the law, the lemmas, and existence are unmoved.

C3d — Interpretation of κ's magnitude (partially closed)

Summary. The measured plateau prefactor (4.91) is a product of five interpretable factors. The steric factor (2) follows from C1; single-gap CFD across the full deep-CIF aspect-ratio range settles two more: no in-plane funnelling ($K_{\text{in-plane}} = 1$, a one-sided null with zero measured bulge) and a depth factor matching the analytic duct law to $\leq 0.6\%$, and in doing so reverses the earlier "apertures funnel more" intuition and retires the "3-D funnelling" reading of the knee. What remains is a single lumped residual, a degenerate product of per-gap over-extraction and cumulative finite-particle displacement, both tied to the unpublished gap count; it is explicitly not recoverable from the device-level filtrate fraction F. An independent branch-flow group likewise finds its measured cutoffs above prediction and splits the excess into the same two mechanism classes, corroborating the picture.

Claim. The plateau prefactor decomposes as

$$\kappa_{\text{app}} \;=\; \kappa_{\text{steric}} \cdot K_{\text{in-plane}} \cdot K_{\text{depth}} \cdot \sqrt{c}\,C \;=\; 2 \cdot 1 \cdot R(\alpha) \cdot \sqrt{c}\,C.$$

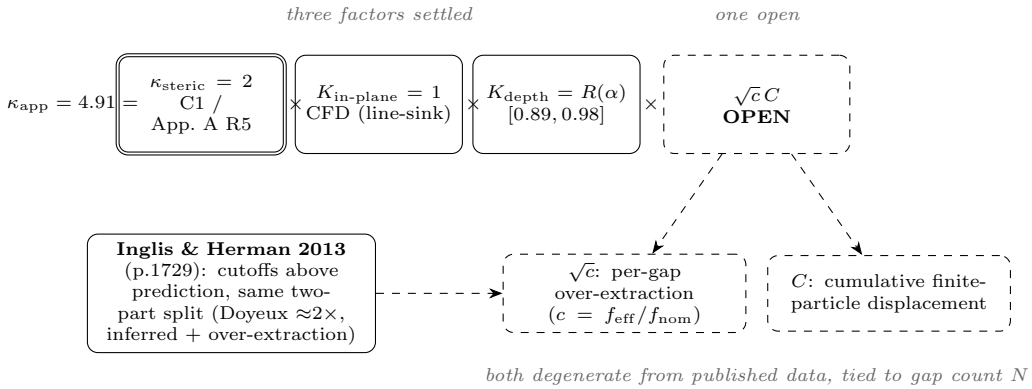


Figure 5: Decomposition of the plateau prefactor (C3d). Three factors are settled: $\kappa_{\text{steric}} = 2$ (doubled border, deductive, from C1 / Appendix A Result 5), $K_{\text{in-plane}} = 1$ and $K_{\text{depth}} = R(\alpha)$ (both from single-gap CFD), leaving a single open quantity, the lumped residual $\sqrt{c}\,C$ (dashed), which splits into per-gap over-extraction $\sqrt{c}$ and cumulative finite-particle displacement C, both degenerate from the published data and tied to the unpublished gap count N. An independent branch-flow group reports the same two-part split, corroborating the interpretation.

Three factors are settled: $\kappa_{\text{steric}} = 2$, $K_{\text{in-plane}} = 1$ (CFD), $K_{\text{depth}} = R(\alpha) \in [0.89, 0.98]$ (CFD + duct law), leaving the single lumped product $\sqrt{c}\,C$ as the one open quantity. With $R(\alpha)$ folded in, the excess over steric is depth-folded and α-independent, $\kappa_{\text{app}}/\kappa_{\text{steric}} = 4.9138/2 = 2.4569$; dividing out the measured $K_{\text{depth}}^{\text{CFD}}$, $\sqrt{c}\,C = \kappa_{\text{app}}/\left(\kappa_{\text{steric}} K_{\text{depth}}^{\text{CFD}}\right) = 2.7574/2.5999/2.5044$ across $\alpha = 0.80/1.40/2.69$ (each value × its $K_{\text{depth}}^{\text{CFD}}$ returns 2.4569).

Scheme. Numerical verification (single-gap CFD: a deductive computation with no critical questions) for the settled factors + Argument from an Established Rule + Argument from Expert Opinion (Doyeux) + Defeasible Modus Ponens closing the Zweifach–Fung undercutter that stands on it (`c3d_zf_refuted`) + Abduction for the open residual.

Premises.

P-C3d.1 ($\kappa_{\text{steric}} = 2$).
From C1/CQ5 and Appendix A Result 5 (Inglis $d_{\text{crit}} = 2b$ + Dinh main-text $a < w$ / $a > w$).

P-C3d.2 (plateau $\kappa_{\text{app}} = 4.91$).
From C3b.

P-C3d.3 ($K_{\text{in-plane}} = 1$; CFD).
The single-gap Stokes campaign across $\alpha = 0.80/1.40/2.69$ (Runs C/B/A) measures $K_{\text{in-plane}} = 1.000$ at every fraction and station (max bulge $0.0000\,\mu$m); the dividing streamsurface *thins* toward the gap (line-sink behaviour), a one-sided null (δ_{bulge} is a maximum, so $K = 1$ only if the line never bulges; funnelling would read $K > 1$). One documented exception: Run A at 2×10^{-4}, $K_{\text{in-plane}} =$ n/a (a low-fraction post-processing limit, flagged not forced) [Methods §4.3]. `[code: B9]`

P-C3d.4 ($K_{\text{depth}} = R(\alpha)$; CFD + duct law).
The CFD depth factor matches the analytic rectangular-duct law to $\leq 0.6\%$ (CFD 0.891/0.945/0.981 vs analytic 0.8879/0.9500/0.9794; per-aspect-ratio deviations 0.349%/0.530%/0.160%, max 0.531% on Run B); the cross-validation `duct.K_depth`$(100, 80) = 0.8879$ against CFD $R(\alpha{=}0.8) = 0.891$ has COMSOL independently confirming the code's duct module at a real device aspect ratio, to 0.349%. `[code: B9; Appendix A Result 9; glossary A.4/A.12]`

P-C3d.5 (the residual: arithmetic).
With $\kappa_{\text{steric}} = 2$ and $K_{\text{in-plane}} = 1$, the residual reduces to the single lumped product $\sqrt{c}\,C$, reportable two equivalent ways: the depth-folded (α-independent) excess $\kappa_{\text{app}}/\kappa_{\text{steric}} = R(\alpha)\,\sqrt{c}\,C = 2.4569$, or, dividing out the measured $K_{\text{depth}}^{\text{CFD}}$, $\sqrt{c}\,C = \kappa_{\text{app}}/\left(\kappa_{\text{steric}} K_{\text{depth}}^{\text{CFD}}\right) = 2.7574/2.5999/2.5044$ across $\alpha = 0.80/1.40/2.69$.

Here $\sqrt{c}$ is per-gap over-extraction ($c = f_{\text{eff}}/f_{\text{nom}}$, a per-device constant from the local gap-to-channel resistance ratio) and C is cumulative finite-particle displacement over many gaps; the two are degenerate from published data, both tied to the unpublished gap count N. A sobering bound: if $C = 1$, then $\sqrt{c} = \sqrt{c}\,C$ and $c = (\sqrt{c}\,C)^2$ is of order 6–8 (≈ 6 with the depth factor folded in, up to $\sim$8 with it explicit): the true per-gap extraction would be several-fold the nominal design fraction, which is why the split cannot be hand-waved. `[code: B6]`

P-C3d.6 / D-Q1 (Doyeux 2011, verbatim): the direction, and the standing objection to it.

This is the premise the Scheme field names as Argument from Expert Opinion, and the graph carries it as `c3d_doyeux_direction`. Doyeux *et al.* set out to test the classical reading of branch partitioning, on which particles are attracted to the *high*-flow branch, and report the reverse: "such an attraction does not occur but is, on the contrary, directed towards the low flow rate branch" [14, abstract]. In CIF the side channel *is* the low-flow branch, so the displacement is toward extraction; it therefore raises the cutoff rather than lowering it, and it grows with particle radius, which is the size-dependence C3c needs (P-C3c.4).

The Zweifach–Fung objection, and why it does not bite. The graph carries `zweifach_fung_objection` as an undercutter on this premise, and it is the first question a reader of the blood-flow literature will raise: the Zweifach–Fung effect is the long-observed *enrichment* of particles in the high-flow branch, which looks like a direct contradiction. Doyeux answer it in the same abstract, by separating two effects rather than denying either: "the enrichment in particles in the high flow rate branch is shown to be mainly a consequence of the initial distribution in the inlet branch" [14, abstract]. A finite sphere's centre cannot approach a wall closer than one radius, so the near-wall streamtube is already depleted of centres before the bifurcation; the enrichment is that inlet depletion, and it is compatible with a particle-scale attraction of the opposite sign. Both hold at once, and only the attraction is a force on the particle. The graph closes the undercutter at `c3d_zf_refuted` (Defeasible Modus Ponens, warranted). Inglis & Herman state the same contrast independently, in the sentence quoted next.

P-C3d.7 / IH-Q3 (Inglis & Herman 2013, p. 1729, quoted continuously): independent two-part corroboration.

"The experimentally observed cutoff diameters are significantly higher than the cutoff sizes determined from our design process. This is in agreement with Doyeux *et al.* who showed that contrary to the situation in the Zweifach–Fung effect, particles are attracted to the low flow rate side of a branch. Through experiment and modelling Doyeux *et al.* found that excluding a particular size particle from a branch may require a streamline width that is as much as 50% smaller than the particle radius. A second contributing reason is that the devices, as fabricated, were approximately 48 μm deep, as opposed to the 50 μm as designed. This increases the main channel resistance more than it does the branch channel resistance because the branches are only 25 μm wide. This increased main channel resistance will mean more fluid flow in the branches, increasing the critical size at each branch" [13, p. 1729].

This passage is quoted *whole*: no ellipsis, and nothing elided but the two reference superscripts, which is why no ellipsis marks them. Two glyphs are set in the appendix's house form because the OCR layer cannot distinguish them: the depth and width units are the paper's micrometres (the layer flattens μ to m), and the dash in Zweifach–Fung is set as an en dash, as the bibliography sets it. $\rightarrow$ an independent branch-flow group reports cutoffs above prediction and decomposes the excess into the same two mechanisms. Their *first* reason is the Doyeux direction, stated with its own anti-Zweifach–Fung contrast, so the corroboration at P-C3d.6 is theirs as well as Doyeux's; the finite-particle factor is $\approx 2\times$, inferred from their "50% smaller than the particle radius." Their *second* is over-extraction from increased main-channel resistance, the $\sqrt{c}$ factor, and they give the reason for it: a shallow mould raises the *main* channel's resistance disproportionately *because the branches are narrow to begin with*, so flow redistributes into the branches. That is the post-CFD CIF picture in kind, a finite-particle knee + $\sqrt{c}$ over-extraction, and not flow funnelling.

Critical questions.

CQ1: Computation / measurement validity (CFD).

The CFD result is trustworthy: $K_{\text{in-plane}} = 1$ is a one-sided null with max bulge $0.0000\,\mu$m across all runs/stations; $R(\alpha)$ reproduces the analytic duct law to $\leq 0.6\%$ and reproduces a known code value at a real aspect ratio; $\delta_{\text{skim}} \propto \sqrt{f}$ to $R^2 \geq 0.9999$; all wired to frozen regression stamps against `REFERENCE_OUTPUT*.txt`. Scheme: numerical verification (deductive computation). $\rightarrow$ *answered: yes* (standing caveat: raw COMSOL fields are not archived; the reference tables + post-processing scripts are

the reproducibility surface; a cheap $0.5\times$ /$2\times$ mesh-convergence point would close the obvious referee question).

CQ2: Direction (does the aperture funnel more than DLD posts?).
No. The aperture funnels less. The earlier intuition (CIF wall-apertures should funnel more than DLD posts, a weak $\sim 1.2\times$ Davis prior) is reversed by the CFD: $K_{\text{in-plane}} = 1$, below $1.2\times$. The aperture is a line-sink and the captured layer does not bulge. Scheme: numerical verification overruling a prior argument from analogy, C2's `c2_funnel_rival`, which the graph carries as defeated. → *answered: reversed; no funnelling.*

CQ3: Can the residual be read out of F?
No. The over-extraction is not "fixed by F." The rigorous outlet identity $F = 2R_c/(R_s + 2R_c)$ (0.5592 and 0.7189 at Dinh's V1 and V3) is outlet-geometry-only and N-independent (C4); 0.3516 is a cumulative-yield number (the product Nf_{gap} that $\ln(1 - F) = -2Nf_{\text{gap}}$ returns at $F \approx 0.505$), not F, and is historical rather than a live value (its provenance and arithmetic are given under C4's cautionary tale). The cutoff depends on the local per-gap c, narrowable by a local resistance-network calculation (the `network` module is the seed), not by F (reading κ out of F inverts the C4 separation lemma). Scheme: Established Rule (C4) blocking a category error. → *answered: no.*

Residual risk. Two open exceptions, named in the graph. The first is `degenerate_from_published_data`: identifiability, meaning that the two factors cannot be separated from what the papers actually report. The second is `more_coherent_explanation`, the f_{eff} rival, shared with C3c. The $\sqrt{c}\,C$ residual is open and tied to N; resolved by finite-particle work (the C factor) + a local resistance-network calculation and/or micrograph gap count (the $\sqrt{c}$ factor), not by the single-gap solve (now done) and not by the F-identity (decoupled). A sibling $\sim 5\%$ 3-D-junction uncertainty rides on $\sqrt{c}$. *Exhaustiveness*, as distinct from identifiability, is answered downstream: within the model's premises, C9d closes the residual list to three model gaps.

Status. Partially closed: $K_{\text{in-plane}}$, κ_{steric}, $R(\alpha)$ settled; $\sqrt{c}\,C$ open.

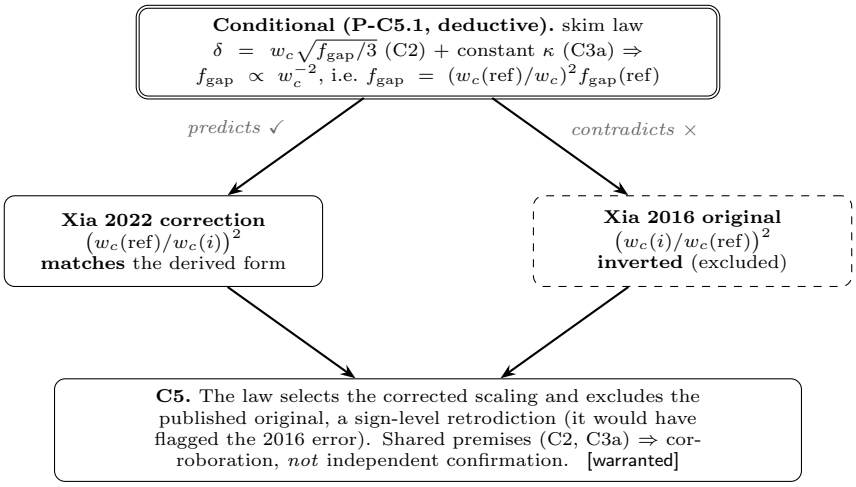


Figure 6: Argument graph for C5 (verification-form Evidence-to-Hypothesis). The deductive conditional (doubled border) predicts a single cross-width schedule, $f_{\rm gap} \propto w_c^{-2}$. The 2022 author correction matches it; the 2016 published original is its reciprocal and is excluded. Because the *form* follows from Xia's own two assumptions (C2 + C3a) and the author correction is a same-lineage self-correction, the confirmation is sign-level (the *direction*) rather than an independent verification. Border style follows the overview map: doubled = deductive, single = warranted/corroborating, dashed = the excluded rival.

C5 — Consilience (sign-level): the skim law recovers the corrected Xia (2022) scaling and excludes the 2016 original

Summary. On the plateau, where the apparent prefactor $\kappa_{\rm app}$ is constant, the skim law forces the centre-channel width into the per-gap fraction as $f_{\rm gap} \propto w_c^{-2}$ at fixed cutoff. This is exactly the 2022 author-corrected Xia design equation and the reciprocal of the since-corrected 2016 original. Agreement on the *form* is a shared-premise consistency check: it reuses the parabolic lamina width of C2 and the proportionality of C3a, which are Xia's own two assumptions, and the author correction is a same-lineage self-correction, so neither is independent confirmation. Agreement on the *direction* is the load-bearing result: the law contradicts the 2016 equation as published and agrees only with the 2022 correction, a retrodiction (it would have flagged the 2016 error).

Claim. With κ held constant (the plateau / proportionality regime),

$$d_{\rm crit} = \kappa\, w_c\sqrt{\tfrac{f_{\rm gap}}{3}} \quad\Longrightarrow\quad f_{\rm gap} \propto w_c^{-2}, \qquad \text{i.e.} \qquad f_{\rm gap}(i) = \left(\frac{w_c({\rm ref})}{w_c(i)}\right)^2 f_{\rm gap}({\rm ref}),$$

which is the 2022 author-corrected Xia design equation (their eqn (1)).

Scheme. Argument from Evidence to a Hypothesis, verification form. The *conditional* (P-C5.1) is that if the skim law holds, the cross-width schedule must be $f_{\rm gap} \propto w_c^{-2}$. The *evidence* (P-C5.2, P-C5.3) is that the corrected literature reports that schedule while the erroneous original reported its reciprocal, so the evidence discriminates rather than merely fitting. The *conclusion* is that the skim law is corroborated in direction.

Premises.

P-C5.1 (derived conditional; exact, code-checked).
Holding the lamina width $\delta = w_c\sqrt{f_{\rm gap}/3}$ fixed (equivalently, $d_{\rm crit}$ fixed *at constant* κ) and solving for $f_{\rm gap}$,

$$w_c\sqrt{\tfrac{f_{\rm gap}}{3}} = w_c({\rm ref})\sqrt{\tfrac{f_{\rm gap}({\rm ref})}{3}} \implies f_{\rm gap} = \frac{w_c({\rm ref})^2}{w_c^2}\, f_{\rm gap}({\rm ref}) \quad \text{(exact)}.$$

The constant-κ step is the plateau / proportionality assumption (identical to C3a and to Xia's assumption (b)); it is the premise CQ1 and CQ4 address, made explicit here rather than left tacit. *Where the* 3

comes from. $\delta = w_c\sqrt{f_{\mathrm{gap}}/3}$ is the deep-channel limit of a finite-aspect-ratio form, $\delta = w_c\sqrt{f_{\mathrm{gap}}/c_2(\alpha)}$ with $c_2 \to 3$ as $\alpha \to \infty$; the depth factor $K_{\mathrm{depth}} = \sqrt{3/c_2(\alpha)}$ is that same rectangular-duct series, evaluated by `K_depth` in P-C2.5. It is folded into the *apparent* prefactor rather than applied to the data (§1.1), so the depth factor sits *inside* the constant-κ_{app} assumption rather than beside it: K_{depth} does vary along a taper, and that variation shows up as variation in κ_{app}, which is the premise CQ4 examines. No separate aspect-ratio premise is needed here. `[code: scaling.f_gap_of_wc` → $f_{\mathrm{gap}}(\mathrm{ref})w_c(\mathrm{ref})^2/w_c^2$`; verify_symbolic asserts it equals` $(w_c(\mathrm{ref})/w_c)^2 f_{\mathrm{gap}}(\mathrm{ref})$`; verify_numeric confirms cutoff-invariance from Xia's OR-7 reference values (`$w_c(\mathrm{ref}) = 100\,\mu\mathrm{m}$`, `$f_{\mathrm{gap}}(\mathrm{ref}) = 5.8 \times 10^{-4}$`), evaluated at` $w_c = 150\,\mu\mathrm{m}$`, the RS-7 taper's minimum centre-channel width, to` 10^{-18}`; Appendix A Result 7]`

P-C5.2 (corrected Xia scaling; verbatim).
The 2022 author correction states that in eqn (1) "the terms $w_c(i)$ and $w_c(\mathrm{ref})$ are inverted," and that it "should read"

$$f_{\mathrm{gap}}(i) = \left(\frac{w_c(\mathrm{ref})}{w_c(i)}\right)^2 f_{\mathrm{gap}}(\mathrm{ref}),$$

exactly the direction the derived conditional (P-C5.1) forces [3].

P-C5.3 (the 2016 original was the inverted form; verbatim).
The published 2016 eqn (1) reads

$$f_{\mathrm{gap}}(i) = \left(\frac{w_c(i)}{w_c(\mathrm{ref})}\right)^2 f_{\mathrm{gap}}(\mathrm{ref}),$$

the wrong direction as w_c narrows along the taper ($300 \to 150\,\mu$m, with $w_c(\mathrm{ref}) = 100\,\mu$m): at fixed cutoff a narrower channel must divert a *larger* fraction, so f_{gap} must rise as w_c falls, whereas the 2016 form makes it fall. The 2022 author correction repairs it in exactly the direction P-C5.1 predicts; the erroneous form is the published eqn (1) [2, eqn (1)], and the correction reproduces it [3, p. 1].

Critical questions.

CQ1: Could the evidence be explained without the hypothesis? (alternative-explanation).
On the *form*, yes: $f_{\mathrm{gap}} \propto w_c^{-2}$ follows directly from Xia's two assumptions, the parabolic profile (C2) and the proportionality of cutoff to lamina width (C3a), so the form agreement restates one relationship rather than supplying a second, independent witness. Answer (by established rule / derivation): shared-premise consistency, disclosed; the form carries no independent confirmatory weight.

CQ2: Is the evidence reliable and independent of the hypothesis's source? (evidence-reliability).
Reliable: a published author correction, transcribed from the original. But *not* third-party independent: it is a self-correction by the Gifford/Xia/Shevkoplyas lineage whose devices and data this analysis builds on. The confirmatory reading is therefore *the law is consistent with the field's accepted (corrected) form and inconsistent with the since-corrected 2016 original*, not *an independent group verified the law.* Answer: reliable but same-lineage; weight bounded accordingly.

CQ3: Was the prediction risky; does the evidence discriminate? (severity).
Yes, in direction: the law's w_c^{-2} contradicted the *published* 2016 eqn (1) and matched only the later correction, so the evidence excludes the then-published rival. The risk is against the literature; given the shared premises the direction is determinate, so the confirmation is modest but real. Answer: discriminates on sign, a successful retrodiction of the corrected direction; equivalently, the law would have flagged the 2016 error.

CQ4: Does the conditional hold without hidden assumptions? (conditional-robustness).
The conditional assumes κ_{app} independent of w_c at fixed cutoff: the plateau, where κ_{app} is flat (CV 4.0%). RS-7, the one RS device whose taper Xia prints, meets that only where it is narrow: across its $300 \to 150\,\mu$m taper the cutoff-to-width ratio runs $d/w_c = 0.023$–0.047 (RS-8 and RS-9 would run 0.027–0.053 and 0.030–0.060 on the same taper; their starting widths are not printed), and 0.07–0.09 is the ratio at the $100\,\mu$m reference width. Off the plateau, on C7's fit of $\kappa_{\mathrm{app}}(d/w_c)$, the implicit law holds the cutoff only if the gap fraction falls more slowly than w_c^{-2}: at $300\,\mu$m the corrected form supplies about half of what a $7\,\mu$m cutoff requires, and RS-7's sections wider than about $236\,\mu$m fall below the saddle-node C7 derives. The direction

does not depend on the plateau: the required gap fraction still falls as w_c widens, and the inverted form errs the other way, about 42× too much at 300 μm. Answer: the direction holds across the taper; the exact w_c^{-2} form needs the plateau, which RS-7 reaches only on its narrow sections.

Status. Warranted as a sign-level cross-check: the law selects the corrected scaling and excludes the published original. It is *not* independent confirmation, because the form agreement reuses the premises of C2 and C3a and the author correction is a same-lineage self-correction. The confirmatory residue is therefore the *direction* (the law would have flagged the 2016 error), which holds across RS-7's whole taper; the exact w_c^{-2} form needs the plateau, which RS-7 reaches only on its narrow sections (CQ4).

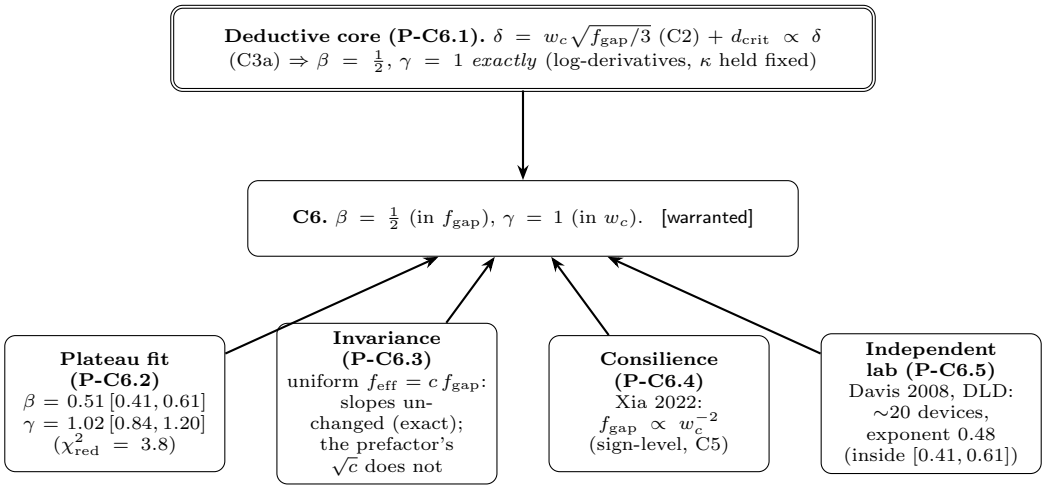


Figure 7: Argument graph for C6. The exponents are a deductive corollary of the C2 skim-width identity (doubled border), confirmed by the three-width plateau fit, corroborated in sign by the Xia 2022 cross-width scaling and in exponent by Davis's independent DLD device family, and, uniquely among the law's parameters, *exactly invariant* to a uniform per-gap over-extraction c (the confound the prefactor absorbs through $\sqrt{c}$). The measured $\beta \approx \frac{1}{2}$ positively selects the thin-lamina $\sqrt{f_{\text{gap}}}$ law over the linear law and over the rejected single global fit.

C6 — Cutoff-law exponents: $\beta = \frac{1}{2}$ (in f_{gap}), $\gamma = 1$ (in w_c)

Summary. The cutoff law's exponents are $\frac{1}{2}$ on the per-gap fraction f_{gap} and 1 on the channel width w_c. They are not free-standing empirical findings: they are the direct corollary of the skim-width identity proved in C2 ($\delta = w_c\sqrt{f_{\text{gap}}/3}$) together with the proportionality of C3a ($d_{\text{crit}} \propto \delta$). The plateau fit returns (0.51, 1.02); the combined width–fraction scaling is corroborated by the Xia 2022 corrected relation $f_{\text{gap}} \propto w_c^{-2}$ (C5); and β is recovered independently, in another laboratory and another device family, by Davis's DLD correlation (P-C6.5). Because β and γ are slopes in a log–log regression, they are *exactly* invariant to any *uniform* misestimate of the per-gap fraction; this is the one confound the prefactor's physical decomposition ($\kappa_{\text{app}} = \kappa_{\text{steric}} \cdot K_{\text{in-plane}} \cdot K_{\text{depth}} \cdot \sqrt{c}\,C$, C3d) does *not* survive (its $\sqrt{c}$ absorbs a uniform c), whereas the exponents do.

Claim. In the operating law

$$d_{\text{crit}} = \kappa_{\text{app}}(d/w_c,\,\text{depth})\,w_c\sqrt{f_{\text{gap}}/3}, \qquad \delta = w_c\sqrt{f_{\text{gap}}/3},$$

the exponent of the per-gap fraction f_{gap} is $\beta = \frac{1}{2}$ and the exponent of the channel width w_c is $\gamma = 1$.

Scheme. Defeasible Modus Ponens on the C2 hydrodynamic derivation (deductive), with Argument from an Established Rule for the log–log shift-invariance, combined with Argument from Example twice over (the weighted fit to the Gifford Fig. 3 data, and Davis's independent DLD device family) and cross-checked against C5, whose own scheme is Evidence to a Hypothesis in verification form, over the Xia 2022 corrected cross-width scaling. The named schemes fix the matching critical questions below.

Premises.

P-C6.1 (derivation; inherited from C2 + C3a).
Holding κ fixed (the plateau), the $\sqrt{f_{\text{gap}}}$ dependence of the law gives $\partial \ln d_{\text{crit}}/\partial \ln f_{\text{gap}} = \frac{1}{2}$ and the linear w_c dependence gives $\partial \ln d_{\text{crit}}/\partial \ln w_c = 1$, both *exactly*. The $\sqrt{f_{\text{gap}}}$ is itself the leading term of the exact skim cubic $f_{\text{gap}} = 3x^2 - 2x^3$ ($x = \delta/w_c$) established in C2, so β and γ are corollaries of an already-proven identity rather than independent claims. `[code: scaling.verify_symbolic() asserts beta == 1/2, gamma == 1; Appendix A Result 6]`

P-C6.2 (plateau fit).
The error-bar-weighted log–log fit on the plateau ($d/w_c \geq 0.035$, $n = 8$) returns

$$\beta = 0.51\ [0.41,\ 0.61]\ \text{(model } \tfrac{1}{2}\text{)}, \qquad \gamma = 1.02\ [0.84,\ 1.20]\ \text{(model 1)}, \qquad \chi^2_{\text{red}} = 3.8.$$

Both model values $\frac{1}{2}$ and 1 lie inside their 95% confidence intervals. The intervals are conservative: the reported covariance is scaled by χ^2_{red}, i.e. the standard errors are inflated by $\sqrt{3.8} \approx 1.95$, exactly the factor that would bring the plateau χ^2_{red} to unity. `[code: B3/B4]`

P-C6.3 (invariance to a uniform f_{eff} mis-estimate).
Write the true per-gap fraction as $f_{\text{eff}} = c\, f_{\text{gap}}$ with c a device-level constant. Then $d_{\text{crit}} = \kappa\, w_c \sqrt{f_{\text{eff}}/3} = \sqrt{c}\,\kappa\, w_c \sqrt{f_{\text{gap}}/3}$: a uniform c rescales the *prefactor* ($\kappa \to \sqrt{c}\,\kappa$) and leaves the exponents of f_{gap} and w_c untouched. Equivalently, in the log–log regression a factor c shifts only the *intercept* ($\ln \kappa$) and not the *slopes*; the invariance is algebraically exact, not approximate, and is confirmed numerically as identical at $c \in \{1, 2, 3.5\}$ (the plateau $\beta = 0.51$, $\gamma = 1.02$ in every case). This is the confound the prefactor's decomposition does *not* survive (its $\sqrt{c}$ absorbs c, C3d), whereas the exponents do. `[code: B6, ''EXPONENTS and KNEE robust to c'']`

P-C6.4 (consilience with the Xia 2022 corrected scaling; shared-premise, C5).
The Xia 2022 Author Correction states (verbatim) that the original equation inverted the width terms and "should read: $f_{\text{gap}}(i) = (w_c(\text{ref})/w_c(i))^2 f_{\text{gap}}(\text{ref})$." With $\beta = \frac{1}{2}$ fixed by the parabolic profile, requiring a matched cutoff across widths forces $w_c \sqrt{f_{\text{gap}}} = \text{const}$, i.e. $f_{\text{gap}} \propto w_c^{-2}$, precisely the corrected relation. This is a consilience, *not* an independent test: the w_c^{-2} form follows from Xia's own two assumptions ("approximately parabolic" profile; diverted-streamline width "roughly proportional to the effective c.d."), the same physics C2/C3a rest on. Its evidential value is in *sign*: the law agrees with the *corrected* equation and excludes the originally published (inverted) form. The empirical cross-width measurement of γ remains the three-width Gifford fit of P-C6.2. `[Xia 2022 Author Correction; Xia 2016 p. 10 (Sci Rep PDF pagination); C5; Appendix A Result 7]`

P-C6.5 (independent-lab cross-check; a different device family).
Davis fits a DLD critical-diameter correlation to data from "about 20 different devices" spanning $G = 1.3$–$38\,\mu$m and $\varepsilon = 0.005$–0.5, with polystyrene beads 0.9–22 μm, and obtains $d_{\text{crit}} = 1.4\, G\varepsilon^{0.48}$ [15, eqn (2-22)]. The exponent 0.48 lies inside this node's plateau interval [0.41, 0.61], and it is load-bearing in its own paper rather than incidental: "equation 2-22 was used to design all the" thresholds reported there. This is the one support for β that survives the withdrawal of the entire CIF derivation, sharing with it neither laboratory, device family nor data; it does not test γ, because his correlation is proportional to the gap G by its form. *Its limit is the source's own, and it is the limit this node would predict*: Davis records that against his measured points "the parabolic relationship underestimates the critical size," attributing the gap to "short, rounded posts" rather than to the exponent. The cross-check is therefore an agreement of *exponent* and not of prefactor (his recomputed $\kappa_{\text{app}} = 2.459$–$2.659$ sits above the exact cubic's 2.040–2.449, C3c P-C3c.3), which is exactly the split P-C6.3 asserts: a uniform mis-estimate moves the prefactor and never the slopes. `[Davis 2008 eqn (2-22), §2.4 (pp. 22--24); graph node c6_davis; Appendix A Result 6]`

Critical questions. The exponents $\beta = \frac{1}{2}, \gamma = 1$ are a deductive corollary of the C2 identity and carry no critical questions *as identities*; the questions below test the *regime* in which they apply and the empirical fit, not the derivation.

CQ1: Is the rule (the derivation) correctly applied and sound?
Yes. $\beta = \frac{1}{2}$, $\gamma = 1$ are a proven symbolic identity (`scaling.verify_symbolic()`), being the leading-order consequence of the exact cubic $f_{\text{gap}} = 3x^2 - 2x^3$ proved in C2; the symbolic and finite-difference exponents agree and reproduce byte-for-byte.

CQ2: Do the exponents hold globally, or only on a restricted subset?
Only on the plateau, by construction, and this is a strength, not a concession. A *single* global power law across all 15 points is *rejected* by its misfit ($\chi^2_{\text{red}} = 30.6$; cf. C3c). The blended slopes that such a fit returns ($\beta \approx 1.21$, $\gamma \approx 1.55$) are the artifact of forcing one straight line through a curve with a knee; they are therefore *not adopted* as the law's exponents. `[code: global_powerlaw_misfit(with_slopes=True), which returns the misfit and these two slopes from one regression]` The global fit is reported alongside the plateau fit precisely so it can be seen to fail: shown to be rejected, not suppressed.

CQ3: Could a different flow regime yield different exponents?

Yes, and the observed value *discriminates* between the laws they give. In a shallow, Hele-Shaw channel ($h \ll w_c$) a lamina much wider than the depth ($\delta \gg h$) samples a depth-averaged plug across the width, $f_{\mathrm{gap}} \propto \delta/w_c$, hence $\delta \propto f_{\mathrm{gap}}$ and $\beta = 1$, a *distinct* law; but β is $\frac{1}{2}$ whenever $\delta \ll h$, at any aspect ratio, and CIF's laminae are orders of magnitude thinner than the channel depth (Appendix A Result 6). The plateau fit lands at $\beta \approx \frac{1}{2}$, with $\beta = 1$ far outside its confidence interval $[0.41, 0.61]$, so the data positively selects the thin-lamina ($\sqrt{f_{\mathrm{gap}}}$) law over the linear one. CIF's moderate aspect ratio ($\alpha = h/w_c \approx 0.5$–$2.7$, not asymptotically deep) enters only through the depth factor $R(\alpha) \in [0.81, 0.98]$ (C2) folded into the apparent prefactor, never through the exponent. For β that is exact, since R does not depend on f_{gap}. For γ it is not an identity, since α falls as w_c widens across the fit: folded into the prefactor, where C2 puts it, $R(\alpha)$ leaves $\gamma = 1.02$; applied to the lamina, it would move γ to 1.24 (interval $[1.04, 1.45]$, excluding 1) at the assumed $80\,\mu$m depth, the shallowest Gifford reports, but keep 1 inside the interval at the 125 or $150\,\mu$m it also reports (§1.3).

CQ4: Is $\beta = \frac{1}{2}$ an artifact of the plateau cut or of the figure digitization?

No, on three independent counts. *(i) Threshold:* sweeping the plateau cut over $d/w_c = 0.030$, 0.035, 0.040 and 0.045 keeps $\beta \in [0.51, 0.59]$ (the first two select the same 8 points). *(ii) Digitization noise:* perturbing each f^* within its digitized σ over 2000 draws gives β median 0.51 with $\mathrm{P}(0.4 < \beta < 0.7) = 1.00$. *(iii) Independent re-digitization:* three independent digitizations of Fig. 3 (the primary one and two re-digitizations, whose group labels were discarded and recovered by sort-and-pair) give $\beta \in [0.50, 0.51]$ and reject the global law every time ($\chi^2_{\mathrm{red}} \in [30.6, 31.4]$).

CQ5: Is the $n = 8$ plateau sample representative?

Yes, and the answer no longer leans on the cross-paper comparison it once cited. The Xia reference triples (OR-7, RS-8, RS-9, all at $w_c(\mathrm{ref}) = 100\,\mu$m) do run through the plateau law to $\kappa_{\mathrm{app}} = 5.03/5.16/5.32$, close to the Gifford plateau $\kappa_{\mathrm{app}} \approx 4.91$, but C3b's CQ2 establishes that the comparison is circular and, worse, non-diagnostic (its agreement is the depth correction), so it is not evidence of anything about the sample. The answer instead rests on what is internal to this node and on C6's own corroboration: the threshold sweep (CQ4 (i)), the digitization Monte Carlo and the re-digitizations (CQ4 (ii)–(iii)), and the fact that these exponents are established by *shift-invariance* rather than by the plateau's absolute value, so a representativeness worry about $n = 8$ would have to move β and γ, not κ_{app}, and the sweep shows it does not. Above all, P-C6.5 answers the question from outside the sample altogether: Davis's ∼20 devices are an unrelated laboratory and a different device family, and they return an exponent inside this node's interval.

Scope note. P-C6.3 establishes robustness to a *uniform* (device-level) f_{eff} mis-estimate only. A *size-dependent* $f_{\mathrm{eff}}(d)$ would shift the slopes and is the separate alternative addressed (bounded but not eliminated) in C3c, where it would manifest as the knee rather than as a shift of the plateau exponents. C6 does not claim to exclude it; that is C3c's burden.

Status. Warranted; robust to a uniform f_{eff} mis-estimate. Residual risk: none on the exponents themselves. The size-dependent $f_{\mathrm{eff}}(d)$ possibility is delegated to C3c; the prefactor's physical decomposition is the subject of C3d and is not part of this claim.

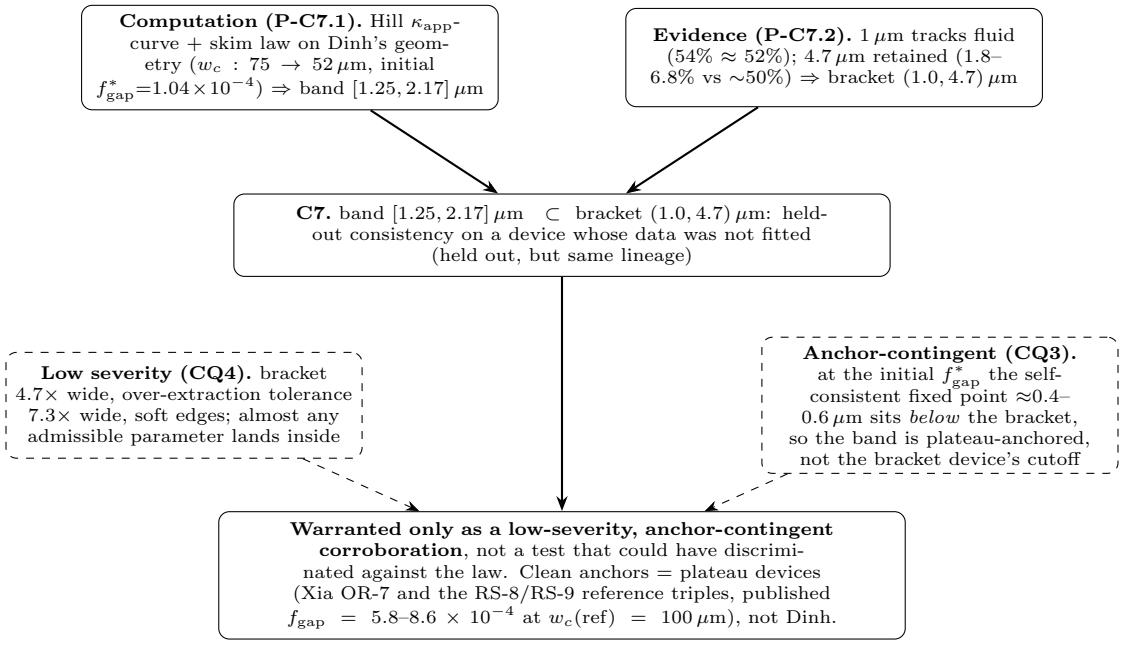


Figure 8: Argument graph for C7. The computed band (Computation, P-C7.1) and the measured bracket (Evidence, P-C7.2) agree: the band is contained in the bracket. Two dashed caveats downgrade what that agreement is worth: the test is low-severity (CQ4, a 4.7× soft bracket) and anchor-contingent (CQ3: at the device's initial gap fraction the self-consistent cutoff lies below the bracket, so the band is pinned to the Gifford plateau rather than iterated to the bracket device's own fixed point). The conclusion is corroboration, not a discriminating test. Border language follows the overview: single = warranted/corroborating, dashed = open/adjudicated caveat; dashed arrows carry the downgrade.

C7 (graph) — The Dinh held-out consistency band

Summary. Applying the fitted $\kappa_{app}(d/w_c)$ collapse curve and the skim law $\delta = w_c\sqrt{f_{gap}/3}$ to Dinh's tapered geometry yields a 1.25–2.17 μm cutoff band that lies inside the measured (1.0, 4.7) μm rigid-bead bracket. Because Dinh's *bracket device* (the $f^*_{gap} = 1.04 \times 10^{-4}$ design, the only one with rigid-bead data) operates *at or below the knee* at its initial gap fraction (held there, the self-consistent cutoff sits at the sub-steric foot, *beneath* the bracket; under the schedule Dinh describes but does not print, read with the Xia scaling Dinh 2024 cites, in its corrected form (C5), it rises into the bracket, so what is shown is the containment of the plateau-anchored band, not of the device's own cutoff), and the bracket is 4.7× wide with soft edges, this is a *low-severity, anchor-contingent consistency check, not a discriminating prediction.* The appropriate anchors for future finite-particle work are plateau devices, not Dinh.[1]

Claim. The $\kappa_{app}(d/w_c)$ collapse curve, transferred to Dinh's geometry through the skim law $\delta = w_c\sqrt{f_{gap}/3}$, places the bracket device's cutoff in the band [1.25, 2.17] μm, which is contained in the measured rigid-bead bracket (1.0, 4.7) μm.

Scheme. A deterministic computation for the band itself (no critical questions), combined with Argument from Evidence to a Hypothesis (verification form) for the held-out match (WRM 2008). The governing critical question of the verification half is severity: *could the band have landed outside the bracket?* (CQ4).

Premises.

P-C7.1 (computed band).
Fitting a Hill form $\kappa_{app}(u) = \kappa_{max}\, u^n/(u^n+u_{1/2}^n)$ ($\kappa_{max} = 5.568$, $u_{1/2} = 0.01651$, $n = 1.791$; $u = d/w_c$) to the κ_{app} collapse, and evaluating $\delta = w_c\sqrt{f_{gap}/3}$ at Dinh's geometry (w_c : 75 → 52 μm, initial $f^*_{gap} = 1.04\times10^{-4}$, gaps 19 μm; from "*All filtration gaps in the linear section were 19 μm-wide. The middle channel narrowed*

[1]Naming caution: this argument node **C7** is distinct from *Result 7* of the algebraic Appendix A (the cross-width scaling $f_{gap} \propto w_c^{-2}$, which backs node **C5**); Appendix A contains no empirical held-out node.

from 75 μm near the inlet to 52 μm near the outlet in all designs. Depth of all channels was 140 μm" [5, p. 917, Fig. 2 caption]), gives, holding f_{gap} at that value, $\delta = 0.442\,\mu$m ($w_c = 75$) and $0.306\,\mu$m ($w_c = 52$), and the band $[1.25, 2.17]\,\mu$m (regression-stamped, and reproduced to the printed precision by the reference code `cif_cutoff.py`).

The band's width is an ambiguity, not a gradient. A tapered device is designed to hold one cutoff, not a range of them; Xia ramps f_{gap} precisely so the lamina is taper-invariant (C5; [2, p. 9 (Sci Rep PDF pagination)], corrected in [3, p. 1]). Dinh states the same design rule for its own devices: once the initial values are chosen, "*the values of $f_{\text{gap}}(i)$ are iteratively adjusted with each subsequent gap i by incrementally increasing the fluidic resistance of the middle channel and/or decreasing the fluidic resistance of the side channels to maintain the desired size cut-off throughout the device*" [5, p. 916], citing Xia's 2016 paper for the method (its ref. 28). What the paper does not print is the schedule itself, only the initial $f^*_{\text{gap}} = 1.04 \times 10^{-4}$ for this device, one of three initial values its designs span [5, p. 917, Fig. 2 caption]. The band is computed holding f_{gap} at that initial value, which makes $\delta \propto w_c$ and spreads the prediction across the taper. Of the band's $1.74\times$ width, $75/52 = 1.44\times$ is that spread and only $1.21\times$ is the near-knee κ_{app} ambiguity, where the fitted prefactor has ramped off the plateau to a local $\approx$4.08. Under the Xia-scaled reading, which is the one Dinh describes, the lamina is the same at every gap, so the cutoff at the plateau prefactor is a single value and the band collapses to [1.80, 2.17] μm, the κ_{app} ambiguity alone; the iterated cutoff is not single-valued (end of the CQ3 backing). The wider band is retained as the conservative envelope, and it is the band the graph records: it contains the Xia-scaled reading, and taking the narrower one would assume that a schedule the paper describes but does not print meets its stated intent exactly. *The rule is stated; the schedule is not, and that is why the band is wide.*

*The channel depth ($h = 140\,\mu$m) is a property of the device but does **not** enter the band:* the skim law is depth-free and the κ_{app} curve is fit on the parabolic δ; depth enters only the interpretation (CQ2). `[code: cif_cutoff.py B5 (kcurve); skim.delta_leading; glossary A.7]`

P-C7.2 (Dinh 2024, [5, ESI, Fig. S1(A) and (B)]; bead data, verbatim).
For the same device ($f^*_{\text{gap}} = 1.04 \times 10^{-4}$, $30\,\mu$L/min): $1\,\mu$m beads give "*fluid volume recovery in filtrate was 52% and bead recovery in filtrate was 54%*" (tracer-like, *below* cutoff, which is how Dinh 2024 reads the same panel: for $a < w$ the split should "*approximate that of the fluid volume*", with Fig. S1A given as the example [5, p. 917]); $4.7\,\mu$m beads give "*fluid volume recovery in filtrate: 50%, bead recovery in filtrate: 1.8%*" at $106\,\mu\text{L}^{-1}$, and stay far below the fluid baseline across the whole sweep (1.8/3.5/6.8% bead against 50/52/48% fluid at $106/799/1714\,\mu\text{L}^{-1}$; *retained*, *above* cutoff). $\Rightarrow$ the measured cutoff lies in $(1.0, 4.7)\,\mu$m. `[Dinh 2024, ESI Fig. S1(A) and (B)]`

Critical questions.

CQ1: Is the calculation correct?
Yes. The executable reference reproduces the band to the printed precision: its regression registry asserts `Dinh band low/high` $= 1.25/2.17$ to the two decimals printed. $\rightarrow$ *answered: yes.*

CQ2: Are there interfering factors?
Two, of unequal magnitude. *(a) Per-gap over-extraction (dominant).* If the true per-gap fraction is $f_{\text{eff}} = c\,f_{\text{nom}}$, then $\delta \propto \sqrt{c}$ and the band scales by $\sqrt{c_{\text{Dinh}}/c_{\text{Gifford}}}$. *(b) Depth (secondary).* The fitted κ_{app} absorbs Gifford's finite-depth factor $R(\alpha_{\text{Gifford}})$; transferring it to Dinh's higher aspect ratio carries a mismatch $R(\alpha_{\text{Dinh}})/R(\alpha_{\text{Gifford}})$ of 1.09–1.21 across Dinh's two stations ($\alpha = 1.87$ and 2.69) and the calibration array's three widths ($\alpha = 0.53$, 0.64, 0.80), and (because the band uses the depth-free parabolic δ) this does not enter the band formula, only its interpretation. The interval over which the band remains inside $(1.0, 4.7)$ is $c_{\text{Dinh}}/c_{\text{Gifford}} \in [0.64, 4.69]$, where $0.64 = (1.0/1.25)^2$ and $4.69 = (4.7/2.17)^2$: *this is a tolerance (sensitivity) interval, not an independent plausibility range.* The honest robustness statement is therefore that the band exits the bracket only if the over-extraction ratio falls below $0.64\times$ or above $4.69\times$. Note what does *not* establish that it will not: bounding each device separately, below by $c \gtrsim 1$ and above by Gifford's stated per-gap bound ($f_{\text{gap}} < 5 \times 10^{-4}$, [1, p. 4498]), caps the Dinh factor only at $5 \times 10^{-4}/1.04 \times 10^{-4} = 4.8$, which is *outside* the window's 4.69 edge, and leaves the ratio unbounded below 0.64 for the same reason. What makes the ratio plausibly near unity is not those bounds but that the two devices' over-extraction has the same origin, the local gap-to-channel resistance ratio, at comparable gap widths ($19\,\mu$m against Gifford's $\sim 20\,\mu$m) and comparable channel widths. That is a *tolerance* argument resting on a similarity premise, not an independent confirmation. Finally, the concentration-dependent $4.7\,\mu$m leakage (rising $1.8 \rightarrow 6.8\%$ as

bead concentration rises 106 → 1714 μL^{-1}) is crowding (bead–bead interaction), not a cutoff shift. Dinh 2024 reads the same sweep the same way: "*As the particle concentration increases, the interactions between particles also increase and, therefore, the recovery of large particles in the filtrate should also increase*" [5, p. 917]. → *answered: bounded.*

CQ3: Is the band the self-consistent cutoff? (anchor dependence).
No. Because the calibration curve returns κ_{app} *as a function of the cutoff*, the operating law is *implicit*, $d_{\mathrm{crit}} = \kappa_{\mathrm{app}}(d_{\mathrm{crit}}/w_c)\,\delta$, and the cutoff is the fixed point of $u = \sqrt{f_{\mathrm{gap}}/3}\,\kappa_{\mathrm{app}}(u)$. Solved honestly at the bracket device's initial gap fraction, held constant as the band holds it, the only self-consistent value is the sub-steric foot ($d_{\mathrm{crit}} \approx 0.41$–$0.59\,\mu$m), *below* the bracket and inconsistent with the 1 μm bead tracking the fluid. On that reading the reported band lands in the bracket only because it is read off the Gifford plateau anchor: its lower edge takes the near-knee curve value $\kappa \approx 4.08$ at the outlet station and its upper edge the plateau anchor $\kappa = 4.91$ at the inlet, the curve value being the same at both stations because κ_{app} is evaluated at $\kappa_{\mathrm{plateau}}\sqrt{f_{\mathrm{gap}}/3}$, which does not depend on w_c. It is not iterated to that device's own fixed point. Along the linear taper, where Dinh's schedule raises the gap fraction, the self-consistent value rises into the bracket instead, when the rise is read with the Xia scaling Dinh 2024 cites, in its corrected form (C5); on either reading the band is the plateau-anchored value and not the iterated one, which is what this question records. The diagnostic, including why the threshold is specific to that gap fraction and what the schedule changes, is worked out in the subsection below (Fig. 9). → *disclosed; conclusion downgraded to anchor-contingent.*

CQ4: How severe is the test? (could it have failed?).
Barely. The bracket is 4.7× wide and the over-extraction tolerance is 7.3× wide, so almost any admissible parameterisation lands inside; the bracket edges are moreover *soft* (bead-population averages, not sharp cutoffs). The robustness asserted in CQ2 and the weakness asserted here are two readings of the *same* insensitivity. Passing this test is weak corroboration of the law. → *answered: low severity.*

Design choice (promoted from caveat). The clean anchors for the knee are *plateau* devices (Gifford's larger beads, and Xia's OR-7 with the RS-8 and RS-9 reference triples, all at the 100 μm reference width), where a stable cutoff fixed point exists on the plateau. The RS devices hold that width only as a reference: they scale the gap fraction as $(100\,\mu\mathrm{m}/w_c)^2$ (eqn (1) as corrected; C5) along a wider taper (RS-7's runs 300 → 150 μm [2, p. 10 (Sci Rep PDF pagination)]; RS-8's and RS-9's starting widths are not printed), so on RS-7's geometry only the narrower sections are plateau. Any Track-B finite-particle study should run on plateau geometries, not on Dinh. (C3b's CQ2 downgrades Xia's $\kappa_{\mathrm{app}} \approx 5.17$ to a lead, on grounds of circularity and non-diagnosticity. That does not bear here: what places these anchors on the plateau is their *published* $f_{\mathrm{gap}} = 5.8$–8.6×10^{-4} at $w_c(\mathrm{ref}) = 100\,\mu$m [2, pp. 6 and 10 (Sci Rep PDF pagination)], read straight off the paper, and not the prefactor computed from them.)

Residual risk. None *open*: the graph records `open_exceptions: []` for this node, and every critical question above is answered. The mapping is one-to-one: CQ2 and CQ3 are the node's two Walton exceptions (`doubly_sensitive_depth_and_feff` and `anchor_contingent_fixed_point_below_bracket`), CQ4 is its assumption (`sufficient_correlation`), and CQ1 belongs to the computation half (`c7_dinh_computation`), which carries no critical question at all. What remains is not risk but *weakness*, and it is the weakness the status word already carries: anchor-contingency (CQ3), double confounding with f_{eff} dominant and depth secondary (CQ2), and low severity against a 4.7× soft bracket (CQ4). Three answered questions whose answers bound how much this band can be asked to carry, which is why it is a band and not a test.

Status. Warranted only as a low-severity, anchor-contingent consistency band: a corroboration of the cutoff law on a held-out device, not a test that could have discriminated against it.

Why the band is anchor-contingent: the implicit cutoff (backing for CQ3)

The operating law fixes the cutoff as $d_{\mathrm{crit}} = \kappa_{\mathrm{app}}\,\delta$, and for the bracket device at its initial gap fraction $\delta = w_c\sqrt{f_{\mathrm{gap}}/3}$ is known (0.31–0.44 μm). The subtlety is that κ_{app} is not a free constant: the calibration curve was built by giving each Gifford device its own apparent prefactor $\kappa_{\mathrm{app}} = d_{\mathrm{crit}}/\delta$ and plotting it against *that device's own* normalised cutoff d_{crit}/w_c. The curve therefore returns κ_{app} *as a function of the cutoff*, and the operating law becomes *implicit*: the cutoff appears on both sides, $d_{\mathrm{crit}} = \kappa_{\mathrm{app}}(d_{\mathrm{crit}}/w_c)\,\delta$. Writing $u = d_{\mathrm{crit}}/w_c$

and dividing by w_c turns this into a one-line fixed-point equation whose root is the predicted cutoff,

$$u \;=\; \sqrt{\tfrac{f_{\mathrm{gap}}}{3}}\,\kappa_{\mathrm{app}}(u) \;=:\; g(u), \qquad \sqrt{\tfrac{f_{\mathrm{gap}}}{3}} = 0.00589.$$

On the plateau this has a clean stable root; the question is whether the bracket device does at its initial gap fraction, held constant along the taper as the band holds it.

It does not. Starting the natural way, the plateau value $\kappa_{\mathrm{app}} = 4.91$ predicts $d/w_c = 0.029$; but 4.91 is the mean over the plateau bin $d/w_c \geq 0.035$, and the fitted curve itself does not reach it until $d/w_c \approx 0.051$. At 0.029 the curve returns 4.08, which predicts a *smaller* cutoff ($d/w_c \approx 0.024$), where the curve is smaller still (≈ 3.7), so the estimate drifts down at every step ($0.029 \to 0.024 \to 0.022 \to \cdots$). The only self-consistent value is the sub-steric foot ($\kappa_{\mathrm{app}} \to 1.34$), $u^* \approx 0.0079$, i.e. $d_{\mathrm{crit}} \approx 0.41$–$0.59\,\mu$m (the bare Hill fit, having no foot, drives the iteration to 0 instead). Either way the self-consistent cutoff lands *below* the measured bracket and would predict the $1\,\mu$m bead retained, contradicting its observed fluid-tracking (54% ≈ 52%).

This is specific to the *bracket* device's initial gap fraction. A self-consistent cutoff *above the foot* exists only once the rising curve clears the knee, that is, only once g touches the diagonal: $\sqrt{f_{\mathrm{gap}}/3}\,\kappa_{\mathrm{app}}(u) = u$ together with $\sqrt{f_{\mathrm{gap}}/3}\,\kappa'_{\mathrm{app}}(u) = 1$. A tangency is a derivative condition, so it is solved on the full-precision fit. It puts the threshold at $f_{\mathrm{gap}} = 1.041 \times 10^{-4}$, with the tangency itself on the rising curve at $d/w_c \approx 0.0145$. Read as the plateau prediction at that gap fraction, $\kappa_{\mathrm{plateau}}\sqrt{f_{\mathrm{gap}}/3}$, the threshold is $d/w_c = 0.02894$, which numerically coincides with the $d/w_c \approx 0.03$ knee C3 and C3c already name, though it is a starting value for the iteration and the knee is a place on the curve. The two independent re-digitisations of Fig. 3 give 1.042 and 1.046×10^{-4}. Xia's OR-7 (5.8×10^{-4}, plateau prediction $d/w_c = 0.068$) sits far above it. **At its initial gap fraction the bracket device sits on it**: its plateau prediction is $d/w_c = 0.02893$ against a threshold of 0.02894, so it is below on all three digitisations, but by less than 0.6% on every one. That is the quantitative content of "the bracket device operates below the knee," and it is a knife edge rather than a margin. What does *not* turn on the margin is the conclusion CQ3 records, that the band is read off the plateau anchor rather than iterated to a self-consistent cutoff: that holds on either side of the knee. What does turn on it is the descent to the foot, which holds on all three digitisations and would not if the device sat a fraction above the threshold. What the knife edge does forbid is reading the threshold as a comfortable separation, and it puts a near-fixed point in the descent at $d/w_c \approx 0.0145$ ($d_{\mathrm{crit}} \approx 0.75$–$1.09\,\mu$m), where g comes within 4×10^{-6} of the diagonal. That near-fixed point is the tangency point itself, where a stable root would appear were the gap fraction a hair higher, so the three steps displayed above understate how slowly the estimate falls to the foot. (The Gifford *calibration* array deliberately runs down to $f_{\mathrm{gap}} = 6.4 \times 10^{-5}$, returning measured thresholds f^*_{gap} as low as $\sim 10^{-4}$ [1, p. 4500], which is exactly why the curve *has* a foot to fall onto; its plateau-anchoring enrichment devices sit at $f_{\mathrm{gap}} \geq 2.2 \times 10^{-4}$ [1, p. 4501, Fig. 5 caption].) At that gap fraction the reported band $[1.25, 2.17]\,\mu$m therefore lands inside the bracket only because it is read off at the plateau (the first step above), not iterated to the bracket device's own sub-bracket fixed point. That dependence on the anchor, not the containment itself, is what CQ3 records, and it is why the clean anchors are the plateau devices, not Dinh.

Dinh's other two designs are on the far side of the threshold. The same caption reports $f^*_{\mathrm{gap}} = 1.12$ and 1.28×10^{-4} beside the 1.04×10^{-4} of the bracket device [5, p. 917, Fig. 2 caption], and both clear the threshold: each acquires a stable fixed point, at $d_{\mathrm{crit}} = 1.02$–$1.47\,\mu$m and 1.26–$1.81\,\mu$m respectively, inside the measured bracket, and the iteration started at the plateau converges to it rather than drifting to the foot. Both roots still lie below the plateau edge 0.035, so neither design is a clean anchor either: a stable fixed point is not a plateau. The anchor-contingency recorded here is therefore a property of the one device the bead bracket was measured on, at its initial gap fraction, not of the paper. Nor can those two stand in for it: the rigid-bead runs were made on the 1.04×10^{-4} device alone [5, ESI, Fig. S1].

Held constant is the band's reading, not Dinh's schedule. Everything above holds f_{gap} at f^*_{gap} along the whole taper, as the band does. Dinh's schedule does not: the gap fraction is adjusted gap by gap "*to maintain the desired size cut-off throughout the device*" (P-C7.1), and in the linear section that is done by "*progressively increasing the width of the side channels and decreasing*" that of the middle channel [5, p. 916]. Under the Xia scaling that design sentence cites, in its corrected form (C5), $f_{\mathrm{gap}}(i) = f^*_{\mathrm{gap}}\,(75\,\mu\mathrm{m}/w_c)^2$, the gap fraction clears the threshold once the middle channel has narrowed to $w_c \approx 74.98\,\mu$m, and from there the iteration started at the plateau settles on a stable root at $d_{\mathrm{crit}} \approx 1.09$–$2.02\,\mu$m along the linear section, rising from the tangency point to the outlet and lying inside the bracket. The descent to the foot, and with it the contradiction with the $1\,\mu$m bead, is then confined to the stretch before the middle channel begins to narrow, where f_{gap} still equals f^*_{gap}; how much it weighs depends on that stretch's share of the extracted fluid, which the paper does not print.

What the schedule does not change is what CQ3 records: on either reading the band is the plateau-anchored value rather than the iterated one, and the paper prints the rule but not the $f_{\mathrm{gap}}(i)$ that would let the iterated value be computed from Dinh alone.

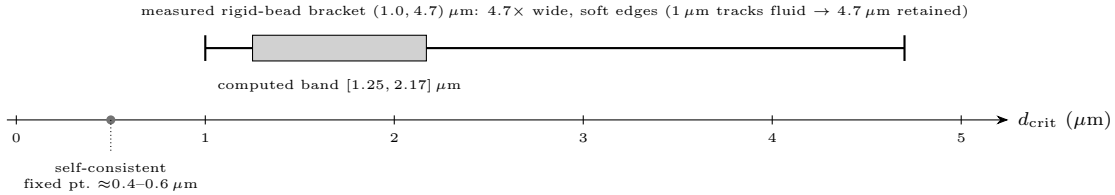


Figure 9: Interval diagnostic for C7. The computed band $[1.25, 2.17]\,\mu$m (shaded) is nested inside the measured rigid-bead bracket $(1.0, 4.7)\,\mu$m, so the held-out check passes, but the self-consistent fixed point at the device's initial gap fraction (≈ 0.4–$0.6\,\mu$m, the filled marker) sits *below* the bracket entirely. At that gap fraction the band agrees with the bracket only because it is anchored to the Gifford plateau, not iterated to the bracket device's own (sub-bracket) cutoff; and the bracket is 4.7× wide with soft ($1\,\mu$m-tracking vs $4.7\,\mu$m-retained) edges. All three facts (containment, the sub-bracket fixed point, and low severity) are visible at once.

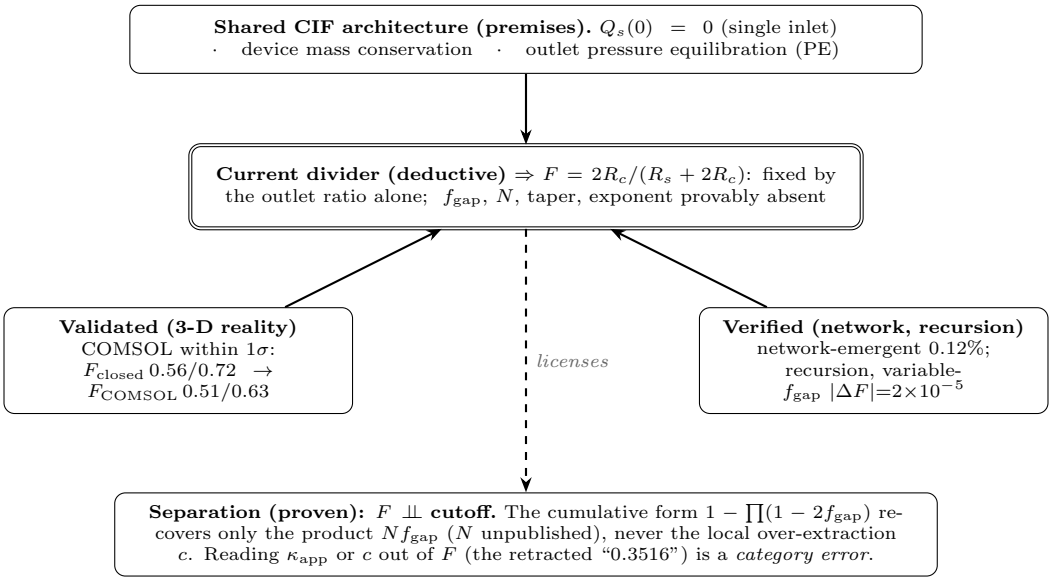


Figure 10: Argument graph for C4. The outlet identity is *deduced* (doubled border) from three shared-architecture premises, *validated* against 3-D reality by COMSOL, and *verified* twice: as the emergent behaviour of the full network, and against the iterated 1-D recursion, most stringently on the variable-f_{gap} schedule. Its payoff is the separation: because F sees only the outlet ratio (or, written as a cumulative product, only the product Nf_{gap}), it carries no information about the local per-gap over-extraction c that sets the cutoff. Border language follows the overview: doubled = deductive, single = warranted; the dashed arrow marks the licensed boundary (what the lemma permits and forbids).

C4 — Separation lemma: $F \perp\!\!\!\perp f_{gap}$ (the outlet identity carries no cutoff information)

Summary. The device-level filtrate fraction F, the share of inlet flow leaving through the two side outlets, is fixed by the outlet resistance ratio alone, $F = 2R_c/(R_s + 2R_c)$, with the per-gap schedule, the gap count N, the taper, and the cutoff-law scaling exponent all provably absent. This is *proven* symbolically, *validated* against 3-D COMSOL on two Dinh variants (the 1-D residual closing to within 1σ of the published values), and *recovered* twice: as the emergent behaviour of the full resistor network, and by the iterated 1-D recursion, including the strongest case, a variable-f_{gap} schedule where f_{gap} changes at every gap yet F is unmoved to within 2×10^{-5}. The same fraction can also be written as a cumulative product, $F = 1 - \prod_i (1 - 2f_{gap}(i)) \approx 1 - e^{-2Nf_{gap}}$, which *is* f_{gap}- and N-dependent; the two forms agree only because a given schedule fixes the outlet geometry. The lemma's content is therefore precise: at fixed outlet geometry, F is independent of *how* the device reached it, and F carries no information about the local per-gap fraction (or the over-extraction c) that sets the cutoff. Recovering κ_{app} or c from F is a category error: the inversion this project retracted.

Claim. At fixed outlet geometry (w_c, w_s), $F = 2R_c/(R_s + 2R_c)$ is fixed by the outlet resistance ratio alone. It is independent of the per-gap schedule $\{f_{gap}(i)\}$, the gap count N, the taper, the cutoff-law scaling exponent, and the inlet flow rate. That holds exactly under the three premises of P-C4.1; a finite tapered network realizes it to within a residual of the order of the gap fraction, which vanishes as the ladder is refined (P-C4.3). Up to that residual, F carries no information about the local per-gap fraction, or the over-extraction c, that sets the cutoff. Recovering κ_{app} (or c) from F is a category error.

Scheme. Argument from Established Rule (deductive): the two-resistor current-divider rule (hydraulic–electrical analogy [9, 10]) applied to the CIF outlet under its shared architectural premises (WRM 2008). The lineage applies the rule at the outlet itself: for devices with 450 μm final side channels and a 100 μm-wide, 125 μm-deep centre channel, Gifford puts the ratio of the two side channels' cumulative output flow to the centre channel's at theoretically $\sim$23 : 1, calculated from the duct-resistance series (his eqn (5)), "*provided the hydrostatic pressures in the output collection areas are kept equal*" [1, p. 4500]: a flow split fixed by the outlet resistances alone, at equal outlet pressures.

Premises.

P-C4.1 (the rule and its antecedents).
The current-divider form follows deductively from three premises shared across the CIF lineage: (1) the boundary condition $Q_s(0) = 0$ (sample enters the centre channel only, a single inlet); (2) device mass conservation $Q_c(0) = Q_c(N) + 2Q_s(N)$; (3) outlet pressure equilibration $Q_s(N)R_s = Q_c(N)R_c$. With $F \equiv 2Q_s(N)/[Q_c(0)+2Q_s(0)] = 2Q_s(N)/Q_c(0)$ (using premise 1), three algebraic steps give $F = 2R_c/(R_s+2R_c)$; equivalently, parallel-combining the two side branches ($R_s \parallel R_s = R_s/2$) reaches the identical result from the same premises by an independent algebraic route. The two are the same physics in two notations, the hydraulic–electrical analogy mapping one onto the other, so their agreement is a consistency check rather than independent evidence; its value is that the current-divider form makes F's dependence on outlet geometry alone immediately visible. The limits are physical ($F \to 1$ as $R_s \to 0$, $F \to 0$ as $R_s \to \infty$, $\partial F/\partial R_s < 0$), and f_{gap} is symbolically absent. `[Routt 2026 §II; code: network.F, network.verify_symbolic, which asserts f_gap not in network.F.free_symbols (separation lemma C4)]`

P-C4.2 (3-D physical validation, with both distances stated).
The closed form predicts $F_{\text{closed}} = 0.5592$ (V1)/0.7189 (V3), above the published $0.505 \pm 0.033/0.649 \pm 0.036$ [5, p. 919, Fig. 4 caption] by 10.7% (V1) and 10.8% (V3); 3-D outlet-bifurcation physics closes the residual ($F_{\text{COMSOL}} = 0.5070/0.6281$), and the *1-D-vs-3-D* residual it closes is 9.3% (V1) and 12.6% (V3), a different comparison from the one above, and the two should not be quoted for each other. The glossary requires both σ-scales wherever one is given, so both are here. On the raw-SD scale the 3-D solve is within 1σ ($+0.06/-0.58$); on the stricter SEM scale it is within 2σ ($+0.12/-1.16$). The *calibration-free* 1-D closed form sits $+1.64\sigma$ (V1) and $+1.94\sigma$ (V3) above the measurement: consistent within 2σ, but **not a match**, and the claim this node makes is that the yield law needs no device-specific fitting, not that the 1-D form reproduces the measurement. The residual is itself an outlet-geometry effect (the V1/V3 asymmetry $\alpha_{3D} = 0.811$ vs 0.660), so the correction introduces no f_{gap} coupling. The same caption bears on the Claim's flow-rate clause directly: Dinh run the V1 device at two inlet flow rates, and "*Volume recovery in filtrate was 50.5 ± 3.3% for 30 μL min^{-1} and 49.9 ± 1.8% for 90 μL min^{-1}*" [5, p. 919, Fig. 4 caption]. A threefold change in flow moves the yield by less than either standard deviation. `[Routt 2026 §IV-B; COMSOL Multiphysics 6.3; glossary A.6 validation block]`

P-C4.3 (network-emergent and schedule-independent).
The full Kirchhoff solve's emergent F equals the symbolic identity at the outlet to 0.12% across ramp shapes (bridge N3); the closed form matches the iterated recursion to $|\Delta F| \leq 2.35 \times 10^{-4}$ across four constant-f_{gap} devices and to $|\Delta F| = 1.97 \times 10^{-5}$ on the variable-f_{gap} Xia RS-7 case, the most stringent test, since f_{gap} enters every recursion step yet drops out of the closed form. This is the direct empirical content of "$F \perp\!\!\!\perp f_{\text{gap}}$": varying f_{gap} at every gap, at a fixed outlet, leaves F at the closed form to the figures above. The separation itself is symbolically exact, and the full network of bridge N3 realizes it to 0.12%; the recursion figures above are a different comparison, against a 1-D recursion that assumes equilibration at every gap [1, p. 4497]. The network's departure is the taper's own: an untapered ladder realizes the identity to rounding, while on the linear tapers probed, Dinh's own geometry among them, the network holds an outlet pressure imbalance that moves F by between about a quarter of and about four times the gap fraction, depending on the taper and on the thickness of the wall each gap crosses, which the papers do not state; it falls as $1/N$ when the same taper is spread over more rows, so it vanishes in the continuum limit. The decoupling test departs further: by at most 1.7×10^{-3} from its own run-outs' divider across its sweep, the leak of the run-out mismatch it carries, which falls as its rung-to-outlet conductance ratio rises (C8, P-C8.3). CQ2's 48× margin is on a different axis, the pressure-equilibration half-life against the device's length, and is not a bound on that departure: the test's ladder clears the half-life criterion and carries the departure anyway. The graph keeps the symbolic separation and the executable test apart, giving the test a conclusion of its own, `f_indep_fgap_executable`, distinct from the symbolic `f_indep_fgap` this node argues. `[Routt 2026 Table I, §8.4; code: _emergent_F_from_solve, verify_against_notebook N1--N3]`

Critical questions. The current-divider identity is deductive *given* its premises and carries no critical questions of its own; the questions below test those premises (pressure equilibration; $Q_s(0) = 0$) and the empirical accuracy of F's *magnitude*, never the separation $F \perp\!\!\!\perp f_{\text{gap}}$, which is symbolically exact.

CQ1: Exception (does the 1-D residual couple F to f_{gap}?).
No. That residual (9.3% at V1, 12.6% at V3; the realized-yield factor $\eta = 0.903$ is the representative single

magnitude, matching V1 closely rather than holding at both) concerns F's own magnitude accuracy, not the separation, and it closes via 3-D outlet-bifurcation fields that are themselves geometry-dependent (P-C4.2); the separation is symbolically exact. A *sibling* 3-D-junction effect, junction hydrodynamics of the same *kind* as the physics that sets F's residual, does perturb the per-gap f_{eff} the 1-D ladder estimates, contributing an *estimated* ∼5%-scale uncertainty on $\sqrt{c}$ (to be bounded by a 3-D junction solve), distinct from F's residual and not entering F. *Sibling* is a statement of kind, not of identity: whether this yield-side residual and the cutoff-side residual in fact share a common three-dimensional origin is an *open question*, deferred to future work in the glossary, and nothing in either appendix turns on the answer. The two are treated throughout as corrections on independent axes. The graph carries that estimate as the one open exception on this node, `sibling_3d_junction_uncertainty_on_sqrt_c`, attached to F's *magnitude* and not to the separation. → *separation holds.*

CQ2: Is the PE premise satisfied?

Yes; PE is tested, not assumed. It is the assumption the lineage's recursion rests on: Gifford treats "*the gaps between the flow channels of the array as nodes which allow equalization of pressure across the width of the device*" [1, p. 4497], and reports that "*This assumption has been validated by observations of fluid flow in this geometry*" [1, p. 4498]. The companion preprint's diagnostic fits the half-life $k_{1/2}$ of the cross-width pressure imbalance and requires $k_{1/2}/N_{\text{total}} < 0.05$, at least twenty half-lives within the device. Its five reference devices are Gifford 2014's bead device, Dinh 2024's three variants (V1, V2 and V3) and Xia RS-7: five devices over three published papers. In the preprint's computation all five clear the criterion with a safety margin of at least 48× (Xia RS-7 tightest). Those half-lives depend on the thickness of the wall each gap crosses, which the papers do not state: the preprint's code takes it equal to the gap's width, and its 3-D COMSOL models of V1 and V3 use the same wall, so the agreement of their outlet half-lives with the 1-D network's, to 2.3% (V1) and 4.7% (V3), tests the network and not the wall. With the thicker walls the design code uses, every half-life is longer and every device still clears the criterion. What the criterion measures is the decay of an imbalance along the ladder; it does not see one the outlet row itself holds, such as the decoupling test's or a taper's (P-C4.3). The code's `ARG_MATH.REFERENCE_DEVICES`, read by the Reynolds-number gates, is a different list of five (two Gifford widths, Lezzar, a Dinh row, and a row labelled Xia RS-7 that carries Dinh's outlet geometry); the preprint reports the criterion for its own five, not for that list. `[Routt 2026 §III-A7 and Table II (the §8.7 diagnostic); code: reference implementation §8.7, pe_consistency_check (gap_p = gap_size)]` → *premise holds for the imbalance the criterion measures, on all five devices.*

CQ3: Is $Q_s(0) = 0$ a genuine rule?

Yes; it is a device-level architectural boundary condition, a single centre-channel inlet with dual side outlets, shared by every device in the published lineage (Gifford, Xia, Lezzar, Dinh), not a modelling choice. `[Routt 2026 §II]` → *rule established.*

Why this lemma exists (the cautionary tale). The canonical equation set carries two expressions both labelled F: the outlet identity $F = 2R_c/(R_s + 2R_c)$ (f_{gap}-free) and the cumulative product $F = 1 - \prod_i (1 - 2f_{\text{gap}}(i)) \approx 1 - e^{-2Nf_{\text{gap}}}$ (f_{gap}- and N-dependent). They are the same quantity only at the self-consistent outlet but expose different variables, and conflating them is exactly the error this project retracted. The retracted "$f_{\text{eff}} \cdot N = 0.3516$ fixed by F" line read Nf_{gap} out of F via $\ln(1 - F) = -2Nf_{\text{gap}}$: that relation belongs to the *cumulative* form, recovers only the product Nf_{gap} (N unpublished), and never touches the local over-extraction c that sets the cutoff; moreover, 0.3516 is that product, a cumulative-yield number, not the outlet F at all. The inversion fails on three counts.

Where the figure comes from, and why it carries no macro. 0.3516 is historical: it is the value the retracted construction $f_{\text{eff}} = 0.3516/N$ asserted, recorded with that construction in the glossary's revision record. It is deliberately not given a value macro. The generated value block holds the project's *live* values, and a retracted figure that tracked them would rewrite the record of what was retracted; two of its four appearances are inside verbatim quotations of the retracted line, where substitution is disallowed in any case. The 0.505 the check that follows is fed is kept literal for the same reason. It is Dinh's measured V1 yield, and it is also the *input* the retracted construction was built on, since $-\ln(1 - 0.505)/2$ is where 0.3516 came from; the input to a retracted computation is as historical as its output. Substituting the value macro would print the same characters today, and would leave a frozen figure paired with a live antecedent that the next regeneration could move on its own. It is checkable where it stands, which is what a macro would have bought: $\ln(1 - F) = -2Nf_{\text{gap}}$ returns $Nf_{\text{gap}} = 0.3516$ at $F \approx 0.505$, one residual-width below the band the closed form predicts for the two validated

variants (P-C4.2), while reading the same number as $F/2$ in the small-f limit puts $F \approx 0.703$, near the top of that band. One number yielding two values of F nearly 0.20 apart is the arithmetic signature of the conflation this lemma exists to block. The lemma is invoked only to license the step this appendix states elsewhere as *narrow the per-gap over-extraction c by a local resistance-network calculation, not via F* (C3d CQ3; C7 CQ2).

Status. The separation $F \perp\!\!\!\perp f_{\text{gap}}$ is **proven** (symbolically exact; network-emergent; schedule-independent). F's absolute magnitude is independently **validated** by 3-D COMSOL, within 1σ on the raw-SD scale and within 2σ on the stricter SEM scale, while the calibration-free 1-D form sits $+1.64\sigma/+1.94\sigma$ from the measurement (P-C4.2). The sibling $\sim$5% uncertainty on $\sqrt{c}$ is an estimate, to be bounded by a 3-D junction solve; that it is a junction effect of the same *kind* as F's is not a claim that the two residuals share an origin, which is open (CQ1).

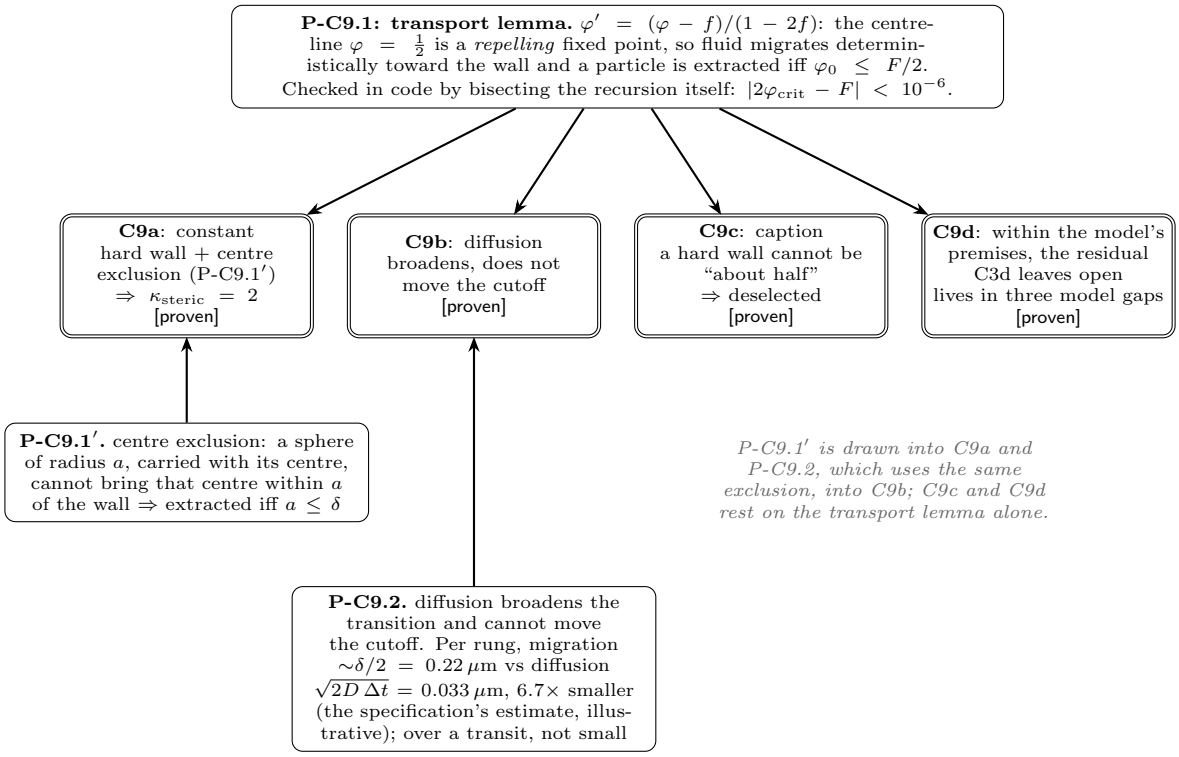


Figure 11: Argument graph for C9. One transport lemma, two auxiliary premises, four deductive consequences. Border style follows the overview map: every consequence carries the *doubled* border of a proven identity, and there is no dashed box anywhere: this is the only section of the appendix in which *every* conclusion is deductive. C9a is the result that matters for the chain, since it converts C1's asserted $\kappa_{\text{steric}} = 2$ into a derived one; C9b, C9c and C9d are dividends of the same solve. The two auxiliary premises are drawn *below* their conclusions, as the deductive strand is in Figure 2, because each feeds one consequence rather than all four. Note that the frozen argument graph records only the transport lemma as a premise of `c9_transport`: P-C9.1′ is the geometric step its CAES encoding folds into the conditional, and it is drawn here because a figure that hid it would make C9a look stronger than it is.

C9 — Transport lemma: why the steric factor is exactly 2

Summary. The steric prefactor $\kappa_{\text{steric}} = 2$ enters C1 as an *asserted* constant, carried by convergent testimony. It can be *derived*. Track a fluid element through a gap: its flux fraction measured from one wall obeys $\varphi' = (\varphi - f)/(1 - 2f)$, so the centreline $\varphi = \frac{1}{2}$ is a *repelling* fixed point, fluid migrates deterministically toward the wall, and a particle is extracted iff $\varphi_0 \leq F/2$. That repulsion is what makes the selection a hard *wall constraint* rather than a statistical position criterion, and with one line of rigid-sphere geometry (a centre cannot come closer to a wall than its own radius) it pins $\kappa_{\text{steric}} = 2$ exactly. The lemma is not a restatement of C1: it *upgrades* C1's testimonial premise to a derivation, and the same solve pays for itself three more times. Diffusion is shown to broaden the transition without moving the cutoff, the Fig. 1 "about half" caption is deselected on *physical* grounds independent of C1's textual adjudication, and, within the model's premises, the residual C3d leaves open is pinned to three model gaps. The node is deliberately *not* load-bearing: no support path runs from $\kappa_{\text{steric}} = 2$ to the thesis, because C9 strengthens a premise rather than adding a branch.

Notation (one collision to forestall). This section uses the curly φ for the *flux fraction* of a fluid element measured from one wall, the quantity the transport map advances, and reserves the straight-stroke ϕ for the *wall porosity* named in C9d's residual list below. C3d's own decomposition never mentions porosity, which is part of what C9d adds. The two are unrelated; the specification distinguishes them the same way.

Claim. The streamline-transport solve derives $\kappa_{\text{steric}} = 2$ from the flow model, taking from the literature only the geometric exclusion of P-C9.1′ and no measurement, no calibration and no reported result; and in doing so it settles three further questions that C1 and C3d leave open.

Scheme. Defeasible Modus Ponens for C9a (the transport lemma together with centre exclusion implies the constant), and Argument from an Established Rule for C9b–C9d (the same solved rule applied to three further questions). All four are *deductive*: every premise they rest on is a closed-form map or a geometric fact, never a measurement, so the scheme's exceptions slot is empty and the nodes are labelled *proven* with no open critical question. The per-rung length scales of P-C9.2 show how narrow the transition is per rung; C9b does not rest on them.

Premises.

P-C9.1 (the transport lemma).
A fluid element entering a gap at flux fraction φ, measured from one wall, leaves at $\varphi' = (\varphi - f)/(1 - 2f)$; equivalently $(\varphi' - \frac{1}{2}) = (\varphi - \frac{1}{2})/(1 - 2f)$. The centreline $\varphi = \frac{1}{2}$ is a fixed point of this map and it is *repelling*: $|\mathrm{d}\varphi'/\mathrm{d}\varphi| = 1/(1 - 2f) > 1$ for every $f > 0$. Fluid therefore migrates deterministically toward the wall, and solving the threshold gives extraction iff $\varphi_0 \leq \frac{1}{2}(1 - P(N)) = F/2$, with the two walls contributing $2 \times F/2 = F$. The code checks this by simulation rather than algebra: the design code's validator M18 finds the critical starting position by bisecting the recursion itself, gap by gap, and requires $|2\varphi_{\mathrm{crit}} - F| < 10^{-6}$ against F from the cumulative extraction. `[spec § 6.2; code: CIF_v2 critical_phi (validator M18); ARG_MATH transport_map, critical_phi]`

P-C9.1′ (centre exclusion: the step from fluid to particle).
The lemma above is about a *fluid element*. The step to a particle is one line of geometry, stated here rather than assumed because without it the factor 2 does not follow. The model carries a particle with its centre on the streamline that centre occupies, and a sphere of radius a cannot bring its centre closer than a to the wall, so it is extracted if and only if $a \leq \delta$, whence $d_{\mathrm{crit}} = 2\delta$ and $\kappa_{\mathrm{steric}} = 2$. Dinh state the same criterion for their devices: "*we would expect particles with radii exceeding the width of the extracted flow lamina ($a > w$) to remain in the middle channel*" [5, p. 917]. This is the same exclusion premise C1 takes from Yamada & Seki and from Inglis & Herman; the difference is that here it does no evidential work of its own. It is a property of rigid spheres, not a finding about CIF, and it is the *only* thing C9a needs beyond the solved map and the model's carrying of the centre, whose error is the finite-particle displacement C9d lists. `[spec § 6.2, consequence 2]`

P-C9.2 (diffusion broadens the transition and cannot move the cutoff).
Per rung, deterministic migration moves a particle by $\sim\delta/2 = 0.22\,\mu$m while cross-streamline diffusion moves it by $\sqrt{2D_{\mathrm{diff}}\Delta t} = 0.033\,\mu$m, a factor 6.7 smaller. These are the specification's estimates; it does not give their inputs, and nothing here recomputes them. Over a whole transit the spread is not small: the specification puts it at 1–4 δ for calibration beads, and first-order for sub-micron particles or long transits. What diffusion cannot do is move the cutoff. A sphere of radius $a > \delta$ cannot bring its centre inside the lamina however it wanders, so diffusion broadens the transition below $a = \delta$ and leaves the cutoff where exclusion puts it. `[spec § 6.2, consequence 1, and § 20 (diffusion)]`

Critical questions.

CQ1: Does the lemma merely restate C1?
No, and the distinction is the point. C1 establishes $\kappa_{\mathrm{steric}} = 2$ by *convergent testimony* across the Shevkoplyas lineage, Yamada & Seki, and Inglis & Herman, and the rule it leans on is itself stated, not derived: Inglis writes "*The critical particle diameter, D_c, for determining which path a particle will follow is then phenomenologically given by*" $D_c = 2b$ [12, p. 656]. C9 establishes the same constant by *solving the flow model*, taking from those sources only P-C9.1′, a geometric property of rigid spheres that no measurement and no laboratory is being trusted for. The label does not change (C1 was already warranted), but what the premise is worth does: a reader who declines every cited *finding* still has the constant, because what remains to be granted is that a sphere's centre cannot sit closer to a wall than its own radius. → *answered: it upgrades the warrant, not the label.*

CQ2: If C9 is the stronger result, why is the thesis not argued through it?
Because it strengthens a premise rather than adding a branch. On the graph's own support edges there is no path from $\kappa_{\mathrm{steric}} = 2$ to the thesis: the constant is consumed by exactly two claims, the mechanism (C1) and the residual decomposition (C3d), and neither carries through to the thesis either. This is a design decision, not an omission: making a granted premise sturdier changes how much confidence it deserves without changing a single label. → *answered: deliberately not load-bearing.*

CQ3: Does the deselection of the caption double-count C1's adjudication?

No: the two routes are independent, and each is on its own sufficient. C1 deselects the Fig. 1 caption on *textual* grounds (the caption and the main text are formally inconsistent; the main text is selected, backed by Inglis's $D_c = 2b$). C9 deselects it on *physical* grounds: a hard wall constraint cannot be "about half" of anything. Withdrawing either route alone leaves the deselection standing and moves no label; withdrawing *both* flips `caption_deselected` to *out* and takes $\kappa_{\text{steric}} = 2$ with it (four labels move: `caption_deselected`, $\kappa_{\text{steric}} = 2$ and its negation, and C3d's `residual_decomp`). That is the precise sense in which the redundancy is real rather than decorative. → *answered: independent and individually sufficient.*

Residual risk. None for the four conclusions: each is a deductive consequence of the recursion, which the code checks by simulating it, and of centre exclusion. P-C9.2's per-rung figures are the specification's estimates and illustrate the transition's width; C9b does not rest on them. The residual that C9d *pins* is C3d's, not C9's.

Consequence for C3d (worth reading across). Within the model's premises, C9d closes the list of places the residual C3d leaves open can live: the flat-lamina idealisation, the wall porosity ϕ, and finite-particle displacement: of the premises the lemma uses, these are the ones that can fail once its declared scope (the thesis's Leg 5) is granted, and a source outside the model, such as how a cutoff is read from data, is not on the list. C3d first divides out the factors it measures ($K_{\text{in-plane}} = 1$, and the depth factor $K_{\text{depth}} = R(\alpha)$, which corrects the flux fraction of a flat slab in a finite-depth duct and is not one of the three gaps), then names what remains by its effects: per-gap over-extraction $\sqrt{c}$ and cumulative finite-particle displacement C. C9d names the model gaps that can produce that product, and the wall porosity is named only here. C9d therefore answers the *exhaustiveness* question C3d leaves open, within those premises, while leaving C3d's *identifiability* question ($\sqrt{c}$ and C degenerate from the published data, being tied to the unreported gap count N) exactly where it was.

Status. Proven: four deductive consequences of one solved transport map, with no open critical question.

C8 (graph) — The design map: targets to geometry, and three dormant defeaters

Summary. Because cutoff and yield are governed by separate laws (C4), and the plateau schedule holds the cutoff at every row (P-C8.1), the map from performance targets to device geometry has a *lower-triangular* Jacobian: changing the gap-row count N leaves the cutoff untouched ($\partial d_{\mathrm{crit}}/\partial N = 0$) while moving the yield, through the terminal width, and so the outlet ratio, that N itself sets. The zero comes from the invariant-lamina schedule on the plateau (P-C8.1): S1 holds the lamina at every row and the plateau prefactor is constant, so the cutoff $\kappa_{\mathrm{app}}\,\delta^*$ does not depend on N. C4 supplies what makes the two targets answer to separate laws: the yield law carries no per-gap parameter. Neither supplies the diagonal; the diagonal is non-zero because every row skims a positive fraction, so the retained flow $\prod_i(1-2f_{\mathrm{gap}}(i))$ falls strictly with N. The inverse map therefore exists: *locally*, and throughout the declared envelope. A triangular Jacobian gives local invertibility and nothing more; this appendix does **not** claim a global one-to-one map over an unrestricted domain, and the distinction is stated here rather than left for a reader to supply. That conclusion is then reached a second time, by construction rather than by argument: the design code's forward march, which never inverts the design map, computes the width schedule for five cutoff targets and finds $N \cdot D^2$ invariant to 0.29% while N itself moves by a factor of 4.5. The row count is simply the price of the target *pair*. Forward substitution then supplies the practical inverse for a device built to the resulting schedule. Realizability, however, is *conditional*, and the argument says so before a reader has to ask. Three defeaters would break the realization if their antecedents held: the device running below the knee, a run-out mismatch, and a row count pinned by a length budget. All three are *withheld*: their antecedents are not granted, so they lie dormant and the design stands. But withheld means "not established for this scoped design," not "shown never to occur," and published designs are worked counterexamples to unconditional application: at their wide ends the plateau law puts Dinh's three designs at $u = d/w_c = 0.029$–0.032 (the bracket device at 0.029), and Xia's RS-7, on the taper Xia prints, lower still at 0.023 (C5), below the plateau floor u_{lo} (set here to the plateau-bin edge 0.035), and, on the cubic pair the appendix argues on, the implementation's guard refuses them. This is the leg on which the headline claim inherits its conditionality.

Claim. The target-to-geometry map is *locally* invertible throughout the declared envelope, and forward substitution realizes both the cutoff and the yield target on a device built to the resulting schedule inside that envelope.

Scheme. Two schemes reach C8a, which is why it is drawn with two incoming routes. Defeasible Modus Ponens for the deductive one: a lower-triangular Jacobian with non-zero diagonal implies a local inverse, which is an entailment, not an inference to the best explanation. Evidence to a Hypothesis for the executable one (P-C8.2): the design code's forward march is run, it computes the width schedule for every target without inverting anything, and the hypothesis is confirmed in the defeasible sense. That scheme's own critical question, whether the invariance is an artifact of the march's model (its F is the cumulative product $1-\prod_i(1-2f_{\mathrm{gap}}(i))$, with no outlet to equilibrate), is discharged by measurement in P-C8.3, on a network that has one, not assumed away. That network is a uniform ladder and not the marched taper: the one attempt to put the taper through the network inverse did not converge, and was reported rather than tuned. C8b then takes Argument from an Established Rule: the forward-substitution rule is applied to a case, and the scheme's exceptions slot is where the three defeaters live. The rule is the lineage's own design practice, which is what makes it established rather than proposed here. In Gifford the width of each side channel at gap i is "*ultimately calculated merely by the desired value of f_{gap}, and their width at the previous gap*" [1, p. 4497]: a forward recursion, not an inversion of the design map, whose only solve at each row recovers a width from the duct-resistance series [1, pp. 4498–4499], as the march's does (P-C8.2). Xia state the order that C8a's triangular Jacobian licenses, the cutoff's parameter first and the yield after it: "*Once the desired value of f_{gap} (and dimensions of the centre channel, posts, and gaps) are selected, a CIF-based device can be rapidly patterned numerically, and structured into a paperclip-like format (Fig. 1a) to achieve the preferred amount of particle concentration/filtration in a compact footprint*". Along the device the side channels widen to a final width that "*is determined by the degree of particle concentration (or, equivalently, the amount of filtration) preferred for a given application*" [2, p. 2 (Sci Rep PDF pagination)].

Kernel declaration (required before any of this means anything). The glossary's pairing rule is directive: κ_{app} is meaningless without the kernel it was fitted with. Two kernels are in use: the leading-order skim branch g : $\delta = w_c\sqrt{f_{\mathrm{gap}}/3}$ with prefactor $\kappa_g = 4.9138$, and the lamina-flux kernel F_{lam}, exact for a flat

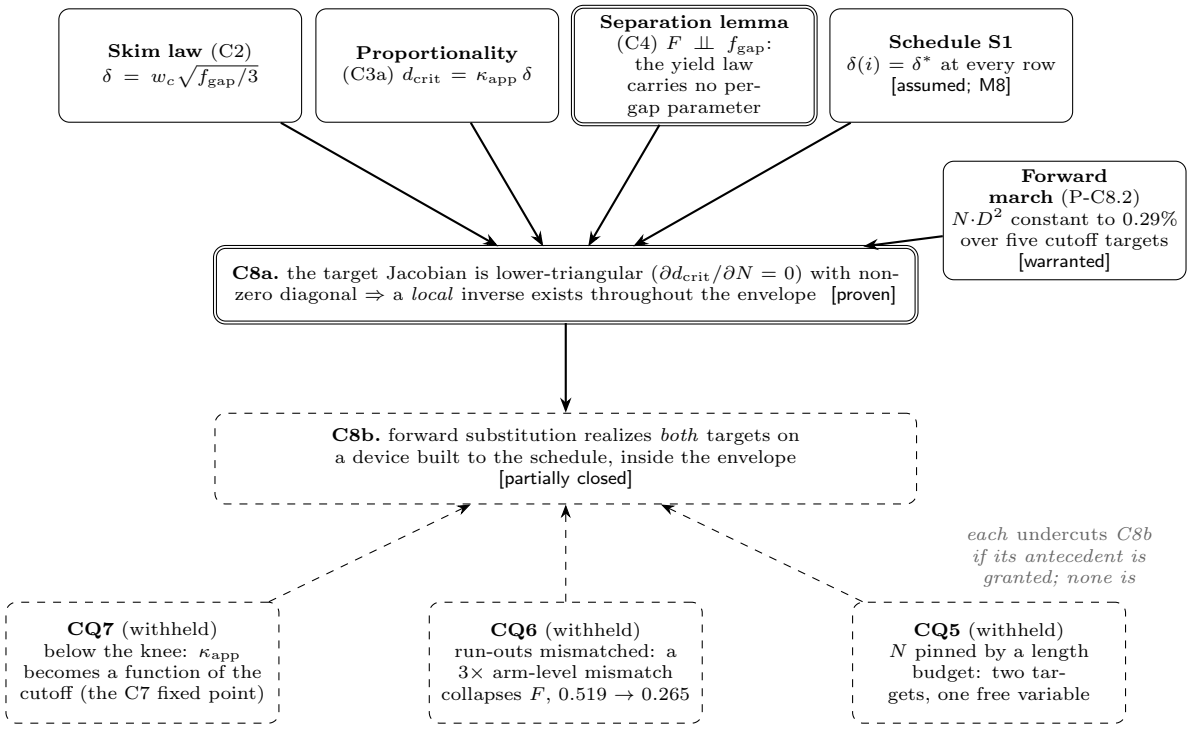


Figure 12: Argument graph for C8. The inversion (C8a, doubled border: proven) is reached twice over. The deductive route is the top row: the skim law, proportionality, the separation lemma and the S1 schedule together make the target Jacobian lower-triangular. The second route, entering from the right, is an executable forward march that never inverts the design map: run at five cutoff targets, it computes the width schedule that realizes each, which reaches the same conclusion by construction, as Evidence to a Hypothesis rather than as an entailment. The two routes share the physics and not the inference, so an error in the triangularity reasoning would leave the march untouched. Only then does forward substitution realize both targets on a device built to the resulting schedule (C8b). The three dashed boxes are the envelope boundaries; their dashed arrows are *undercutters*, not corroboration: each would withdraw C8b's warrant if its antecedent were granted, and none is granted here. C8b itself is dashed because its status is *partially closed*, exactly as C3d is dashed in Figure 1. Note that the undercutters attach to C8b and not to C8a: C8a is claimed only inside the envelope, so what the boundaries qualify is the realization, never the inversion.

lamina in an unperturbed duct (which the skimmed fluid is not), whose prefactor $\kappa_F = 5.798$ is not exact even in that sense: it is transferred from κ_g at an assumed 80 μm calibration depth. A pair must be declared and used at all three call sites: the inversion, the schedule, and the scale $d_{\text{crit}} = \kappa_{\mathcal{K}}\,\delta$. **This appendix argues on the cubic branch with** κ_g, as the glossary directs while the calibration depth remains unmeasured: every prefactor fitted and every cutoff computed in C1–C7 is a cubic-branch number, and C8 uses the other pair only where it says so, in the design code's march (P-C8.2) and the kernel comparison below.

That declaration matters here more than anywhere else in the appendix, because the design code carries both pairs: its `design` function defaults to the cubic pair, while its physics layer, the one C4's design-map test marches (P-C8.2), runs on F_{lam}. The two are not interchangeable and the mismatch is *not* detected automatically: at Dinh's geometry and the bracket device's gap fraction, comparing the exact inversions of the two kernels as the glossary does, the pair choice alone moves the cutoff by +14.1% at the inlet row, rising to +15.6% at the outlet, and crossing a prefactor onto the other kernel's lamina shifts it a further +18.0% or −15.3%. Switching costs +26.9% on the row count at the assumed 80 μm calibration depth, falling to +3.9% at 150 μm: figures the glossary flags as *ungated*, coming from a calibration-depth sweep the derivation record does not reproduce, so they are carried here as an order of magnitude rather than as verified constants. So the claim C8b makes is that forward substitution realizes both targets *with one declared pair used throughout*; a design that inverted on one kernel and scaled on the other would be a silent modelling error of that size, and it is the specification's pairing rule, not this argument, that prevents it. The glossary records the transfer bias as an uncertainty band

rather than applying a correction whose calibration cannot be verified, and this appendix does the same.

Premises.

P-C8.1 (the invariant-lamina schedule).
The canonical schedule S1 holds the diverted lamina constant along the array, $\delta(i) = \delta^*$ at every row. This project's own design specification states it as "*S1 — invariant lamina (canonical). Exact at every aspect ratio*" (§ 7); the quotation is from an internal document, not the literature, and is quoted only so that the schedule the argument relies on is fixed in writing. It is this invariance, on the plateau where the prefactor is a constant, that makes the cutoff a property of the *schedule* rather than of the row index, and so makes $\partial d_{\mathrm{crit}}/\partial N = 0$ exact rather than approximate. Below the plateau it does not: κ_{app} then depends on d/w_c, so a lamina held constant no longer holds the cutoff. C7's computation for Dinh's *bracket device* is the worked case: under the schedule Dinh describes, read in the corrected Xia form, the leading-order lamina stays at $0.442\,\mu$m while the self-consistent cutoff rises from 1.09 to $2.02\,\mu$m along the linear section. The plateau guard (CQ7) keeps a design in the plateau bin, where the prefactor is taken as constant; C7's smooth fit of $\kappa_{\mathrm{app}}(d/w_c)$ reaches the plateau value only at $d/w_c \approx 0.051$, so just above the floor the premise is exact on the bin's model rather than on that fit. The schedule is an assumption of the design, and the validator that tests it exactly is M8. `[spec § 7 (S1); validator M8]`

P-C8.2 (the forward march: a second route to C8a).
This premise does not invert the design map, and that is the point: the march *computes* $f(i)$ from δ^* rather than solving for it, so it reaches the same conclusion as the Jacobian argument by a different *inference*. The two are not physically independent: the march is the design code's physics layer, on the pair its own lineage spec declares (the lamina-flux kernel F_{lam} with $\kappa_F = 5.798$, the Kernel declaration above), which is the lamina physics of the skim law at finite depth. But they fail differently, which is what makes the redundancy worth having: an error in the triangularity reasoning would not move the march, and a convergence failure in the network inverse cannot reach a computation that never calls it (the march's only solve, for the channel width at each row, is a bisection on a monotone function). Sweeping the cutoff across five targets while holding the yield target fixed leaves $N{\cdot}D^2$ constant to 0.29% ([14764, 14776, 14781, 14791, 14807], tolerance 2%) while N itself spans 1446–6562, a factor of 4.5, as the specification's closed form for the canonical taper requires ($N \propto D^{-2}$, § 13); the converse sweep meets every yield target while the row count moves. Most of this holds by construction and is not evidence: the march fixes δ^*, and so the cutoff, at every row (S1), so $\partial d_{\mathrm{crit}}/\partial N = 0$ is its input rather than its finding; it stops at the first row that meets the yield target, so meeting the target is its stopping rule, and the converse sweep's $< 10^{-3}$ with it; and its taper step is the closed form's own, so $N \propto D^{-2}$ follows to leading order. What it does test is that the computation is feasible across the targets, with the taper not exhausted before the yield target is met, and that on the exact kernel the row count stays 0.26–0.55% above the closed form, which is written for the leading-order schedule (S1′). That is the sense in which it reaches C8a a second time: the inverse is computed rather than argued, and the computation agrees with the specification. The row count is therefore the price of the target *pair*, and the two targets are independently specifiable. The row counts are F_{lam}-pair numbers, not cubic-branch ones: the pairing rule applied, one declared pair used throughout. The march also holds its prefactor constant and applies no plateau guard, and it runs on the $75\,\mu$m lineage inlet, where only its largest target, $3.2\,\mu$m, reaches the plateau floor $u = d/w_c \geq 0.035$; the other four, the converse sweep's fixed cutoff among them, are targets the plateau guard would refuse. What it corroborates is therefore the triangular structure of the constant-prefactor law, which is what C8a asserts inside the envelope, and not a realization below the knee. `[code: tests/test_C4_design_map.py]`

P-C8.3 (decoupling, executable, and the discharge of P-C8.2's critical question).
The separation lemma is not assumed here but measured: the filtrate fraction F stays at the value the outlet resistance ratio fixes while the *measured* f_{gap} spans 3.63×10^{-5} to 1.93×10^{-3}, a range of 1.73 decades. Across that sweep the worst residual is 1.7×10^{-3}, at the weakest outlet equilibration reached (PE = 59×), inside the $O(1/\mathrm{PE})$ bound the test sets there, 2.6×10^{-3}, and it falls to 2.3×10^{-5} at PE = 5093×. The sweep moves PE and f_{gap} with one knob, the rung width, so that fall alone cannot say which of the two the residual follows. Holding the rung, and so f_{gap}, fixed and scaling only the outlet resistances, from the test's own up to sixteenfold, settles it: the residual times PE stays constant to under 1% at every rung of the sweep, while across the sweep, where f_{gap} moves 1.73 decades, it moves only from 0.10 to 0.12. The residual is a finite-PE effect, not a coupling, and a graded negative control, coupling the outlet to f_{gap} at the 2% level, is caught. It exists because the test carries a run-out mismatch: its run-outs impose $F = 0.5$ on a ladder

that would, left to itself, send 2.9% of the flow down its rails, so the outlet row holds a pressure imbalance, and $F(1-F)$ times that imbalance is the residual to first order; with run-outs matched to the ladder's own split it falls to the solver's floor. The PE here is the test's own axis, a rung-to-outlet conductance ratio; the 48× margin of C4's CQ2 is the preprint's, on a different axis (the pressure-equilibration half-life against the device's length), so the two numbers are not comparable, and the half-life does not bound this residual. The ladder's half-life depends on its channel and rung conductances alone: $k_{1/2} = 0.366$ rows at the weakest rung, 109 in the test's 40 rows, which clears the preprint's criterion. Yet the outlet row's imbalance does not fall as rows are added, although the preprint's exponential model would have it fall: that model covers an imbalance carried in from upstream, and this one the outlet holds.

One caveat travels with this premise and the glossary states it explicitly: that 1.73-decade span is a *deliberately widened numerical sweep* built to demonstrate decoupling over a decade, not a range any built device occupies. Gifford's calibration array caps at $f_{\mathrm{gap}} = 5.76 \times 10^{-4}$ and no published lineage device exceeds 1.0×10^{-3}. At the sweep's top, $f_{\mathrm{gap}} = 1.93 \times 10^{-3}$, the flux-rescaled gap Reynolds number would be ≈ 0.010, at the $\mathrm{Re}_{\mathrm{stokes}}$ gate, which is where the creeping-flow premise stops having room. The decoupling result is therefore comfortable across every band a device occupies and marginal only at the top of the test sweep, which is the honest way to report it. `[code: tests/test_C4_decoupling_v2.py; glossary A.9]`

Critical questions. Three, all *open by design*, and numbered CQ5–CQ7 rather than CQ1–CQ3 because these are the numbers the argument graph gives them (`c8_def_pinned`, `c8_def_runout`, `c8_def_knee`), and renumbering them here would break the correspondence this appendix exists to maintain. Their ground atoms sit in the graph's withheld block, which records them as possible without granting them: each computes *out* until its antecedent is asserted, so each defeater lies dormant. Closing them by fiat would assert that a CIF device is always above the knee, always run-out-matched and never length-budgeted, and the first of those is false of a published device.

CQ5: Is the row count free? (`pinned_row_count`)

Not always. The inversion consumes N as the free variable that meets the yield target. If a device-length budget pins N externally, two targets meet one free variable and the system is over-constrained. Nothing in the physics prevents this; it is a packaging constraint, and the lineage has met it. In Xia's words the recursion "*enables patterning devices of any total length that is needed*", and the next sentence reads "*In practice, however, one must partition a lengthy CIF device into 'legs' in order to fit its overall footprint onto a standard substrate mould*" [2, p. 2 (Sci Rep PDF pagination)]. It is the cheapest of the three to check before a design is committed: the specification checks it as G9, each leg's length and the array's total against the leg budget. The design code does not enforce it. Its G9 compares the array's total with the budget and reports the result without failing, and only for the validator battery's own short test device; `design` itself refuses nothing on length, and a design's report prints the length beside the budget. → *open, withheld: an envelope boundary, not a defect.*

CQ6: Are the run-outs matched? (`runout_mismatch`)

They must be built to match the schedule $\rho(N)$, and the tolerance is tight: a 3× mismatch collapses the filtrate fraction from 0.519 to 0.265, a factor of two in delivered yield (reading the mismatch at arm level, as the specification defines it; trebling only one of the two side run-outs gives 0.418). The specification calls this the most sensitive input its previous version left un-gated, and now gates it at design time: G13 is a pre-flight assertion, before the march, that the run-out resistances match the outlet ratio. The design code has no such pre-flight check, since its G13 runs on the validator battery's own short test device. It meets the condition by construction: it builds both run-outs at the last row's widths over one length, so their ratio is that row's resistance ratio, and the march's closure check refuses any row whose ratio misses its target, the outlet ratio at the last row. What no gate sees is the device as built. → *open, withheld: gated by the specification at design time (G13) and met by construction in the design code; a fabrication tolerance after it.*

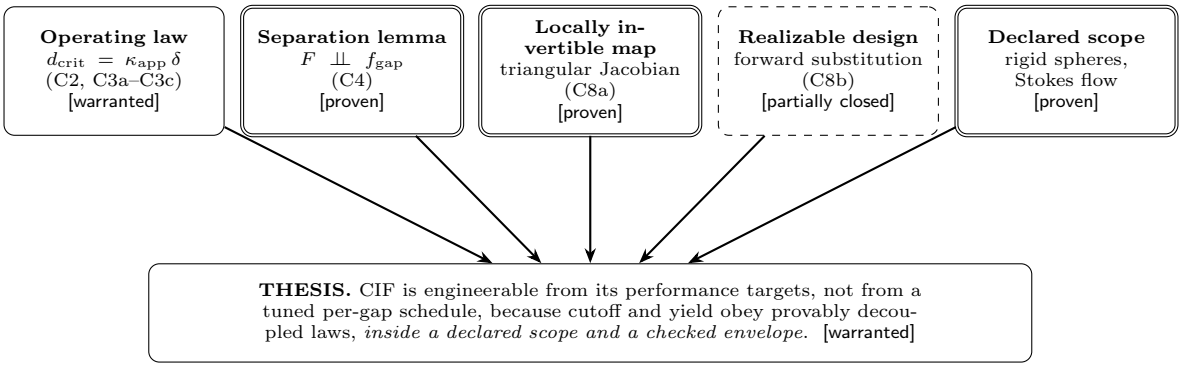
CQ7: Is every row on the plateau? (`below_knee`)

This is the one with a worked counterexample, and C7 has already done the work. If a row runs below the knee, κ_{app} becomes a function of the cutoff and the operating law turns *implicit*, $d_{\mathrm{crit}} = \kappa_{\mathrm{app}}(d_{\mathrm{crit}}/w_c)\,\delta$; the first triangular row is then the fixed point of Figure 9 rather than a direct solve. The published lineage taper does exactly this at its wide end: Dinh's bracket device, a 75 μm inlet read at $d_{\mathrm{crit}} = 2.17\,\mu$m, gives $u = d/w_c = 0.029$ against the plateau floor u_{lo}, set here to the plateau-bin edge 0.035, and the plateau

guard refuses it. Dinh's other two designs give 0.030 and 0.032 on the same reading; those fall in the gap in Gifford's calibration data between $u = 0.0294$ and 0.0353, where the bin edge could sit anywhere, but their own stable roots, which C7 reports as cutoffs of 1.02–1.47 μm and 1.26–1.81 μm across the taper, are $u \approx 0.020$ and 0.024, below it wherever it sits. That 2.17 μm is the *upper* edge of C7's band, and the device's own self-consistent cutoff at that end is lower still: at its initial gap fraction only the sub-steric foot is self-consistent (0.59 μm at 75 μm), and under the schedule Dinh describes, read in the corrected Xia form, the first root above the foot, at 1.09 μm, appears only once the channel has narrowed to 74.98 μm (C7). The margin below the floor is therefore if anything understated here, whether the gap fraction is held at its initial value or follows the schedule. C7 is thus not merely a corroborating band but a *counterexample to unconditional application* of C8. Nor is the 75 μm lineage alone in this. On the taper Xia prints, RS-7 runs $d/w_c = 0.023$ at its 300 μm wide end (C5 CQ4), further below the floor than any of Dinh's three designs, and the guard refuses it on the same test. One caveat rides along, the one the Section 2 row states: the *need* for the guard is derived, from C7, but its *value* is not. u_{lo} is the plateau-bin edge, an empirical boundary nobody derived; what it is *for* is checked (above it the calibration points collapse onto one κ_{app} about 22 times more tightly than below it), but the number itself is an external input, so the guard is only as sharp as that number. → *open, withheld: bounded by a code-enforced guard, and shown on published designs.*

Residual risk. The three envelope boundaries above, none excluded and each named: `below_knee`, `runout_mismatch`, `pinned_row_count`. The *inversion* carries no residual inside the envelope it is stated for: C8a is proven, and it survives every rival in the graph including the one still open, the size-dependent f_{eff} of C3c.

Status. Partially closed. The map is proven *locally* invertible; its realization is warranted *inside the declared envelope*, with three named boundaries dormant rather than excluded. The headline claim inherits this conditionality, and states it.



An accrual adds no inference beyond the conjunction, so the thesis carries no residual of its own; what it inherits from Leg 4 it inherits as a condition*, which is why the node is warranted and its statement is conditional. Border style follows the overview map: doubled = proven, single = warranted, dashed = partially closed.*

Figure 13: The thesis and its five legs. Every arrow is a premise of a single accrual node: remove one leg and the thesis loses a premise, which is what distinguishes these five from the corroborating branches (C5, C7, and C6's Davis cross-check) that have no support path to the thesis at all; nor, as C9's CQ2 records, does $\kappa_{\mathrm{steric}} = 2$. The fifth leg is the scope declaration, drawn beside the other four rather than beneath them because it is a premise in the same sense: an argument that did not declare its regime would not be a weaker argument for this thesis, it would be an argument for a different and false one. Note that the thesis node is drawn with the single border of a *warranted* claim even though Leg 4 is dashed: an accrual inherits its premises' conditions as the conditions of its own statement, and this statement says so in its own words.

The thesis, and the fence around it

Summary. Everything above is argued in service of one conditional sentence: *within a declared physical regime (rigid spheres in Stokes flow), and provided a design stays inside its admissible operating envelope, a CIF device can be designed directly from its two performance targets, a size cutoff and a yield, because the laws governing those two targets are* provably decoupled. One word in that sentence has to be pinned before anything else: throughout this appendix *yield* means the filtrate fraction F, the fraction of *fluid* sent to the filtrate channels, and never particle recovery. The two are different quantities and the separation lemma is about the first of them. The thesis is an *accrual*: it rests on five premises at once, each established in its own section. The conditions are not fine print. They are load-bearing (the realizability leg is formally partially closed on exactly these envelope conditions), and the scope is carried as a claim in the argument rather than as a sentence only in the abstract, because an examiner reads an unstated exclusion as an oversight and a stated one as a decision.

Claim. CIF can be engineered from its performance targets instead of from a tuned per-gap schedule: its yield follows a closed-form law with no fitted constant and its selectivity a derived law with a single calibrated prefactor, and the two are provably decoupled, so the size threshold and the filtrate fraction are independent design *inputs* rather than coupled outcomes. All of it holds inside a validity envelope that the design tool checks: it refuses a design that leaves the plateau or the Reynolds gates, and it reports, without refusing, a design over its length budget (C8, CQ5).

Scheme. Accrual. This is a local tag rather than a WRM 2008 scheme: the node's premises are other *nodes*, not atoms, and it carries no inference of its own beyond the conjunction. Its weight is experimental, the weakest evidential tier among its legs, and it is evaluated under preponderance of the evidence.

Premises. Five, each a claim argued in its own section.

Leg 1: the operating law (C2, C3a–C3c).
$d_{\mathrm{crit}} = \kappa_{\mathrm{app}}(d/w_c,\ \mathrm{depth})\, w_c\sqrt{f_{\mathrm{gap}}/3}$, with κ_{app} calibrated rather than universal. Itself an accrual of exactly four premises: the skim law (C2), proportionality (C3a), the plateau value (C3b), and the *existence* of the knee (C3c). C1 and C9 are deliberately *not* listed: they establish and then derive $\kappa_{\mathrm{steric}} = 2$, which underpins the mechanism and C3d's decomposition but appears in none of those four premises and reaches this node by no path. C3d is likewise absent: it interprets the prefactor's magnitude rather than supplying it.

Leg 2: the separation lemma (C4).
$F = 2R_c/(R_s + 2R_c)$ depends only on outlet geometry, so yield carries no cutoff information and the two targets are genuinely two knobs.

Leg 3: the locally invertible design map (C8a).
The target Jacobian is lower-triangular with non-zero diagonal, so the inverse exists locally throughout the envelope.

Leg 4: realizability (C8b).
Forward substitution realizes both targets on a device built to the resulting schedule, inside the envelope, with three named boundaries dormant. This appendix reports no device fabricated to such a schedule: the leg rests on the inversion and three granted assumptions (C8).

Leg 5: declared scope.
Rigid spheres in Stokes flow at nearly constant viscosity: the regime in which the skim cubic, the current-divider identity and the steric criterion are all derived. Deformability, cell crowding and whole blood are *declared* out of scope, not overlooked.

Scope, stated as a claim rather than a disclaimer. Three limitations are carried inside the argument, each sourced.

Platelets are discoid, not spherical.
The motivating application is also the limitation. Dinh describe platelets as "1 μm-*thick oblate spheroids (discoids) ranging from 2 to* 7 μm *in diameter*" [5, p. 913] and name "*the characteristic tumbling motion oblate spheroids undergo when flowing near the channel walls*" as a separation challenge in its own right [5, p. 921]. Gifford concede the same ("*approximately discoid*") and, tellingly, determined the platelet f_{gap} by *observation* rather than from the cutoff formula [1, p. 4502]. The diameter entering the cutoff law is therefore *orientation-dependent*, the rigid-sphere idealisation is an approximation the founding paper already worked around empirically, and platelet-specific validity is **not** established by this argument. That is precisely why the thesis is scoped to rigid spheres.

The Stokes regime is gated in code, not merely assumed, and the channel gate can bind.
The numbers, not just the thresholds. Taking the whole flow through the narrow outlet section, a worst case that bounds what the code computes from the local flow, the *channel-scale* Reynolds number of a water-like fluid is ≈5.2 at 30 μL min^{-1}, rising to ≈16 at 90 μL min^{-1}; platelet-rich plasma, more viscous than water, runs lower in proportion. The Reynolds number *at the gap* is obtained by rescaling the *flux*, not the length: a gap carries only the diverted fraction f_{gap} of the channel flow, so $\mathrm{Re}_{\mathrm{gap}} \approx f_{\mathrm{gap}}\,\mathrm{Re}_{\mathrm{channel}} \approx 10^{-3} \times 5.2 \approx 5.2 \times 10^{-3} \ll 1$, evaluated at the *top* of the published lineage range rather than at a comfortable margin below it. The gates are $\mathrm{Re}_{\mathrm{stokes}} = 10^{-2}$ on the gap and lamina and $\mathrm{Re}_{\mathrm{max}} = 10$ on the channel. The design code computes both from the geometry and the local flow and refuses a design that violates either (its M14); the physics auditor separately checks the drawn device's inlet against its own, stricter channel limit.

Two things are worth saying plainly rather than leaving to be found. First, **the channel gate can bind**: for a water-like fluid, ≈16 at 90 μL min^{-1} exceeds $\mathrm{Re}_{\mathrm{max}} = 10$, and the design code refuses that operating point even when it uses the local flow, so the envelope is a real constraint on throughput rather than a formality. For platelet-rich plasma the answer turns on a viscosity no paper here states: the same point passes above about 1.4 mPa s and is refused below it, as at the 1.2 mPa s the preprint's code takes for plasma. Dinh ran V1 with platelet-rich plasma at exactly this rate (P-C4.2), so whether the design code would refuse the lineage's own operating point turns on that same unstated number; if it does, it refuses on the channel gate alone, since the gap and lamina gates, on which the selection physics rests, stay at about 3×10^{-3} or below there, under their 10^{-2}. Second, the gap figure depends on which estimate is used: the flux argument above is the one the scope gate evaluates, and a *length*-rescaling estimate at the same velocity gives different numbers (≈1.3 using the 19 μm gap width, ≈0.02 using the outlet lamina width), which must not be conflated with it. The Stokes premise on which the whole skim derivation rests is therefore checked, and checked by a stated method. `[code: CIF_v2.py reynolds, design (M14); physics auditor (P8); reference implementation, VISCOSITY_PLASMA; glossary A.9]`

Two software contributions are out of *this argument's* scope, deliberately.
The targets-to-photomask tool (Prospectus C7) and the machine-enforced validity envelope (Prospectus C8)

carry no argument nodes. They are delivered as verified software, warranted by their acceptance suite and the physics auditor rather than by a defeasible inference. The silence is a decision, recorded here so that it reads as one. *Naming caution:* those are the *prospectus* numbers, not this appendix's; the collision is declared under "C7 and C8 are the *graph's* numbers" (§1.3).

Critical questions.

CQ1: Is the conditionality doing real work, or is it a hedge?
Real work. The realizability leg is *partially closed* on three named envelope boundaries, and one of them has a published counterexample: the lineage taper, which runs below the plateau floor at its wide end (C8 CQ7). An unconditional version of this thesis would be false of a device that already exists. → *answered: the conditions are load-bearing, and one is demonstrably binding.*

CQ2: Does the one live open question threaten the thesis?
No, and this is checked rather than argued. The single undefeated mechanistic rival, a size-dependent $f_{\text{eff}}(d)$ as the cause of the knee, moves exactly four labels when granted: the rival and its mechanism come *in*, the finite-particle reading and the specific residual decomposition go *out*. The thesis, the operating law, the exponents and the *existence* of the knee are all untouched, because a size-dependent f_{eff} is itself a form of size-dependence and so cannot defeat the variation it would merely reattribute. The four-label blast radius is asserted and re-checked on every build. → *answered: the engineering thesis survives whichever way the mechanism resolves.*

CQ3: Does the thesis depend on the appendix's chosen weights?
Barely, and the dependence is measured rather than assumed. Collapsing all six evidential tiers to any common value above the proof-standard threshold moves *no* label; reversing the ranking entirely (expert testimony above deductive proof) moves *one* of 151. The labelling therefore depends on the tier weights through a single binary distinction, whether an argument supporting a beyond-reasonable-doubt or clear-and-convincing conclusion clears the threshold, and not through the ordering at all. → *answered: the ordinal ranking is inert; only the threshold does work.*

Residual risk. Inherited, not new: the three envelope boundaries of Leg 4, and the open evidentiary questions of Legs 1 and 2. The thesis adds no residual of its own: an accrual carries no inference beyond the conjunction of its premises.

Status. Warranted, conditionally: every leg is established within its own section; the weakest of them, realizability, is partially closed on three named and dormant envelope boundaries; and the headline claim inherits that conditionality in the same breath as it is stated.

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